



**GREEN
CLIMATE
FUND**

Meeting of the Board
1 – 4 July 2018
Songdo, Incheon, Republic of Korea
Provisional Agenda Item 17

GCF/B.20/10/Add.07

8 June 2018

Consideration of funding proposals – Addendum VII

Funding proposal package for FP088

Summary

This addendum contains the following three parts:

- a) A funding proposal summary titled “Biomass Energy Programme in the South Pacific” submitted by Korea Development Bank;
- b) No-objection letters issued by the national designated authority(ies) or focal point(s); and
- c) Environmental and social report disclosure;

These documents are presented as submitted by the accredited entity and the national designated authority(ies) or focal point(s), respectively. Pursuant to the Comprehensive Information Disclosure Policy of the Fund, the funding proposal titled “Biomass Energy Programme in the South Pacific” submitted by Korea Development Bank is being circulated on a limited distribution basis only to Board Members and Alternate Board Members to ensure confidentiality of certain proprietary, legally privileged or commercially sensitive information of the entity.

Table of Contents

Funding proposal submitted by the accredited entity	1
No-objection letters issued by the national designated authorities or focal points	80
Environmental and social report disclosure	60



Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: Biomass Energy Programme in the South Pacific

Country/Region: Fiji, Papua New Guinea (PNG)

Accredited Entity: Korea Development Bank (KDB)

Date of Submission: December 18, 2017

Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	FINANCING / COST INFORMATION
Section C	DETAILED PROJECT / PROGRAMME DESCRIPTION
Section D	RATIONALE FOR GCF INVOLVEMENT
Section E	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section F	APPRAISAL SUMMARY
Section G	RISK ASSESSMENT AND MANAGEMENT
Section H	RESULTS MONITORING AND REPORTING
Section I	ANNEXES

Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"[FP]-[Agency Short Name]-[Date]-[Serial Number]"

List of Abbreviations

AMA	Accreditation Master Agreement
BAU	Business as Usual
CPI	Consumer Price Index
DSCR	Debt Service Coverage Ratio
E&S	Environmental & Social
EBITDA	Earnings Before Interest, Taxes Depreciation, and Amortization
EoI	Expression of Interest
EPC	Engineering, Procurement and Construction
ESIA	Environmental and Social Impact Assessment
ESMF	Environmental Social Management Framework
ESMP	Environmental and Social Management Plan
ESS	Environmental and Social Safeguards
FAA	Funded Activity Agreement
FAO	Food and Agriculture Organization of United Nations
FEA	Fiji Electricity Authority *Renamed EFL (Energy Fiji Limited) as a result of the corporatization
FJD	Fijian Dollar
FY	Fiscal Year
GCF	Green Climate Fund
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GWh	Gigawatt/hour
ICCC	Independent Consumer and Competition Commission
IFC	International Finance Corporation
INDC	Intended Nationally Determined Contribution
IPCC	Intergovernmental Panel on Climate Change
IPP	Independent Power Producer
KDB	Korea Development Bank
LTV	Loan to Value
MoU	Memorandum of Understanding
NCV	Net Calorific Value
NDA	National Designated Authority
NDC	Nationally Determined Contribution
O&M	Operations and Maintenance

PF	Project Finance
PNG	Papua New Guinea
PPA	Power Purchase Agreement
PPL	Papua New Guinean Power Limited
PPP	Public-Private Partnership
PS	Performance Standards
RC	Regulatory Contract
SESA	Strategic Environmental and Social Assessment
SEP	Stakeholder Engagement Plan
SIDS	Small Island Developing States
SPC	Special Purpose Company
SRWC	Short Rotation Woody Coppices
TA	Technical Assistance
TBD	To Be Determined
tCO ₂ eq	Tons of Carbon Dioxide equivalent
UNDP	United Nations Development Programme
USD	United States Dollar
VAT	Value-added Tax
VEP	VAT Exclusive Price

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Biomass Energy Programme in the South Pacific
A.1.2. Project or programme		programme
A.1.3. Country (ies) / region		Fiji, Papua New Guinea (PNG)
A.1.4. National designated authority (ies)		Fiji : Ministry of Economy PNG : Office of Climate Change and Development, Ministry of Environment
A.1.5. Accredited entity		Korea Development Bank (KDB)
A.1.5.a. Access modality		☐ Direct ☒ International
A.1.6. Executing entity / beneficiary		Executing Entity: (Component 1& 2) SPCs (Component 3) KDB Beneficiary: Local communities of Fiji and PNG
A.1.7. Project size category (Total investment, million USD)		☐ Micro (≤ 10) ☐ Small ($10 < x \leq 50$) ☒ Medium ($50 < x \leq 250$) ☐ Large (> 250)
A.1.8. Mitigation / adaptation focus		☒ Mitigation ☐ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		December 18, 2017
A.1.10. Project contact details	Contact person, position	In-suk Park, Head of Climate Finance Team Suwon Yi, Senior Manager
	Organization	Korea Development Bank
	Email address	insuk@kdb.co.kr suwon@kdb.co.kr
	Telephone number	+82-2-787-7209 +82-2-787-7211
	Mailing address	22 Eunhaeng-ro, Youngdeungpo-gu, Seoul 07242, Korea

A.1.11. Results areas (mark all that apply)	
<u>Reduced emissions from:</u>	
☒	Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.)
☐	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☐	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☒	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
<u>Increased resilience of:</u>	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☐	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)
☐	Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.)
☐	Ecosystem and ecosystem services (E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Project / Programme Executive Summary (max 300 words)

Please provide a brief description of the proposed project/programme, including the objectives and primary measurable benefits (see [investment criteria in section E](#)). The detailed description can be elaborated in [section C](#).

The “Biomass Energy Programme in the South Pacific (hereinafter the “Programme”)” originates from the Fijian government’s endeavors to control disastrous invasive alien species – *Spathodea campanulata* (hereinafter ‘African tulip’) – that threatens native biodiversity and ecosystem functions. Also, agricultural land degradation by the uncontrollable invasive species has been a huge threat to livelihoods of vulnerable farming communities. Nevertheless, the government was reluctant to commence eradication of African tulips without finding the way to make economic use of the harvested invasive species which would be otherwise wasted and a disposal headache to the authorities in charge. The Programme finally provides the government a highly effective solution that utilizes African tulips as a biomass fuel source. Responding to the country needs, the Programme will supply low-carbon biomass as reliable base load electric power in Fiji, and expand the Fijian model to PNG, which will contribute to energy paradigm shift in South Pacific Small Island Developing States (SIDS) as a whole.

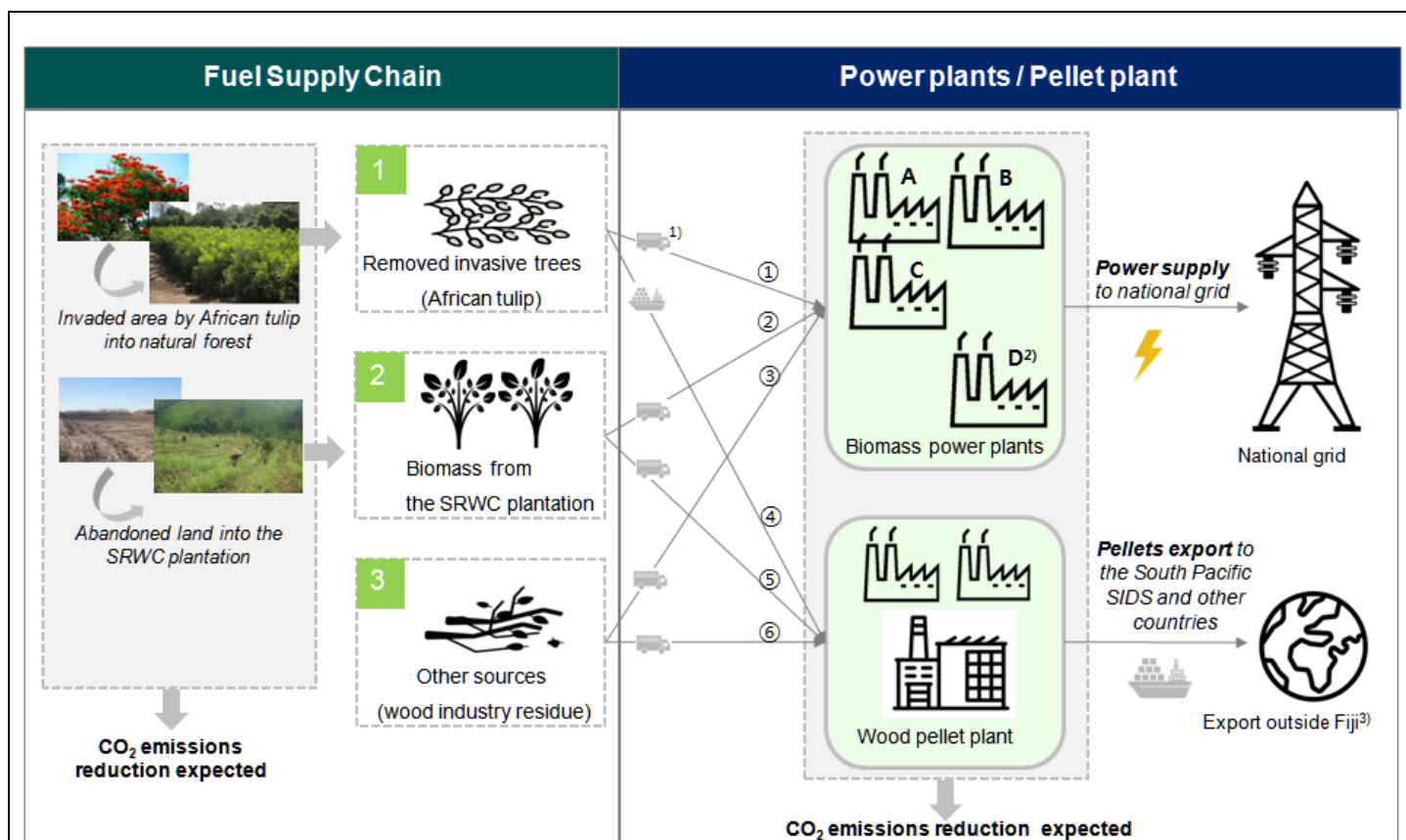
Based on the background above, the Programme is structured by three components in total:

- 1) Development of four biomass power plants;
- 2) Development of one wood pellet plant; and
- 3) Technical assistance

The Programme satisfies two results areas of ‘energy access and power generation’ and ‘forestry and land use’, through which it plays a pivotal role in achieving NDC (Nationally Determined Contribution) goals. Fiji NDC Implementation Roadmap¹, recently presented during COP23, notes the Programme’s biomass energy deployment including sustainable biomass plantation as the largest mitigation contributor; 210,000tCO₂/year mitigation target in biomass sector cannot be reached without the Programme. Most importantly, the Roadmap demonstrates how well the Programme upholds country ownership by integrating needs of its target countries through close consultation with NDA. Fiji will harness the Programme as a steppingstone towards her reaching 100% renewable energy target by 2030.

The Programme will also contribute to establishing sustainable land use management systems through eradication of African tulips and reforestation with native species in forests where invasive species control actions are taken. Moreover, it will do plantations over lands idled and degraded by dwindling sugarcane industry and disruptive weather patterns affected by climate change. These activities fundamentally aim to convert land that has been long left idle, neither having been natural forest nor currently used for agricultural purposes into productive land that can be once again vitalized for ensuring socio-economic benefits to vulnerable local communities. In addition, the Programme ensures to recover local food production chains disturbed by African tulip trees, which finally confirms the Programme as a genuine win-win solution.

¹ GGGI (2017) Fiji NDC Implementation Roadmap 2017-2030 Setting a pathway for emissions reduction target under the Paris Agreement



1) Transport distance from biomass sources to the plants :

- average sourcing distance of ①, ②, ③, ⑤, ⑥ : 15 km, average sourcing distance of ④ : 64 km

2) One biomass power plant in PNG will source biomass from the SRWC plantation(②) and other woody residue(③) at local areas.

African tulip(①) will be used for three biomass power plants and the pellet plant in Fiji, not a biomass power plant in PNG.

3) Part of pellets can be sold to off-grid regions in Fiji.

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	The Programme is expected to be approved via internal approval process of KDB within 120 days after GCF Board Approval.
Expected financial close (if applicable)	Each project of the Programme has a different financial closing schedule. The first biomass project in Fiji is expected to have financial closing in 4Q 2018, while the last biomass project in PNG is expected to have financial closing in 3Q 2022.
Estimated implementation start and end date	Start: 4Q 2018 End : 3Q 2025
Project lifespan	25 years

B.1. Description of Financial Elements of the Project / Programme

Please provide:

- *an integrated financial model in Section I (Annexes) that includes a projection covering the period from financial closing through final maturity of the proposed GCF financing with detailed assumptions and rationale; and a sensitivity analysis of critical elements of the project/programme*

The Programme will be financed through the GCF loan, KDB loan, and private sector's equity investment and loan. The anticipated financial structure of each sub-component of Component1 is as below:

Elements	Percentage	Sub-component1.1	Sub-component 1.2~1.3	Sub-component 1.4	Total
Debt – GCF	32.0%	16.3	18.5	18.5	71.7
Debt – KDB	5.0%	2.5	2.9	2.9	11.2
Debt – Other Investors	33.0%	16.8	19.0	19.1	73.9
Equity – Project Sponsors	30.0%	15.2	17.3	17.3	67.2
Total	100.0%	50.8	57.7	57.8	224.0

The anticipated financial structure of each sub-component of Component 2 will have a similar financial structure with Component1 as below:

Elements	Percentage ²	Sub-component2.1	Sub-component2.2	Sub-component2.3	Total
Debt – GCF	15.1%	13.8	3.5	13.4	30.7
Debt – KDB	15.5%	14.3	3.6	13.7	31.6
Debt – Other Investors	39.4%	36.0	9.1	34.7	79.8
Equity – Project Sponsors	30.0%	27.5	7.0	26.5	60.9
Total	100.0%	91.7	23.1	88.2	203.0

Component 3 is TA to be financed by the GCF grant for capacity building, knowledge sharing, and creating enabling environments for renewables, etc.

Please refer to the Integrated Financial Model attached in Section I as Annex 3 for more details.

- *a description of how the choice of financial instrument(s) will overcome barriers and achieve project objectives, and leverage public and/or private finance*

Since the South Pacific region has its context-based barriers such as low sovereign credit ratings, currency risks, and construction risks, few renewable energy projects have been financed so far by private investors.

The GCF loan financing will be a critical catalyst in triggering private sector investments by lowering currency exchange

² Percentage of each sub-component is slightly different.

and sovereign credit risk, and filling financing gaps. Furthermore, equity investments from tentative project sponsors with sufficient capacity and technology proven by the first pilot biomass project in Fiji have played an anchor role in mobilizing private sector loans by reducing construction risks.

Besides of filling in the financing gap, the GCF's grant paired with loan investment will support Fiji and PNG to formulate an enabling environment where the uptake of renewable energy can be well realized and leveraged by private finance. In fact, private financial investors from Korea and Fiji are now positively considering investment in the Programme given the positioned senior loan paired with grant support from the GCF.

For further details, refer to Section D.1.

- *a breakdown of cost estimates for total project costs and GCF financing by sub-component in local and foreign currency and a currency hedging mechanism:*

The indicative breakdown of cost estimates and the GCF funding amount and other co-financing by (sub) components is as below:

Component	Sub-component	Total	GCF	KDB	Co-financiers		Currency of financing	Currency of disbursement to recipient
					Equity	Other lenders		
Component 1. Development of four biomass power plants	Sub-component 1.1. Biomass power plant #1 in Sabeto, Fiji	50.8	16.3	2.5	15.2	16.8	million USD (\$)	million USD (\$)
	Sub-component 1.2. Biomass power plant #2 in Fiji	57.7	18.5	2.9	17.3	19.0		
	Sub-component 1.3. Biomass power plant #3 in Fiji	57.7	18.5	2.9	17.3	19.0		
	Sub-component 1.4. Biomass power plant #4 in PNG	57.8	18.5	2.9	17.3	19.1		
Component 2. Development of one wood pellet plant	Sub-component 2.1. Pellet production infrastructure	91.7	13.8	14.3	27.5	36.0	million USD (\$)	million USD (\$)
	Sub-component 2.2. Capacity increase for existing transportation infrastructure	23.1	3.5	3.6	7.0	9.1		
	Sub-component 2.3. Biomass power plant for the wood pellet plant	88.2	13.4	13.7	26.5	34.7		
Component 3. Technical assistance	Sub-component 3.1. Technical assistance in Fiji	1.0	1.0				million USD (\$)	million USD (\$)
	Sub-component 3.2. Technical assistance in PNG	3.0	3.0					
Total project financing		431.0	106.4	42.8	128.1	153.7		

B.2. Project Financing Information							
	Financial Instrument		Amount	Currency	Tenor		Pricing
(a) Total project financing	(a) = (b) + (c)		431.0	million USD (\$)			
(b) GCF financing to recipient	(i) Senior Loans		102.4	million USD (\$)	As defined in the Term Sheet (Annex 5)		As defined in the Term Sheet (Annex 5)
	(ii) Grants *		4.0	million USD (\$)	N/A		N/A
	* Please provide economic and financial justification in section F.1 for the concessionality that GCF is expected to provide, particularly in the case of grants. Please specify difference in tenor and price between GCF financing and that of accredited entities. Please note that the level of concessionality should correspond to the level of the project/programme's expected performance against the investment criteria indicated in section E .						
	Total requested (i+ii)		106.4	million USD (\$)			
(c) Co-financing to recipient	Financial Instrument	Amount ³	Currency	Name of Institution ⁴	Tenor	Pricing	Seniority
	Senior Loans	6.2	million USD (\$)	<u>KDB</u>			<u>Pari Passu</u>
	Senior Loans	36.6 ⁵	million USD (\$)	<u>KDB</u>	<u>TBD Based on market and project conditions</u>	<u>TBD Based on market and project conditions</u>	<u>Pari Passu</u>
	Senior Loans	111.0	million USD (\$)	<u>Mirae Asset Daewoo Loan Fund</u>			<u>TBD</u>
	Senior Loans	42.7	million USD (\$)	<u>Other investors</u>			<u>TBD</u>
	Equity	128.1	million USD (\$)	<u>Project Sponsors</u>	<u>N/A</u>	<u>N/A</u>	junior
	Lead financing institution ⁴ : KDB, Mirae Asset Daewoo * Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution						
(d) Financial terms between GCF and AE (if applicable)	In cases where the accredited entity (AE) deploys the GCF financing directly to the recipient, (i.e. the GCF financing passes directly from the GCF to the recipient through the AE) or if the AE is the recipient itself, in the proposed financial instrument and terms as described in part (b), this subsection can be skipped. As stated in the Term Sheet, the agreed Accredited Entity Fee is four per cent (4%) of the total GCF proceeds, subject to finalization of the AMA and FAA. It should also be noted that the Accredited Entity fee is not included in the project budget in section (b) above.						

³ The amount of each financial instrument is indicative, and the ratio of co-financing will be slightly different in each project based on market conditions. The exact amount will be decided after the financial closing of the final project. GCF loan to go proportionately with co-investors including KDB.

⁴ Co-financing institutions and lead financing institutions except for KDB are indicative.

⁵ A portion of senior loan (the actual covered amount) where lenders are protected and covered against the credit risk of the borrower and/or political risk of the project by third party entities (i.e. MDBs/ECAs/private entities)

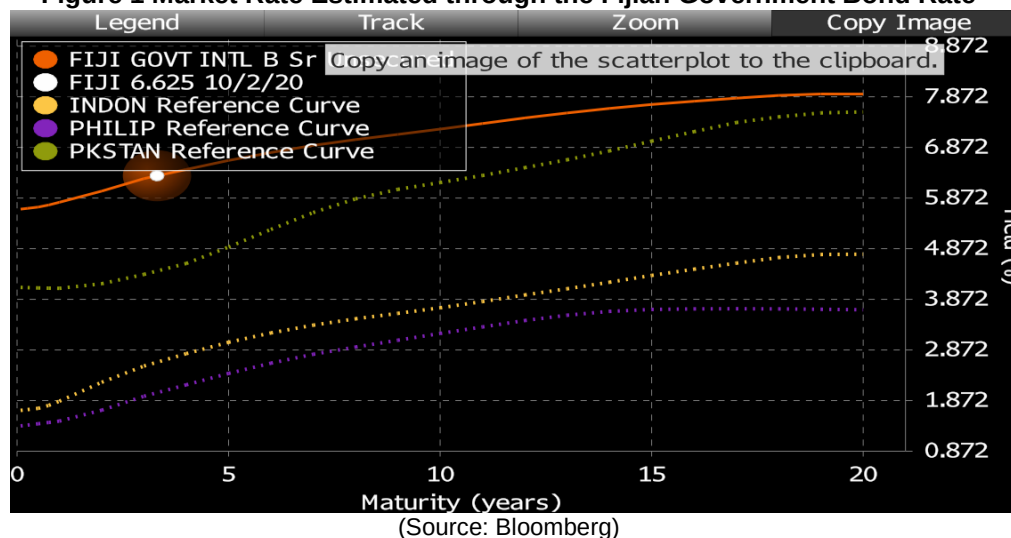
B.3. Financial Markets Overview (if applicable)

How market price or expected commercial rate return was (non-concessional) determined?

The only government bond issued in USD for the past five years in Fiji is a USD 1 billion government bond issued in September 22, 2015 (5-year maturity). The initial interest rate of the government bond was 6.625% and currently, the distribution rate is 6.518%.

Based on the information, it is estimated that the USD market rate is formed at about 7% with 8-year maturity (Assuming no other credit risk except country risk).

Figure 1 Market Rate Estimated through the Fijian Government Bond Rate



Private financial institutions such as local commercial banks and pension funds have been contacted to check their willingness to participate in financing the Programme, and they requested financial terms as below:

- LTV: Less than or equal to 70%
- DSCR: Minimum simple DSCR 1.2x

Please provide an overview of the size of total banking assets, debt capital markets and equity capital markets which could be tapped to finance the proposed project/programme.

Due to only a few renewable energy IPP projects in Fiji and PNG, constraints impeding adequate financing for the Programme exist. In the case of a pilot project implemented by potential project sponsors of the first biomass power plant in Sabeto, Fiji (Sub-component 1.1), foreign commercial banks including those in Korea and local commercial banks in Fiji were contacted for debt financing, but due to political, construction, and foreign exchange risks, financing was unsuccessful. As a result, the corporate financing scheme based on sponsor credits was applied for the pilot project. However, this is not a replicable or expandable financial structure due to credit limit set by institutional internal control measures. In order to encourage participation of other commercial banks, an approach of enhancing stability of the financial structure through the GCF participation is needed.

Please provide an overview of market rates (i.e. 1-year T-Bill, 5-year government bond, 5-year corporate bond (specify credit rating) and 5-year syndicate loan.

Provide examples or information on comparable transactions

Please refer to the information provided for the above question.

Please fill out applicable sub-sections and provide additional information if necessary, as these requirements may vary depending on the nature of the project / programme.

C.1. Strategic Context

Please describe relevant national, sub-national, regional, global, political, and/or economic factors that help to contextualize the proposal, including existing national and sector policies and strategies.

The proposed Programme is designed to address various needs of South Pacific SIDS, namely Fiji and PNG, in an integrated manner. That is, the Programme will enable transformation of South Pacific SIDS into countries with not only sustainable economies, but also climate-resilient environment. National and regional context that could be dealt through the Programme include eradication of invasive plants, utilization of idled and degraded lands, and diversification of the countries' energy portfolios.

<Fiji>

Country context. With a population of approximately 890,000, Fiji is a large tropical island country in the South Pacific Ocean with a land area of more than 18,333 square kilometers. The country consists of 330 islands, and amongst those islands almost 70% of total population resides in its two largest islands, Viti Levu and Vanua Levu. Although Fiji is relatively developed and is a relatively large country among South Pacific SIDS, it confronts various challenges that other neighboring South Pacific SIDS face. In particular, Fiji is highly vulnerable to climate change and extreme weather events which often affect health care, water resources, agricultural activities and infrastructure located near the coastal regions of the country, resulting in slowing down of economic growth. For instance, in 2016, Cyclone Winston⁶ slammed the entire nation, killed more than 40 people, left tens of thousands homeless, and severely damaged schools, housings and public infrastructure. Since the disastrous cyclone outbreak, the Fijian government has developed the cyclone response system, and arranged a cluster team in collaboration with Fiji Red Cross Society. At present, the cluster team acts as a control tower in cyclone emergency.

Expansion of renewable energy capacity. Fiji mainly utilizes hydropower and diesel power plants for electricity generation; the power generation mix for 2015 was 44.90% hydro, 52% diesel and heavy fuel oil, 0.63% wind with the remaining 2.47% provided by few IPPs. However, both hydropower and diesel power generation are unstable sources of energy, as they are highly influenced by external conditions such as seasonal fluctuation in the precipitation rate and global energy price. Therefore, the two energy sources are growingly becoming unreliable, given the increase in economic activities and electricity demand. According to the Fiji Electricity Authority (FEA)'s Power Development Plan, the electricity demand is expected to grow by 3% annually until 2020. Accordingly, total electricity and peak load demand are expected to reach 1,352 Gigawatt-hour (GWh) and 256 MW, respectively in 2020.⁷ In order to reduce the environmental impact of its electricity generation portfolio and to meet the growing electricity demand, Fiji Electricity Authority is looking for potential IPPs and alternative options in favor of renewable energy sources.

Aligned with such needs for more sustainable energy sources, the Department of Energy announced Fiji National Energy Policy 2013-2020 in 2013, which attempts to respond to increasing energy demand, and to decrease imported fossil fuels hindering the country's economic development. The policy aims at raising the share of renewable energy sources in electricity generation to 81% by 2020 and 99% by 2030.

The Fijian government's commitment to the impacts of climate change by reducing CO₂ emission in the energy sector has been demonstrated in Fiji's NDC Implementation Roadmap. The NDC Implementation Roadmap, recently published by the GGGI under the guidance of the Ministry of Economy, suggests medium to long-term actions for the Government of Fiji. The Roadmap includes actions to increase renewable energy capacity, and highlights the role of biomass and waste to energy as the largest GHG mitigation contributor. In summary, these mitigation actions include the followings and items which include this Programme are underlined:

⁶ Cyclone Winston, rated a category 5 cyclone, swept through the Fiji Island in 2016. During this disaster, 43 people were confirmed dead, 156 people injured, and at least two people missing. Communication systems and power grids have been disturbed for more than a year. Due to the limited number and capacity of existing power stations, the cyclone impact on a single power station may cause a cascade impact on the nationwide electricity supply.

⁷ Fiji Electricity Authority (FEA), Power Development Plan (2010)

Table 1 Mitigation Actions in NDC Implementation Roadmap

Period	Contents
Short-Term (2017–2020)	• Grid extension and improvements • Increased RE power generation (12MW biomass, 32MW solar) • Increased sustainable biomass for fuel use (14,000 ha)
Medium-term (2021–2025)	• Grid extension and improvements • Increased RE power generation (28MW biomass/Waste to energy, 26MW solar, 84 hydro) • Increased sustainable biomass for fuel use (5,000 ha)
Long-term (2026–2030)	• Grid extension and improvements • Increased RE power generation (25MW biomass/biogas, 69MW solar) • Increased sustainable biomass for fuel use (14,000 ha)

Forestry and land use in Fiji. Fiji has been suffering from rampant diffusion of invasive trees throughout their lands, as the trees are increasingly threatening agricultural activities and natural ecosystems. Particularly, African tulip, which was purposefully introduced in the 1930s in Fiji as a street tree and household ornaments, has been invading arable lands and threatening biodiversity as it takes up the sunlight and moisture needed for native plants. Recognizing such adverse impacts of African tulip, the Fijian government has identified African tulip as a major threat to the indigenous forest estate as well as to plantations and agricultural land as early as 1999, when FAO implemented a project to identify the level of the problem posed by African tulips. Also, in 2000, the Fijian government held a national workshop on the control of African tulip in coordination with FAO, through which a survey of the tree distribution in Viti Levu Island was conducted. Moreover, recognizing such adverse impacts of African tulip, the government is currently developing a strategy to control and eradicate the species from Fiji.

As an initiative to eradicate the invasive species, in 2015, the Ministry of Fisheries and Forests issued an Expression of Interest (EoI) seeking for interested parties that are willing to assist the Fijian government in controlling and eradicating African tulip in Fiji, as part of a programme titled “Implementing a Ridge to Reef (R2R) approach to Preserve Ecosystem Services, Sequester Carbon, Improve Climate Resilience and Sustain Livelihoods in Fiji” supported by UNDP. The programme’s components included 1) conservation of terrestrial and marine biodiversity, 2) conservation, restoration and enhancement of carbon stocks through sustainable forestry, and 3) integrated natural resources management. Activities for the second component included removal of invasive alien species, notably African tulip. Further details regarding the EoI released by the Ministry can be found in Section E.5.3.

In addition, Fiji has been experiencing decline in the country’s major industries such as sugarcane and pine industry. This has led to ongoing exit of farmers and companies from such industries, and consequently, much of the artificially afforested lands for such traditional businesses have been left idle. Each biomass plant of the proposed Programme will be accompanied with a plantation business, which will utilize ex-sugar and ex- pine plantation areas that are abandoned or severely degraded because of 3 to 4 times pine harvesting. This will serve as a sustainable means for fuel supply of the plant, and will ultimately enable income generation for the local community.

<PNG>

Country context. With a population of approximately 7.5 million, PNG is a large tropical country in the South Pacific regions with a land area of 462,840 square kilometers. PNG is classified as a lower middle income country⁸ and has a GDP per capita of USD 2,744 in 2015. The country’s economy is largely divided into formal and informal sectors. Formal sector is where natural resources such as copper, gold, and oil are exported on a large scale, while informal sector mainly refers to agricultural sector which 85% of the population, majority living in rural areas, depend their livelihoods on. Such high dependence on the limited number of primary industries leaves the country vulnerable to internal and external factors such as total output changes and global commodities downturns. Moreover, the livelihoods of the rural population

⁸ World Bank, PNG, <http://data.worldbank.org/country/papua-new-guinea>

that depend on self-subsistent agriculture are threatened by extreme weather and climate events such as El Nino, as agriculture is a climate-sensitive sector.

Expansion of renewable energy in PNG. According to PNG National Energy Policy 2016-2020, it has about 580 MW of installed capacity, with most of it covered by hydro (230 MW, 39.7%) and diesel (217 MW, 37.4%).⁹

The electricity demand in PNG is anticipated to rise sharply as needs for energy intensive activities such as mining, infrastructure development, and commercial and residential building constructions will increase.¹⁰ Also, the PNG government is aiming to raise the national electrification rate to 70% by 2030, especially by expanding rural electricity infrastructure.¹¹ Under such circumstances, the PNG government is turning to renewable energy sources for which the country has large potential, but is yet to be fully utilized. As an effort in expanding renewable energy capacity, the PNG government initiated the National Strategic Plan 2010-50, also known as the PNG Vision 2050. Among the seven pillars for development, the Environmental Sustainability and Climate Change Pillar focuses on the development of energy sector, and posits that the PNG economy should attain 100% electricity generation from renewable and sustainable sources by 2050. Specific level of mitigation targets will be determined by the availability of external funding.

Aligned with the PNG Vision 2050, the PNG Government has set total electricity capacity targets by 2030 in the Medium Term Development Plan 2011-2015, which is a five-year rolling development plan for the country, as below:

Table 2 Targets of Total Electricity Capacity (MW) in PNG

Technology	2020	2030
Hydro	580	1,020
Gas	280	390
Diesel and Coal	90	60
Geothermal	112	350
Biomass	24	75
others	24	75
Total	1,110	1,970

(Source: Papua New Guinea, Department of National Planning & Monitoring)

Moreover, PNG submitted its INDC (Intended Nationally Determined Contribution) in September 2015, and the final version of NDC (Nationally Determined Contribution) to the UNFCCC in March 2016, which was the first formal submission of an NDC to the UNFCCC parties. The NDC sets out a clear goal of raising the share of renewable energy for electricity generation to 100% by 2050.¹²

Forestry and land use in PNG. Despite covering less than 0.5% of Earth's surface, it is estimated that PNG holds 5-10 % of global biodiversity. Since 74% of PNG is covered with forest,¹³ the country has been paying much attention to seeking ways for effectively managing forests and land use. National Forest Development Guidelines announced in 2009 reflects PNG government's initiatives of achieving economic recovery through the development of the nation's natural resources including the forest resources, while taking into consideration environmental preservation, social and cultural values of the forests in PNG.

⁹ PNG Department of Public Enterprises and Department of Petroleum and Energy, National Energy Policy 2016-2020 (2015)

¹⁰ Department of National Planning and Monitoring, Papua New Guinea Development Strategic Plan 2010-2030 (2010)

¹¹ ANZ, Powering PNG into the Asian Century (2015)

¹² PNG INDC

¹³ World Bank, PNG, Forest area <http://data.worldbank.org/indicator/AG.LND.FRST.ZS?locations=PG&page=2> (2015)

C.2. Project / Programme Objective against Baseline

Describe the baseline scenario (i.e. emissions baseline, climate vulnerability baseline, key barriers, challenges and/or policies) and the outcomes and the impact that the project/programme will aim to achieve in improving the baseline scenario.

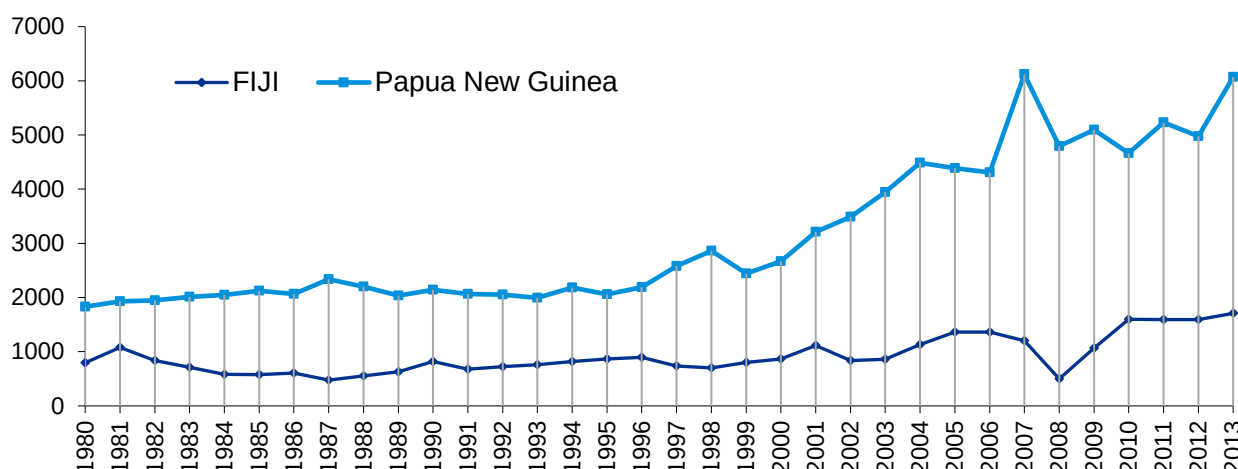
<Emission Baseline>

Fiji contributes minimally to GHG emissions on global scale. Over the span of 30 years, from 1980 to 2013, Fiji has only experienced a subtle increase in the GHG emissions. However, according to the INDC, business as usual (BAU) scenario for increases in total fossil fuel consumption for energy production considering extrapolated population and

economic growth would give total GHG emissions in 2030 from the energy sector of around 2500 kt with an electricity sector GHG emission level of around 500 kt.¹⁴ Despite having relatively small GHG emissions compared to heavy emitting countries, Fiji is one of the most vulnerable countries facing the worst effects of climate change due to its physical location. Fijian communities are experiencing climate change impacts such as eroding shorelines and riverbanks, shortage of water, depletion of fisheries stock, reduction in food production, increase in large-scale flooding increase in outbreaks of vector borne diseases, and sea level rise.

In case of PNG, the GHG emission has been drastically increased last 10 years, reaching 6,073 kt in 2013. According to the INDC, under BAU scenario, GHG emissions in 2030 could reach 18 Mt CO₂ per year.¹⁵ Being one of South Pacific SIDS, just like Fiji, PNG has suffered a lot from extreme weather events related to climate change impacts. These events will eventually lead to inundation of coastal areas and low lying inland. It is also expected to cause loss of wetlands, changes to fisheries, forestry and agriculture sectors, alteration to water resources and land use practices and impacts on health, particularly vector borne diseases such as Malaria and other related water and air borne diseases.¹⁶

Figure 2 CO₂ emissions in Fiji and PNG (kt)



<Programme Objective¹⁷>

This Programme will mitigate 4,736,230tCO₂eq (annually 189,449 tCO₂eq) in Fiji and PNG from 1) transportation of African tulip; 2) development of the SRWC plantation on idled and degraded land, and 3) the plant operation in Component 1 and 2. E.1.2. presents the detailed calculation.

C.3. Project / Programme Description

Describe the main activities and the planned measures of the project/programme according to each of its components.

<Programme overview>

The Programme consists of three components: 1) development of four biomass power plants; 2) development of one wood pellet plant; and 3) technical assistance. The figure below depicts all the courses from securing biomass fuel sources through the biomass power plants and pellet plant to biomass outputs and more detailed description of each component and efforts to increase sustainability are further added on.

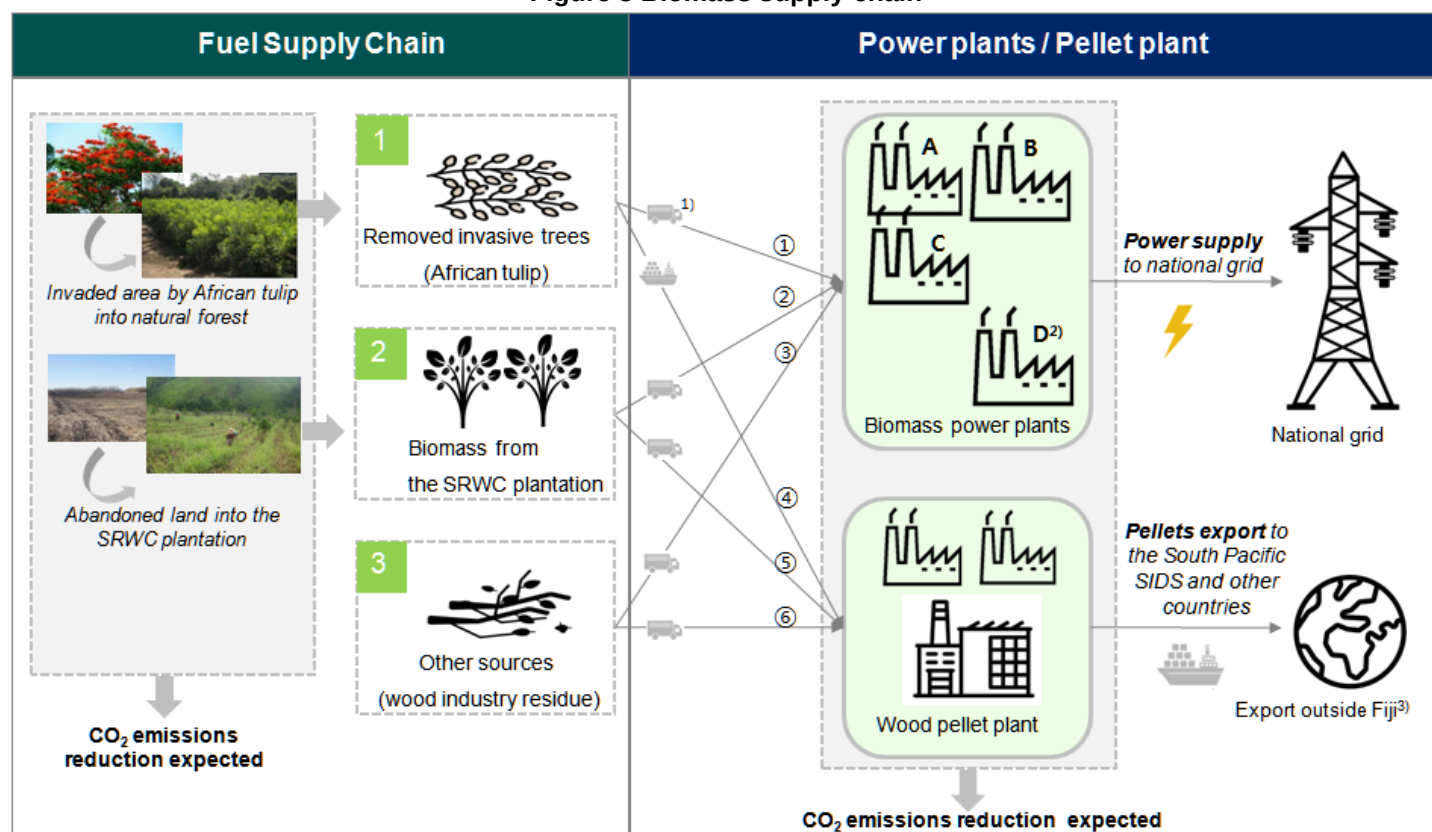
¹⁴ Fiji INDC

¹⁵ PNG INDC

¹⁶ UNDP, Climate Change Adaptation

¹⁷ The precise capacity of each biomass power plant is subject to change depending on the needs and conditions of each project site. Also, the amount of CO₂ emission reduction is an indicative one based on brief analysis, and more detailed analysis will be conducted with TA grant provided by the GCF.

Figure 3 Biomass supply chain



<Component 1> Development of four Biomass Power Plants

Biomass power plants will be developed along with land use for sustainable biomass production in both Fiji and PNG.

Biomass power plant. Component1 plans to build four biomass power plants with each 12MW capacity: three in Fiji; and one in PNG. The first biomass power plant will be developed in Sabeto, Fiji. The construction is expected to start in 4Q of 2018, and to be completed in 4Q of 2020 (the end date of EPC). The rest three biomass power plants (two additional 12MW biomass power plants in Fiji and one more plant in PNG) will be developed in accordance with the implementation schedule (see C.8).

All the biomass power plants of Component 1 are expected to have similar specification. Key details of the specification are shown as below.

- The site area required for each biomass power plant is estimated to be 10~20 ha. Only idle and industrial areas will be considered for the project site.
- Approx. 84,000 tons of biomass fuel per year is needed for each power plant.
- The biomass power plant shall be developed on conventional steam turbine technology including a turbine generator having a nominal gross capacity of approximately 12MW.
- Electricity generated from the plants will be sold to the Fiji Electricity Authority (FEA) and Papua New Guinean Power Limited (PPL).

Sustainable biomass production. Biomass fuel resources have been categorized into 3 different categories: 1) eradication of African tulip; 2) SRWC plantation; and 3) wood residues. Eradicated African tulip and cultivated SRWC are the primary fuel source while wood residues, by-products from harvesting and sawmills, are the secondary fuel source. Three biomass power plants in Fiji will use all the three options as biomass fuel sources, while one biomass power plant in PNG will use only the second and third options as fuel sources except for the first option of African tulip. All biomass fuel resources will be sourced from nearby areas in order to reduce inefficient transportation costs and ensuing emissions, and will be used as a woodchip type (raw chopping biomass).

When it comes to sourcing the three types of biomass fuel, the Programme ensures sustainability of land use management in Fiji and PNG.

(1) **African Tulip.** The first biomass fuel source is harvesting of African tulip which is widely spread in Fiji, where the amount of land that has been invaded is estimated to be 30,000ha¹⁸. The Fijian Ministry of Forests has identified the invasive African tulip tree as a major threat to the indigenous forests as well as to agricultural land as early as 2000. For the past 16 years, however, little has been done to take a coordinated action, with the tree continuing to spread over larger areas. African tulip harvesting presents a great momentum for the Ministry of Forestry to initiate invasive species control activities. In addition, the Programme suggests a solution making use of eradicated African tulip piled in stacks for energy production.

All the harvesting activities will be monitored by the Ministry of Forests for its compliance to Fiji Harvesting Code as well as international industrial best practices. There is no need for land purchase or lease associated with the African Tulip fuel supply operation. Only access rights need to be obtained through an access agreement with landowners in order to conduct the harvesting of Africa Tulip trees. After eradication, decision on the use of the harvested land lies with local landowners' desires: agricultural activities or reforestation. In case of native forestry which takes up the most part of African tulip areas, on the other hands, reforestation will be implemented with carefully selected native species which will be planted in forests where invasive species control actions are taken.

(2) **SWRC plantation.** While other fuel sources are only for the short to medium term supply or only form a small portion of the entire biomass fuel supply, SRWC plantation is designed to supply a significant amount of reliable fuel source to the power plants in the long term. Each power plant is targeting 5,000 ha of land for SRWC plantation, which is focusing on idled and degraded areas that sugar cane and pine trees were once planted in, and that are outside of the present economic supply zone. The plantation areas will be selected among places close to biomass power plants to minimize the transportation distance.

For plantation, *Gliricidia sepium* (hereinafter "Gliricidia") which is well known for its fast growing and easy coppicing nature and high calorific value has been selected. Developing the SRWC (Short Rotate Wood Coppices) plantation with Gliricidia in ex-sugar cane lands will not only contribute to forest and land use management in a sustainable and efficient manner by recovering degraded lands into arable lands, but also help revitalizing local economy for community members.

(3) **Wood residues.** For sustainable biomass supply, wood residues from harvesting and sawmill processing will be considered. Sawmills generate more than 100,000 tons/year of residue which is normally treated as waste and either burnt at the sawmill yard or dumped. The Ministry of Forests encourages sawmill operators not to dump or burn the residue but to find a sustainable use for it. This Programme offers the opportunity to reuse the wasted residues in an efficient way as well as finally gives the economic benefit to local sawmill companies.

Individual fuel agreements between the project SPC and sawmills are to be made for every project. Under the agreement, surplus, if any, goes to corresponding projects. Nonetheless, in the same conditions such as location, transportation efficiency, etc, the supply priority goes to biomass power plants than the pellet plant.

¹⁸ The amount of land that has been invaded is estimated to be 65,000ha, however 35,000ha of this land is for steep areas, forests roads, waterways and buffer-zones, therefore only 30,000ha is available.

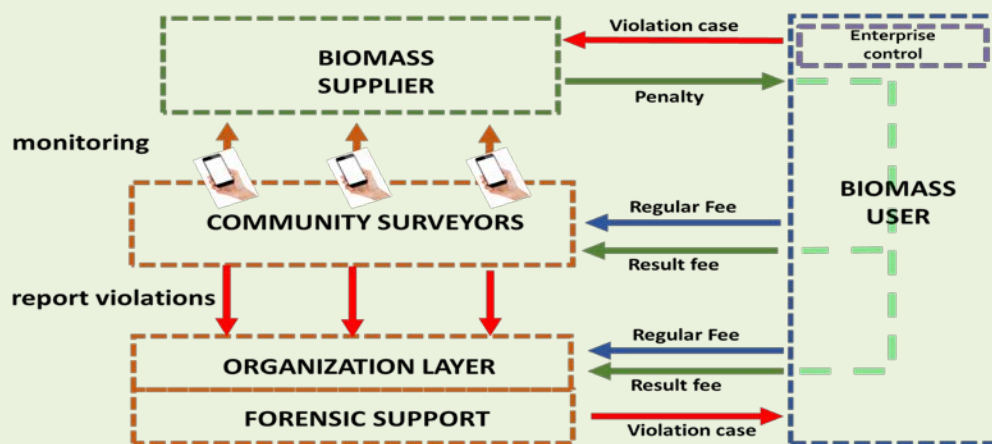
※ Biomass Monitoring System

The Programme contains essential arrangements for monitoring activities in compliance with the GCF standards (see H.2). In addition to such arrangements, the Programme intends to set up a biomass monitoring system that leads to spontaneous local engagement in an efficient manner, and that benefits end beneficiaries of affected communities. The biomass monitoring system will contribute to sustainable biomass supply chain. This monitoring system will be designed after considering local context, applicable technology with affordable cost, and the most appropriate providers in the relevant market.

Technological change occurs at an exponential rate. Also, an innovative solution in a context can be an unattractive solution in another context. Accordingly, the Programme will do its best to devise the most feasible and attractive monitoring technology that corresponds the local circumstances at the timing of the project implementation. For instance, the following methodologies could be considered as one of technology application options.

Firstly, an innovative monitoring technology could use smart-phone or satellite-based data for land-use and biomass fuel resources so as to protect forests from illegal and unsustainable exploitation, to verify the legality and sustainability of commodities entering the relevant supply chains, and simultaneously to improve the economic stability, welfare, and security of forest-based communities. An indicative monitoring methodology could begin with a step that trained community surveyors register approved harvest zones and biomass power plant locations with the use of web or smart-phone application. The social supply chain tracking technology is based on a trusted digital mechanism around registered individuals and communities, where, instead of just tracking the commodities, the relationships through which a flow of the commodities can be tracked.

< Example 1. Establishing a Smart-phone & Satellite System >



A second option is to adopt a drone-based monitoring system with regards to poor telecommunication infrastructure especially in many places where forest activities will be undertaken. Monitoring results on harvesting and plantation could be updated on a quarterly basis at a website. Collected data sets would be shared to the public, except for some confidential data sets. Local people will regularly check the data after registration as community surveyors. Actually, Nabou project is currently trying to go through the feasibility of the drone-based technology as a trial. The following pictures are a capture screen (left) and a monitoring trial activity (right) using drones.

< Example 2. Establishing a Drone-based System >



Above-mentioned technologies are attractive in theory but those should be applicable in local context of vulnerable target countries. In short, the feasibility of such indicative technologies and methodologies should be thoroughly studied, analyzed, and finally proven in terms of applicability and cost for two years before the very first project comes into commercial operation. A series of such technology financing process will be financed by the GCF grant (Component 3). Aside from the two examples, the Programme is open to other technological options if a more proper technology is identified and the technology can be applied within the reasonable boundary of spending. A finally chosen and developed monitoring system could be realized firstly around small areas, and be expanded into farther affected regions with tracking all biomass moves within the supply chain.

Additionally, the supervision of the Fijian Ministry of Forests will be added on to all the procedures of the biomass supply chain in accordance with local regulations in order to deliver added values of the GCF involvement into vulnerable end beneficiaries in local communities

Biodiversity and sustainability in land use management. The Programme contributes to biodiversity and sustainability of Fiji's indigenous environment by eradicating African tulip trees that uncontrollably crowd out native species. With the invasive species control being done, all the likely environmental and social risks will be thoroughly analyzed and properly managed in accordance with the Plantation and Harvesting Guidelines developed by the project sponsor in accordance with both IFC and local standards (see Appendix 6 in ESIA for the first biomass power plant in Sabeto). Once invasive species control operations are done, selected native species will be planted along buffer areas to develop natural forests and to prevent African tulip trees from being re-established, which clearly promotes biodiversity and sustainability of Fiji.

In addition, the Programme ensures no adverse biodiversity impact of SRWC plantation as it will develop plantation in ex-sugarcane or ex-pine plantation areas. The ex-plantation areas are characterized by fragile biodiversity because no indigenous species can be properly settled down in the long idled and severely degraded ex-plantation land. The SRWC plantation, therefore, does not require any clearance of indigenous forests, which confirms no threat nor likely adverse impact on biodiversity at all, and further enhance sustainability by recovering the soil fertility.

In case of PNG, the same criteria utilizing the idle and degraded lands will be applied to select approximately 5,000 plantation areas in PNG and the site-specific guidelines for plantation based on its unique contexts will be developed.

※ Biomass vs. Food Security

The Programme makes sure no possibility to distort the local food production chain including food markets in both Fiji and PNG as land required for the Programme will be chosen among idle or degraded areas with no competitive tension between 'land for food' and 'land for energy'. In short, the change in land use will not cause adverse consequences on the local food security as the project sites have not been used for agricultural purposes for a number of years. Instead, the Programme will address threats to food security by transforming idle and degraded lands into productive and fertile plantation, which anticipates the improvement of livelihoods for local residents as well as the enhancement of food production.

In case of Fiji, a total crop production has fallen due to a declining sugarcane export industry damaged by change of the European Union's sugar subsidy policy. As the ex-sugarcane plantation has been developed for the purpose of export, the Programme's use of ex-sugarcane plantation areas is unlikely to have adverse impacts on the local food production status.

The total land areas acquired for biomass plantation will be 50,000 ha (5,000 ha for a biomass power plant * 3 projects + 35,000 ha for a pellet plant) in Fiji. Since the agricultural land is 425,000 ha*, the land to be used for the plantation is approximately 11.8% of a total agricultural area in Fiji.

At present, 39,000 ha out of 80,000 ha of sugarcane plantation areas and 30,000 ha of 79,000 ha pine plantation areas are being idled and abandoned without productive activities in Fiji. The Programme makes sure that all the 50,000 ha of SRWC plantation areas in Fiji will be chosen only within those idle lands. The plantation area, which is approximately 11.8% of the total agricultural area, will be 0% of the actual agricultural productive area, as it will be all located in the idle abandoned portion of Fiji agricultural land, and it will have direct and multiple positive effects for the local rural economy. The land selection amongst idle areas will be periodically and thoroughly monitored by third party experts under the supervision of KDB.

Furthermore, Fiji is supposed to witness a decline in population since a slight increase of around 25,000 by 2030, according to the United Nations Population Division. Asian Development Bank says 0.5% population growth rate in Fiji, which means little growing needs to increase food production. From a long term perspective, the conversion to productive land use will deliver socio-economic benefits to Fijians so that the local community could afford enough to maintain livelihoods including food security.

PNG expects different dynamics in terms of land use area and population forecast. The total land areas acquired for the plantation of a single project is to be 5,000 ha in PNG. Comparing 1,190,000 ha* of the agricultural land, the portion used for the plantation is less than 0.4% of total agricultural areas in PNG. The identical monitoring activities will be done in PNG. Rather, considering the United Nations Population Division's substantial population growth forecast in PNG for the project lifespan, the Programme will be indeed able to address socio-economic needs of the increasing population in the foreseeable future. A series of socio-economic benefits will contribute to enabling the growing population to improve quality of life as well as sustain their livelihoods.

(*Source : <http://www.fao.org/countryprofiles/en/>)

<Component 2> Development of one Wood Pellet Plant

The wood pellet plant comprises of three sub-components; 1) pellet production infrastructure; 2) capacity increase for existing transportation infrastructure; and 3) biomass power plant for the wood pellet plant. The Component 2 has a strong rationale as a mitigation project that satisfies two result areas of (1) forestry and land use and (2) energy access and power generation.

Firstly, the Component 2 satisfies the result area of forestry and land use. The development of a wood pellet plant

contributes to CO₂ mitigation in forestry and land use by carrying out plantation. Plantation for pellet will be only in idle and degraded lands of 35,000 hectares in Vanua Levu which were abandoned with dwindling sugarcane and pine industries of Fiji. In brief, the pellet plant development and accompanying plantation will contribute to 329,717 tCO₂ mitigation (see E.1. Impact Potential for detail).

Secondly, the Component 2 satisfies the result area of energy access and power generation. The development of a wood pellet plant contributes to CO₂ mitigation in pellet purchasing regions. Priority target areas for pellet provision are South Pacific SIDS – that is the reason for the Programme to have a title 'in the South Pacific', not only Fiji and Papua New Guinea. Furthermore mitigation impacts further reaching out to Asian countries with a surging demand on biomass pellets.

Sub-component 2.1: pellet production infrastructure. This sub-component plans to build a wood pellet production plant for producing pellets of 500,000 tons per year. The plant is composed of chipping, drying, pelletizing, and storing facilities. This sub-component requires around 60ha for such facilities.

Biomass fuel sources for the wood pellet production will be secured in a similar structure with Component1- the sources will be secured through 1) eradicated African tulip, 2) SRWC plantation and 3) wood residues from local sawmills. For the SRWC plantation, Component 2 plans to develop 35,000 hectare of plantation in idle and degraded ex-plantation lands near the pellet plant in Vanua Levu.

Sub-component 2.2: Capacity increase for existing transportation infrastructure. This sub-component plans to renovate the Wairiki Port¹⁹ and its surrounding facilities such as loading conveyor in Vanua Levu for efficient fuel sourcing and smooth transport of produced pellets. This Port and its related facilities are now being utilized by nearby sawmills owned and run by Tropik Wood, a statutory body of Fiji. Related to the Port usage, a concern has been gradually increased by the Fijian government, owing to a low frequency of usage caused by declining local industries. The sub-component aims at an increasing the frequency of its utilization in an efficient way. As part of the renovation work, an additional barge docking point is supposed to be built for the sake of moving harvested African tulip into the pellet plant.

Sub-component 2.3: Biomass plant for the wood pellet plan. This sub-component plans to establish the captive biomass power plant of 24MW capacity attached to the pellet production infrastructure; this sub-component is composed of two sets of main facilities of 12MW biomass power plant. The maximum electricity requirement for the pellet infrastructure operation is 17.33MW. The power generation plant will supply electricity for wood pellet production with any surplus electricity to be sold to the FEA.

Potential buyers. Produced wood pellets optimized for long-haul shipping are used for mostly export purpose, and not for the biomass plants within this Programme. The potential export target regions are SIDS and Asian markets along with consideration of a surging demand, transportation efficiency, sustainability of long-term contracts, and so on.

When it comes to the SIDS market, produced pellets will be supplied to isolated off-grid areas in Fiji as well as to neighboring SIDS with a heavy reliance on imported fossil fuels, together with delivery of small scale biomass facilities as a package. Small-scale biomass power facilities are expected to be supplied into areas without national grid connection in South Pacific SIDS, along with grant resources from international aid agencies.

An initial target area is vulnerable off-grid areas in Fiji²⁰ and the facilities with 100kW capacity are planned to be built. Once the initial trial settles down in Fiji, the modality will be expanded into neighboring SIDS. The pellet will selectively target off-grid communities – representing deprivation and climate vulnerability – scattered all the way throughout South Pacific. Such wood pellet supply to the South Pacific is expected to bring about transformation towards renewable

¹⁹ The tentative location of the pellet plant is in Wairiki jetty in Vanua Levu, which is an unseen place on Google Maps. The site is definitely different from Wairiki in Taveuni Island.

²⁰ Even though produced pellet is mostly sold for export, parts of pellets can be sold to selectively off-grid communities – representing deprivation and climate vulnerability in Fiji. By the way, produced pellet will not be used for on-grid biomass power plants under the Programme.

biomass energy in off-grid communities having no choice but diesel.

Once the initial trial settles down in Fiji, the modality will be expanded into neighboring SIDS. Neighboring SIDS have been actually paying a close attention to benefits the Programme would deliver. A series of consultation with aid agencies are being in progress in order to provide supports for distributing small-scale facilities into the vulnerable in isolated areas.

Especially, wood pellets fit for long-distance shipping will give benefits to other Pacific SIDS such as Tonga, Vanuatu, and Marshall Islands where their soil profiles based on coral islands are too fragile to develop renewable energy facilities and plantation. The produced pellets can replace fossil fuels as the very economic alternative for these coral island countries when it comes to electricity generation. The increasing inter-Pacific trading activities further anticipate the enlargement and diversification of the current small-scale biomass market within the Pacific region as a whole.

The pellets will also be sold in Asian countries with limited availability of readily accessible biomass fuels under long-term supply contracts on a commercial basis. Korea and Japan are the potential export target countries in that they have strengthened policy schemes for supporting biomass.

<Component 3> Technical Assistance

Technical assistance is composed of 1) capacity building and knowledge sharing, and 2) creating the enabling environment for renewables (only PNG), and 3) development of CO₂ calculation methodology. Since the Programme aims at the expansion of the Fijian model to the South Pacific region as a whole, a substantial portion (75%) of grant will be allocated to PNG in order to create the enabling environment for scale-up and replication.

Sub-component 3.1 Technical assistance in Fiji

By the support of the GCF grant, Sub-component 3.1 includes capacity building and knowledge sharing, and development of CO₂ Calculation Methodology in Fiji. For the sake of management of capacity enhancement, Sub-component 3.1 carries out capacity building activities by organizing training sessions at different career levels from the managerial personnel to local engineers, and carries out knowledge sharing activities by transferring technology into various stakeholders at a national level. As part of the capacity building and knowledge sharing element, a proper and feasible biomass monitoring system will be studied, analyzed, and proposed. The Programme will make endeavors to integrate the monitoring system into local entities in change. These activities will enable transfer of state-of-the-art technology to local employees in Fiji, and will raise awareness of the best technologies applied and to achieve a greater success of the Programme.

As well, financial support will be given to the in-depth methodology study on how to measure, report and verify CO₂ savings from the use of biomass fuel stock in a sophisticated way.

Sub-component 3.2 Technical assistance in PNG

In reference to capacity building and knowledge sharing, and calculation methodology for CO₂ savings, Sub-component 3.2 has a similar structure with sub-component 3.1. As planned in Sub-component 3.1, Sub-component 3.2 will deliver capacity building training and knowledge sharing events that include the enhancement of the existing monitoring system of target countries, and devise precise CO₂ measuring methodology on the basis of the PNG context. Although all the activities feature a similar structure and content with those in Fiji, more financial resources will be assigned into these activities because PNG is extremely an unexplored land in comparison with Fiji. The Programme has already pioneered Fiji for biomass pilot, and has obtained baseline information and knowhow by undergoing trial and error. Undoubtedly, more efforts and resources are required for obtaining a comprehensive understanding and making the Programme work in the PNG. In order to replicate the successful Fijian model in PNG, these activities are necessarily needed and the grant portion of the GCF will be used for this in PNG.

Considering the two countries' similar features like the Pacific community, some parts of TA activities – such as biomass monitoring system and CO₂ calculation methodology development, etc. – may be combined and carried out simultaneously if reasonable and necessary.

Additionally, the Sub-component 3.2 focuses further on creating an enabling environment for successfully deploying renewable energy including biomass. In that PNG has experienced few renewable IPPs and no successful biomass IPP so far, the Programme sees a necessity to formulate the nation-wide enabling environment where multi-level stakeholders contribute to and benefit from the Programme.

First of all, the Programme considers supporting the PNG government to make a model framework for successful renewable energy IPPs on the basis of the PNG context. As a first step, the Sub-component 3.2 may support a review on relevant policy, statutory, and other regulatory instruments that are applicable to the local energy market, and may identify all the potential parties involved in IPPs. Based on investigation results, the model framework will be established and the renewable IPP manual will be produced, containing the way how to design renewable IPP projects, and to facilitate a series of contracts and negotiation procedures. More specifically, the model framework will be a useful guide that provides eligibility criteria for selecting IPP stakeholders, and standards for unit costs, procurement, relevant product specifications, power purchase contracts, financing, responsibilities, and so on. Along with the model framework production, the Programme plans to support the enhancement of the institutional governance system in terms of capability and accountability. Therefore, it aims for responsible institutions of PNG to have a full understanding on the model framework, and to utilize that in proper ways as intended.

As another part of the enabling environment making work, the Sub-component 3.2 will conduct investigation for mapping the pattern of land use and dynamics of biodiversity nation-wide across PNG. The mapping output will be part of valuable baseline information when it comes to enhancing the Programme's suitability for the PNG context, and to the PNG government's forthcoming policy making activities. In short, this mapping result will greatly support to make the model framework materialized in the unique PNG context.

The detailed plans of TA are described in the table below, and the indicative procurement plan is attached as Annex 11 in Section I.

Table 3 Detailed plans for Technical Assistance

Item	Activities	Country of Operation
Capacity building and knowledge sharing	• Training session for technical, E&S and gender capacities, and sustainable land management from the managerial personnel to local engineers. • Knowledge sharing events for learning advanced technologies and experiences from international best practices • Capacity building for designing the biomass monitoring system and knowledge sharing for spreading the system out	Fiji and PNG
Creating enabling environments for renewables	• Setting a model framework for renewable IPPs after a thorough review on relevant regulatory instruments: 1) eligibility criteria for IPP stakeholders; and 2) systematic standards for unit costs, procurement, relevant product specifications, supply contracts, responsibilities. • Improving responsible authorities' institutional governance system so that the devised model framework could be properly realized in practice • Mapping the land use patterns and deforestation dynamics as well as local biodiversity features	Only PNG
Development of CO ₂ calculation methodology	• Sophisticated measuring methodology for CO₂ savings will be devised in accordance with varying local contexts and project design details.	Fiji and PNG

Provide information on how the activities are linked to objectives, outputs and outcomes that the project/programme intends to achieve. The objectives, outputs and outcomes should be consistent with the information reported in the logic

framework in section H.

The Programme has five projects in total under the overall framework of three components. Component 1 has four projects (development of four biomass power plants with sustainable land use management) and Component 2 has one project (development of a wood pellet plant with sustainable land use management) while Component 3 plans to conduct technical assistance. The inputs and outputs of each project are described in the figure below.

Figure 4 Input/output of Each Project in the Programme

Input	Component/Project/Sub-component		Output
USD 50.8mil	Component 1	Project 1 ■ 1.1 Biomass power plant #1 in Sabeto, Fiji	■ 12MW biomass power plant
USD 57.7mil		Project 2 ■ 1.2 Biomass power plant #2 in Fiji	■ 12MW biomass power plant
USD 57.7mil		Project 3 ■ 1.3 Biomass power plant #3 in Fiji	■ 12MW biomass power plant
USD 57.8mil		Project 4 ■ 1.4 Biomass power plant #4 in PNG	■ 12MW biomass power plant
USD 91.7mil	Component 2	Project 5 ■ 2.1. Pellet production infrastructure	■ 500,000t/yr wood pellet plant
USD 23.1mil			■ 24 MW biomass power plant
USD 88.2mil			■ 2.2. Biomass power plant for the wood pellet plant
USD 1mil	Component 3	■ Technical assistance in Fiji	■ Capacity enhancement and knowledge sharing
USD 3mil		■ Technical assistance in PNG	■ Creating enabling environments

*Sustainable forest management will be carried out with Project 1~5.

Section H elaborates how project activities of the Programme are designed to meet the target in a way consistent with the GCF's Results Management Framework and Performance Management Framework using the baseline scenario, assumptions, indicator, and means of verification.

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

Describe the quality of the management team, overall strategy and financial profile of the Sponsor (Executing Entity) and how it will support the project/programme in terms of equity investment, management, operations, production and marketing.

KDB will select project sponsors in order to establish a project SPC for each (sub) component in a open and transparent way and the project SPC will perform as an executing entity of each project. The Investment Criteria for selecting sponsors (Appendix 1 of the Term Sheet attached as Annex 5 in Section I) is as follows:

(i) Credit Requirement: Companies having credit ratings that are at least "BBB-" by S&P or "Baa3" by Moody's. In case of Korean sponsors, companies having credit ratings of at least "BBB-" by one of NICE Information Service Co, Ltd., Korea Ratings Co, Ltd. or Korea Investors Services, Inc.; or;

(ii) Track Records Requirement:

(1) Experiences that utilize commercially proven technologies and employ experienced management teams which are related to bio energy industries;

(2) Track records that meet the ongoing operating, monitoring and maintenance requirements for bio energy

industries; and,

(3) Experience of performing and/or to have performed at least one project in South Pacific SIDS related to bio energy industries

- Financial investors (financial institutions or private equity funds as the case may be) can also be invited as sponsors. Credit Requirement or Track Records Requirements are not required for financial investors

Based on the above criteria, KDB has been tentatively considering a consortium of GIMCO, GS Power and Mirae Asset Daewoo as the project sponsors for the first biomass power plant in Sabeto, Fiji (Sub-component 1.1). GIMCO and Mirae Asset Daewoo are also tentatively considered as the project sponsors for the wood pellet plant in Wairiki, Fiji (Component 2). The potential project sponsors were those of the pilot biomass power plant implemented in Nabou, Fiji, and thus are well-equipped with knowledge and experience pertaining to developing IPP projects in Fiji. Background information regarding each tentative partner of the consortium is as below:

GIMCO. GIMCO, established in 2001, is a small and medium-sized enterprise in Korea that specializes in plant facility, renewable energy, biomass co-generation, community energy supply system, and smart grid. For overseas projects, it has numerous track records in various biomass power projects including a 30 MW biomass power plant (engineering) in Philippines and a 1.5 MW biomass power plant (EPC) in Cambodia. It has especially paid attention to implementing small-scale IPP projects and fuel supply projects in South Pacific SIDS that lack power generation infrastructure, albeit their abundant bio-fuel resources such as wood and agricultural residues. In 2017, GIMCO has successfully completed Fiji's first biomass IPP project, and plans to replicate such achievement.

GS Power. GS Power is one of Korea's major IPPs established in 2000. It is in charge of load management during peak hours in the Seoul-Incheon area of South Korea, and operates a total of 950 MW plants and has abundant experience in developing and operating traditional and renewable power generation projects. In terms of financial soundness, the company has generated net cash flow of more than KRW 120 billion on average for the past five years, based on its stable foundation for earnings and low working capital. It has shown a strong financial structure with net borrowings of KRW 71.6 billion, the debt ratio of 79.5%, and the debt to total assets ratio of 11.1% at the end of 2014. GS Power is a subsidiary of GS Group, the 7th largest conglomerate in Korea, and this provides stable credit ratings for the company, thus enabling GS Power to have better access into the capital market.

Mirae Asset Daewoo. Mirae Asset Daewoo, the Programme's financial investor, will contribute to the successful implementation of the Programme based on its solid foundation of capital and experience of investing in various renewable energy projects. Mirae Asset Daewoo has participated in multiple renewable energy projects as principal investor or financial advisory. Moreover, it is in possession of the largest equity capital in Korea. Its revenue in FY2017 was KRW 10,299 billion and the owner's capital was KRW 7,385 billion in 2017. It is part of Mirae Asset Financial Group, which is one of Asia's leading independent financial services companies.

C.5. Market Overview (if applicable)

Describe the market for the product(s) or services including the historical data and forecasts.

Describe the competitive environment including the list of competitors with market shares and customer base and key differentiating factors (if applicable)

< Biomass Market in Fiji and PNG >

Fiji. In terms of securing fuel sources by means of forestry and land use management, there are no other competitors in the market since Eltech, a subsidiary of GIMCO as well as a potential fuel supplier, has been selected as the service provider for an Expression of Interest published by the Fijian Ministry of Fisheries and Forests. The company has obtained full and sole right in eradicating African tulip.

For biomass market, it is difficult to say that an active market for biomass power generation exists in Fiji, considering the fact that the Fiji Sugar Company and Tropik Wood Industries Ltd. are the only two companies that use biomass for power generation, but they utilize biomass only for self-supplying purpose. Therefore, there are no existing competitors regarding biomass power plant. Moreover, if the biomass pilot project previously implemented by the project sponsor of the first biomass power plant in Sabeto, Fiji (Sub-component 1.1) is set aside, there are no ongoing renewable energy projects or operating renewable power plants that can be considered to form a competitive environment.

Although the government is eager to accept private investment, it does not have the sufficient capacity to carry out IPP projects yet. This Programme will contribute to strengthening Fiji's capacity not only for biomass power generation but also sustainable forestry and land use management, and will attract private investment for renewable energy sector.

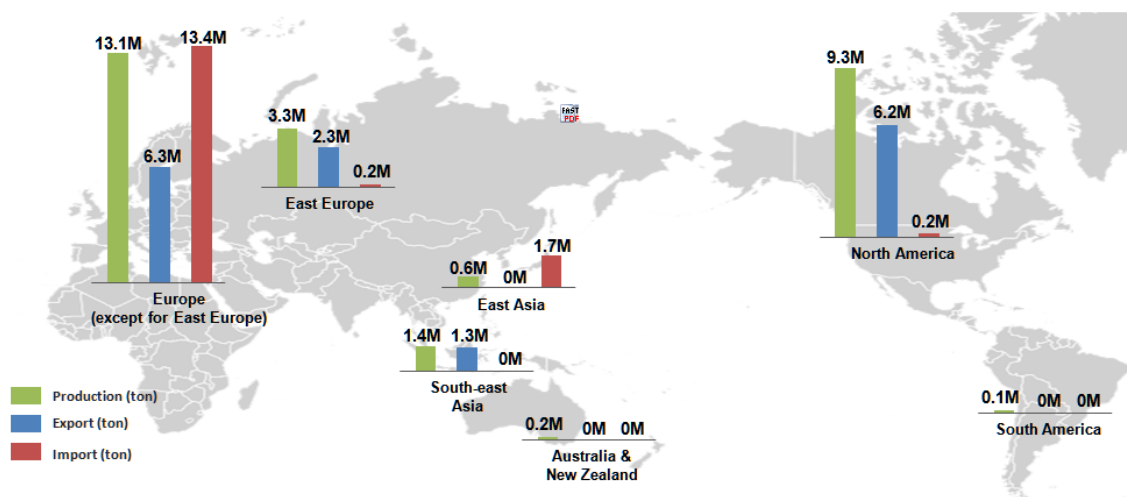
PNG. Biomass market for fuel and electricity trading in PNG is yet to be developed. Currently, a joint venture of Oil Search, the largest oil and gas company in PNG, and Aligned Energy, an international biomass energy consultancy, is establishing a 30 MW biomass power plant that supplies base-load power to the Ramu Grid through trees grown from a dedicated plantation. However, other than the project, the potential of biomass in PNG has not been fully utilized. This Programme will play a meaningful role of expanding biomass energy capacity of the country by means of sustainable forestry and land management.

< Global Pellet Market >

The current global pellet market overview expects global pellet demand to exceed 28 million tons per annum, and indicates Europe and North America as the largest consumer and supplier respectively.

We have been keen to the pellet demand trends globally and the two East Asian countries are being recognized as candidate consumer countries of the produced pellets. Both Korea and Japan have an increasing need for pellets as biomass business grows up on the fragile geographical basis of a biomass fuel source shortage. In Korea, biomass is a rapidly growing sector by the support of RPS (Renewable Portfolio Standard) scheme since 2012. Many utility companies have henceforth included biomass as part of their business strategy. Also, the Japanese government is providing the FIT (Feed In Tariff) program and more assisting biomass business in pursuit of diversifying renewable energy portfolio, so as to address a heavily weighted reality in favor of solar power.

Figure 5 Global Pellet Market (2015)



(Source: FAO Forestry Statistics)

Additionally, biomass needs are growing in the Pacific SIDS with a heavy reliance on diesel and lack of the government-led subsidy instruments. For these countries, biomass is an attractive renewable alternative in reference to the replacement for diesel, the potential as base load power, and more favorable costs. Biomass differentiates itself from other renewable options such as solar and wind as the smartest renewable solution for the South Pacific SIDS; a biomass plant can afford constant electricity generation as base load power and the off-grid biomass power system features flexible variations in terms of size. These characteristics make biomass the most attractive renewable energy in the SIDS context.

Provide pricing structures, price controls, subsidies available and government involvement (if any).

Fiji. In 2015, Fiji Department of Energy developed a standardized Power Purchase Agreement (PPA) with support of the UNDP in order to attract IPPs. Prior to the pilot biomass power plant developed by tentative project sponsor of Sub-component 1.1, there was no successful track record of IPP projects in Fiji. Therefore, the country's PPA framework had not been properly utilized until the project sponsor signed a PPA with the FEA at the rate of FJD 0.3308 VEP per unit. In other words, the project sponsor provided an opportunity for Fiji to newly establish and practice procedures for PPAs. Current IPP tariff rate is also set at FJD 0.3308 VEP per unit.

PNG. A national uniform tariff is applied to supply of electricity. The Independent Consumer & Competition Commission (ICCC) is responsible for regulating competition, controlling prices and protecting consumers in PNG, and is the regulating authority for the electricity industry. The major electricity company PPL submits a Regulatory Contract (RC) for a given period of time that contains tariff calculations to the ICCC and obtains approval. The calculation of power tariff reflects fuel price, PNG CPI, Australian CPI & Exchange Rate, and US CPI & Exchange Rate. Current RC is valid from January 1st, 2013 to December 31st, 2017. The Non-Fuel Weighted Average Price is approximately 0.17 USD/kWh, and the price is adjusted annually.²¹

C.6. Regulation, Taxation and Insurance (if applicable)

Provide details of government licenses or permits required for implementing and operating the project/programme, the issuing authority, and the date of issue or expected date of issue.

Fiji. For eradication of invasive plants and reforestation, a fuel supplier needs to obtain approval of environmental impact assessment from the Ministry of Fisheries and Forests and Department of Environment separately, and shall acquire Harvesting License which will be in accordance with the Ministry of Fisheries and Forests' Harvesting Code. The same applies to afforestation/reforestation, where a fuel supplier will manage plantation in compliance with the Harvesting Code.

For the construction of biomass power plants, a project sponsor is required to obtain 1) legal rights for land development from the iTaukei Land Trust Board, the administrator for all native lands in Fiji, 2) Special Water Rights permission from the Director of Lands for consent for river water extraction, 3) consent from the Department for Town and Country Planning and Rural Local Authority for building application, and 4) approval from the Department of Environment for an Environmental Impact Assessment. Also, during the construction and operation phases, the project sponsor needs to obtain approvals of Environmental Management Plans and waste disposal permits from the Department of Environment.

For the development of a wood pellet plant, the required licenses or permits include 1) land lease agreements for plantation and project facilities and 2) harvesting rights.

PNG. The Electricity Industry Act 2002 requires electric power companies to obtain licenses for electricity generation, transmission, distribution, and/or retail from the ICCC. Also, construction activities such as road development, logging, waste disposal, etc. need to comply with the PNG Logging Code of Practice. Major projects including forestry, mining and petroleum proposals need to acquire environmental permit from the Conservation and Environment Protection Authority to ensure minimal environmental and social effects.

Describe applicable taxes and foreign exchange regulations.

Below is the list of taxes and foreign exchange regulations relevant to the Programme.

<Fiji>

- IPP Tariff Rate
 - Minimum FJD 0.3308 VEP per unit

²¹ PNG Power Limited, Electricity Regulatory Contract (2013)

- Corporate Tax : 20% (single tax rate)
 - 'Tax holiday system' is applied in which companies producing bio-fuel via processing of agricultural products are exempt from corporate tax for initial 10 years.
 - For renewable energy and cogeneration, 'Tax holiday system' is applied in which the corporate tax on P&L generated from new business is exempted for initial 5 years.
- VAT : 15%
 - VAT is applied to most goods and services including construction and development cost.
- Foreign Exchange Regulations
 - Approval from Reserve Bank of Fiji is required for overseas remittance and borrowings.
 - Regulation on debt-to-equity ratio exists for foreign-invested companies and thus the debt-to-equity ratio of less than 300% must be satisfied after dividends.

<PNG>

- IPP Tariff Rate
 - Tariff applications are made annually by PPL, and fuel cost pass through is made on a quarterly basis.
- Corporate Tax : 30% (standard resident corporate), 48% (standard non-resident corporate)
 - Incentives: Depreciation incentives or concessions include following: a qualifying industrial plant not previously used in PNG is eligible for a flexible rate of depreciation up to 100%, but this deduction may not create a tax loss; eligible new plant may be subject to an additional 20% depreciation in the first year of use.
- VAT : 10%
 - VAT is applied to most goods imported into PNG and on any customs or excise duty paid on importation.
 - Land is excluded from Goods and Services Tax, but buildings and other improvements are subject to tax.
- Foreign Exchange Regulations
 - The Bank of PNG (Central Bank) administers exchange controls, including in respect of inward investment, registration of foreign capital, foreign currency and repatriation of capital and earnings.
 - Central Bank approval is required for non-exempt foreign exchange transactions including:
 - a. where a resident company contracts with another resident company in foreign currency.
 - b. where a guarantee or security contract is provided by a resident company in favor of an offshore lender, unless there is a benefit to PNG.
 - c. where a resident company opens and maintains foreign currency accounts within and outside PNG.
- Tax Incentives
 - Infrastructure Tax Credit: Resource companies are allowed a full credit for the cost of any approved infrastructure developments which they undertake for the benefit of the community in which they operate.
 - Rural Development Incentive: This scheme was designed to allow income tax exemption on the net income of new businesses set up in specifically designated underdeveloped areas that are not dependent on the exploitation of natural resources. It provides exemption to businesses in a rural development industry for up to 10 years after the first year of commencement of business. Losses arising from the newly exempt activities are deductible against taxable income from other activities.

Provide details on insurance policies related to project/programme.

In terms of insurance, KDB's requirements to projects will be consistent with international best practice standards for project financing. Specific insurance will be required and reviewed for each project during the due diligence process. Such insurance will be part of security package for the project lenders, including GCF.

C.7. Institutional / Implementation Arrangements

Please describe in detail the governance structure of the project/programme, including but not limited to the organization structure, roles and responsibilities of the project/programme management unit, steering committee, executing entities and so on, as well as the flow of funds structure. Also describe which of these structures are already in place and which are still pending. For the pending ones, please specify the requirements to establish them.

KDB and the GCF will enter into an FAA or FAAs, in the form of a funded sub-participation for the financing of Components 1 and 2, and in the form of a grant for the financing of Component 3. KDB will be an accredited entity for the Funded Activity at the Programme level. The respective SPCs will be the executing entities for each project under Components 1 and 2. The accredited entity shall be the sole executing entity for Component 3.

< Component 1 & 2 >

- ① Subject to the terms and conditions of the FAA, KDB will conduct due diligence, and select individual projects that will meet the Investment Criteria set out in Appendix 1 of the Term Sheet.
- ② KDB, as a lender, will enter into a loan agreement (each "Project Loan Agreement") with a SPC for each project. The Project Loan Agreement will include the Concessional Tranche, the terms and conditions of which will be in accordance with Appendix 2 of the Term Sheet.
- ③ KDB will remain the lender of record of the Concessional Tranche and the GCF will be a sub-participant for 100% of the Concessional Tranche in accordance with the terms and conditions of the FAA.

< Component 3 >

Subject to the terms and conditions of the AMA and FAA, KDB will provide technical assistance in Fiji and PNG. As the Executing Entity of the Programme, KDB will procure professional service providers, and sign TA contracts with them for the implementation of capacity building and knowledge sharing activities, the establishment of a model framework for renewables (only in PNG), and the development of CO₂ calculation methodology in Fiji and PNG. All the TA activities will be managed and supervised by KDB. Disbursement will be done as per Term Sheet (to be reflected in the FAA).

Describe construction and supervision methodology with key contractual agreements.

KDB, as an accredited entity, shall supervise key contractual agreements such as Project Loan Agreement and TA contracts, and provide to the GCF Annual Performance Reports (APRs) stating the implementation status of those agreements during the implementation period. Upon the completion of the implementation period, KDB shall provide a performance report about the Programme results and mitigation impacts on a semi-annual basis.

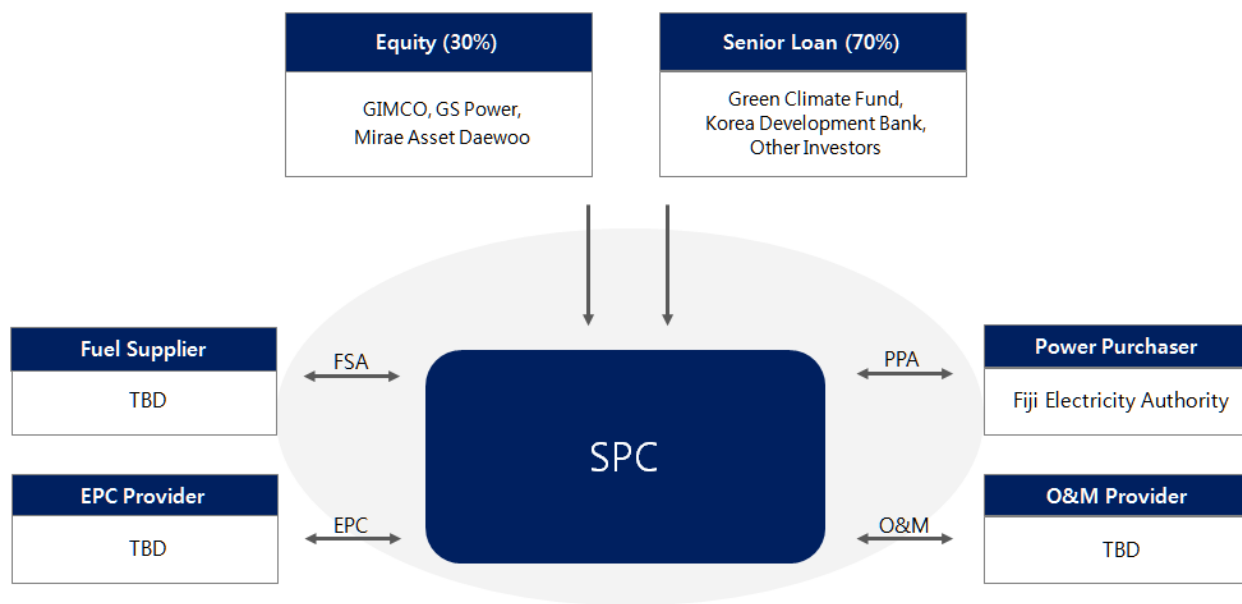
Also, interim and final evaluation reports will include assessment of the contract performance against investment framework criteria. Financial information will be audited annually by independent external auditors, and provided to the Fund. Please refer to H.2 for more details of reporting.

Describe operational arrangements with key contractual agreements following the completion of construction. If applicable, provide the credit analysis of key counterparties of key contractual agreements and/or structural mitigants to cover the counterparty risks.

Each biomass power plant and wood pellet plant will function under the framework of the above implementation structure of the Programme. When KDB selects project sponsors for a project, the project sponsors will establish an SPC as a duly registered entity in target countries of the Programme for project execution. KDB will select proper project sponsors pursuant to the Investment Criteria (Appendix 1 of the Term Sheet attached as Annex 5 in Section I) in an open and transparent manner.

Anticipated implementation of biomass power plants in Component 1 is as below;

**Figure 6 Indicative Implementation Arrangement for the Biomass Power Plant in Sabeto
(Sub-component 1.1)**



For the Sabeto biomass power project, the project site and tentative project sponsors have been identified by KDB and the PPA is under negotiation with the Fijian government. Subsequent biomass power plants in Component 1, including two 12 MW biomass power plants in Fiji (approximately 24 MW in total) and one 12 MW biomass power plant in PNG, are in the inception stage, and will be implemented under the similar structure.

KDB will also play a role as a financial investor to provide senior loans in partnership with the GCF and other private investors. KDB will review the key contractual agreements of the SPCs, and hedge potential risks by suggesting mitigation measures in case a risk factor is found.

In terms of the wood pellet plant, the SPC will be established and some off-takers will also participate for profit stability. The SPC will be provided with fuel through a fuel supply agreement with selected fuel suppliers based on KDB Investment Criteria. The wood pellets produced by the SPC will be supplied to owners of biomass power plants in South Pacific SIDS and, by extension, other countries with limited availability of readily accessible biomass fuel sources, which will create additional value by developing new export market for Fiji.

C.8. Indicative Timetable of Project/Programme Implementation

Please provide a project/programme implementation timetable in [section I \(Annexes\)](#). The table below is for illustrative purposes. If the table format below is used, please refer to the activities as numbered in Section H. In the case of outputs, please mark when all the required activities will be completed.

Calendar Quarters	2018			2019				2020				2021				2022				2023				2024				2025			
	2 Q	3 Q	4 Q	1 Q	2 Q	3 Q	4 Q	1 Q	2 Q	3 Q	4Q	1 Q	2 Q	3 Q	4Q	1 Q	2 Q	3 Q	4Q	1 Q	2 Q	3 Q	4 Q	1 Q	2 Q	3 Q	4 Q	1 Q	2 Q	3 Q	4 Q
Component 1. Development of four biomass power plants			√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√					
Sub-component 1.1. Biomass power plant #1 in Sabeto, Fiji			√	√	√	√	√	√	√	√	√																				
Output 1. 12MW biomass power plant #1 in Fiji			√	√	√	√	√	√	√	√	√																				
Activity 1.1. Project Preparation (Permit and Basic Engineering)			√																												
Activity 1.2. Project Execution (EPC)			√	√	√	√	√	√	√	√	√																				
Sub-component 1.2. Biomass power plant #2 in Fiji																	√	√	√	√	√	√	√	√	√	√	√				
Output 2. 12MW biomass power plant #2 in Fiji																	√	√	√	√	√	√	√	√	√	√	√				
Activity 2.1. Project Preparation (Permit and Basic Engineering)																	√	√													
Activity 2.2. Project Execution (EPC)																		√	√	√	√	√	√	√	√	√	√				
Sub-component 1.3. Biomass power plant #3 in Fiji																	√	√	√	√	√	√	√	√	√	√	√				
Output 3. 12MW biomass power plant #3 in Fiji																	√	√	√	√	√	√	√	√	√	√	√				
Activity 3.1. Project Preparation (Permit and Basic Engineering)																	√	√													
Activity 3.2. Project Execution (EPC)																		√	√	√	√	√	√	√	√	√	√				
Sub-component 1.4. Biomass power plant #4 in PNG																	√	√	√	√	√	√	√	√	√	√	√				
Output 4. 12MW biomass power plant #4 in PNG																	√	√	√	√	√	√	√	√	√	√	√				
Activity 4.1. Project Preparation (Permit and Basic Engineering)																	√	√													
Activity 4.2. Project Execution (EPC)																		√	√	√	√	√	√	√	√	√	√				

[illegible]



Calendar Quarters	2018			2019				2020				2021				2022				2023				2024				2025			
	2Q	3 Q	4 Q	1 Q	2 Q	3 Q	4 Q	1 Q	2 Q	3 Q	4Q	1 Q	2 Q	3 Q	4Q	1 Q	2 Q	3 Q	4Q	1 Q	2 Q	3 Q	4 Q	1 Q	2 Q	3 Q	4 Q	1 Q	2 Q	3 Q	4Q
Component 3. Technical assistance				√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√
Sub-component 3.1. Technical assistance in Fiji				√	√	√	√	√	√	√	√	√																			
Output 8. Capacity enhancement and knowledge sharing in Fiji				√	√	√	√	√	√	√	√																				
Activity 8.1. Project Preparation				√	√																										
Activity 8.2. Project Execution					√	√	√	√	√	√	√	√																			
Output 9. Development of CO ₂ calculation methodology in Fiji						√	√																								
Activity 9.1. Project Preparation						√																									
Activity 9.2. Project Execution						√	√	√	√	√	√																				
Sub-component 3.2. Technical assistance in PNG												√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	
Output 10. Capacity enhancement and knowledge sharing in PNG																√	√	√	√	√	√	√	√	√	√	√	√	√	√	√	
Activity 10.1. Project Preparation																√	√														
Activity 10.2. Project Execution																	√	√	√	√	√	√	√	√	√	√	√	√	√	√	
Output 11. Creating enabling environments for renewables in PNG												√	√	√	√	√	√	√	√	√	√	√	√								
Activity 11.1. Project Preparation												√	√																		
Activity 11.2. Project Execution													√	√	√	√	√	√	√	√	√	√	√								
Output 12. Development of CO ₂ calculation methodology in PNG																	√	√	√	√	√										
Activity 12.1. Project Preparation																	√														
Activity 12.2. Project Execution																		√	√	√	√										

[illegible]

D.1. Value Added for GCF Involvement

Please specify why the GCF involvement is critical for the project/program, in consideration of other alternatives.

The participation of GCF is critical for mobilizing and scaling up climate finance and attaining sustainability in stabilizing forestry and land use management systems and distributing clean renewable energy throughout South Pacific SIDS. Out of the total programme cost of USD 431.0 million, private investment of USD 324.6 million will be financed, given the GCF financing accounting for USD 106.4 million. The scale of financing would not have been feasible if the GCF had not participated.

Barriers for attracting commercial financing. The GCF involvement will create an environment where a large pool of private investors participates in the Programme, and multiply the effect of the initial financing. In fact, South Pacific SIDS have difficulties in attracting private financing due to the regional context-based investment hurdles such as low sovereign credit ratings, currency risks, and construction risks.

As an illustration, when the potential project sponsor had reached out to financial institutes for a pilot(Nabou) biomass project, the sovereign risk of pacific SIDS and relatively new renewable technology of biomass in Fiji were found to be high barriers for private sector financing. All the tapped commercial banks ended up not participating in the Nabou project. In the end, the corporate financing scheme based on sponsor credits was applied instead of project financing scheme. Given the limited financial capacity of self-funded sponsors, they can no longer afford to finance further projects in the Programme. Nevertheless, Nabou project is significantly meaningful in that it showcased a proven biomass technology in the Fijian context. Furthermore, this project garnered a strong support from the Fijian government and the Fijian government recognized the merits of biomass projects in Fiji to incorporate them into their action plans to deliver its NDC.

Against this backdrop, KDB put forth this biomass program in Fiji and PNG, aiming to scale up the proven biomass technology in Fiji and further expand to other parts of the South Pacific islands with the GCF, addressing the specific financing barrier identified in the pilot phase of the project. KDB is structuring this program with ECA or guarantee agencies to cover the sovereign risk for private sector investors while KDB and GCF financed the uncovered portion of the investment. In this way, GCF and KDB fill the gap for the financing and the private sector investors have a comfort by having reputable financial institutions like the GCF and KDB to take a higher risk tranche in the program. Indeed, the local private banks in Fiji are considering their funding the Programme in case that GCF financing in this project is approved and therefore confirmed.

GCF concessionality needed. The GCF participation does not necessarily lead to bringing the private sector's attention into the Programme because private investors still have the hurdle rate of financial terms. From lenders' perspective, a certain level of financial terms should be established in general – such as 70% or less of LTV and 1.2 or more of minimum simple DSCR (see B.3. Financial Markets Overview). Therefore, the GCF participation should satisfy a certain level of concessaionlity to attract commercial lenders. Taking an example of Sabeto, the Programme requested a minimum level of the GCF concessionality that could increase the minimum simple DSCR, the most important factor for investment decision, from 1.06 (non-GCF funding assumed) to 1.2 (generally accepted for the private). Without the GCF's involvement, private lenders' request for fulfilling the financial terms cannot be achieved, which would result in financing failure of the Programme (see E.6.3 for more detail). The Programme believes that the GCF engagement here is exactly in line with the GCF's mandate of crowding in private sector investors in climate actions in SIDS which are otherwise barely financed from the private capital.

Besides filling in the financing gap, the GCF's grant paired with loan investment adds value to the Programme. The GCF grant supports Fiji and PNG to formulate an enabling environment where the uptake of renewable energy combined with sustainable land use management can be well realized and effectively accelerated. In particular, the TA

activities financed by the GCF grant portion mainly aims to increase capacity of local human resources and support knowledge sharing in alignment with global standards. A series of TA activities will enable both target countries to leverage well-trained local personnel and voluntarily meet international-level E&S standards even after the project implementation period. Therefore, the TA activities well accords with the fundamental mandate of the GCF.

D.2. Exit Strategy

Please explain how the project/programme sustainability will be ensured in the long run, after the project/programme is implemented with support from the GCF and other sources, taking into consideration the long-term financial viability demonstrated in E.6.3. This should include a description of strategies for longer term maintenance of physical assets (if applicable).

The Programme well recognizes that devising a viable exit strategy at the outset is critical for achieving long-term sustainability.

First of all, provision of sufficient trainings and capacity building programmes will ensure sustainability of the Programme. Training local employees, landowners, and community members who are engaged in reforestation, forestry plantation, and biomass energy production is important in pursuit of sustainability even after the end of the Programme period. Properly trained local human resources will attain capacity enough to independently manage their lands and biomass power plants in accordance with international best practices as well as national standards in the long run.

In addition, the signing of long term PPA contracts for power generation is a key element for financial sustainability. The first biomass power plant in Sabeto, Fiji (Sub-component 1.1), is to generate stable and predictable revenue for the project sponsor's sound financial management given that the PPA has been guaranteed between the project sponsor and the FEA for 25 years. After twenty five years, the PPA will be automatically extended for another five years in Fiji. Similar degree of financial soundness is expected for other subsequent projects since it will apply the same criteria used for the first project sponsor in selecting project sponsors. Ensuring financial sustainability and bankability through the PPA, every investor will be able to exit with expected return and it will accelerate access to more climate finance from commercial sources.

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's [Investment Framework](#), should be addressed where relevant and applicable. This section should tie into any request for concessionality made in [section B.2](#).

E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation / adaptation impact potential

Specify the mitigation and/or adaptation impact, taking into account the relevant and applicable sub-criteria and assessment factors in the Fund's [investment framework](#).

The Programme is estimated to mitigate 4,736,230 tCO₂eq in total throughout the 25-year lifecycle of a series of projects (annually 189,449 tCO₂eq). The mitigation potential combines all the emission reduction impacts of Component 1 and 2 across Fiji and PNG.

The Programme ensures that the mitigation calculation has been done with most conservative estimates and detailed figures are elaborated in E.1.2. Sophistication will be added on in order to derive more accurate figures of CO₂ mitigation by the support of the GCF grant.

When applicable, specify the degree to which the project/programme avoids lock-in of long-lived, high emission or climate-vulnerable infrastructure.

There is an increasing need for reducing energy dependency on imported fossil fuel as well as for enhancing grid stability with continuous forms of renewable energy. Compared with other types of renewable energy sources, biomass is believed to be highly efficient in replacing diesel as a base-load power source in the regional context. Currently, one of the popular renewable power sources is hydropower and it is used as an alternative source of base-load power. Due to disruptive weather patterns such as El Niño, however, hydropower is becoming more and more unstable in supplying electricity and the problem is getting worse, being exacerbated by climate change. Other renewable energy sources such as solar and wind also have an issue of unsteady electricity provision. Above all, unless there is an addition of high-cost facilities such as energy storage system, they cannot replace diesel power plants.

By building a total of 48 MW biomass power plants in Fiji and PNG, the Programme will prove that biomass is an attractive renewable energy source that can replace diesel and guarantee stable supply of electricity. Meanwhile, the wood pellet plant including 24 MW biomass power plants that would utilize affluent biomass resources will serve as a lubricant of paradigm shift towards clean and sustainable energy across Fiji, and beyond that, throughout the South Pacific region and to Asia that has limited biomass resources. Through such expansion of renewable energy supply, the Programme will play a significant role in GHG mitigation.

E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

GCF core indicators	Expected tonnes of carbon dioxide equivalent (tCO ₂ eq) to be reduced or avoided (Mitigation only) ²²	Annual	189,449 tCO ₂ eq
		Lifetime	4,736,230 tCO ₂ eq
	Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience);	Total	373,869 persons (221,205 in Fiji and 152,664 in PNG) including 183,609 women

²² The amount of annual CO₂ emission reduction is an indicative one based on brief analysis, and more detailed analysis will be conducted with the TA grant provided by the GCF.

	Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Percentage (%)	4.2% (25% in Fiji and 2% in PNG)
Other relevant indicators	- Establishing biomass power plants with a total capacity of 48 MW²³ 500,000 ton/year of wood pellet produced from a pellet plant with a biomass power plant with 24MW in total Increasing number of households with access to clean and low-emission energy: a total of approximately 77,102 households will gain access to clean and low-emission biomass energy in Fiji and PNG (Fiji: approximately 48,298 households through three 12 MW biomass power plants, PNG: approximately 28,804 households through one 12 MW biomass power plant)		

Describe the detailed methodology used for calculating the indicators above.

< **Expected CO₂ Reduction from Direct Aspect** >

Calculation for CO₂ reduction of the Programme has meticulously considered 1) transportation of African tulip, 2) development of SRWC plantation on idle, and degraded lands, and 3) operation of plants in both Component 1 and 2. The calculation methodology has been carefully chosen with the support of external experts and the Life Cycle Assessment (LCA) was applied for estimating expected CO₂ reduction impacts during the lifespan of 25 years (PPA period). Detailed calculation figures are elaborated in the table as below;

Table 4 Indicative CO₂ reduction from direct aspects

Items		Lifespan (tCO ₂ eq)		Sub-total
		Component 1	Component 2	
1. Transportation of African tulip		-581,006	-954,703	-1,535,709
2. SRWC plantation	Plantation	2,245,834	3,930,208	6,176,042
	Harvesting	-287,552	-383,484	-671,036
	Transportation	-1,312,004	-2,032,192	-3,344,196
	Sub-total	646,227	1,514,532	2,160,809
3. Plant operation		4,485,464	592,575	5,078,039
4. Pellet export transportation		n/a	-966,910	-966,910.
Total		<u>4,550,735</u>	<u>185,495</u>	<u>4,736,230</u>

1. Transportation of African Tulip

In reference to African tulip transportation, the mitigation potential calculation considered diesel consumption and ensuing emissions along with the distance and number of round-trips as below. Since eradication of African tulip is to be completed within 10 years, the period of 10 years is an applied figure. Relevant calculation logics are as follows.

²³ The precise capacity of each biomass power plant is subject to change depending on the needs and condition of each project site.

As for Component 1, three biomass power plants in Fiji will use African tulip sourced via inland transportation. 40km is the average distance of African tulip transportation within Viti Levu (as no shipping transportation is required) and 4,000 times/yr of the round-trip is considered for calculation. A biomass power plant in PNG is not considered for African tulip transportation emissions because the plant will not use African tulip as a fuel source.

[Component 1] CO₂ emission = Transportation emissions by trucks (40 km of inland transportation from harvesting areas to three biomass power plants in Fiji) = Diesel emission factor * Diesel consumption of trucks per trip * 40,000 times of round trips/yr * 10 years = 581,006 tCO₂eq

As for Component 2, the wood pellet plant will source African tulip widespread nationwide across Viti Levu (main island) and Vanua Levu (where the pellet plant is to be located). Once the main island's African tulip is all consumed as a result of successful invasive species harvesting activities of the Programme, the pellet plant will use African tulip spread out in Vanua Levu as well. Regardless of this plan, the mitigation potential calculation sought the most conservative measure by adopting the assumption that African tulip will be transported via shipping across the sea into the pellet plant in Vanua Levu. In short, the calculation output is the most conservative estimate in that inland transportation emissions are lower than outland transportation emissions.

For calculation of African tulip transportation emissions in Component 2, 15km and 64km are applied respectively for inland transportation from harvesting areas to the port (in Viti Levu), and for outland shipping transportation. It should be again noted that actual transportation emissions are assumed less than the calculation output because the pellet plant of Component 2 plans to source African tulip from Vanua Levu as well as from Viti Levu.

[Component 2] CO₂ emission = Transportation emissions by trucks (15 km of inland transportation from harvesting areas to the port in Viti Levu before shipping transportation) + transportation emissions by ships (64 km of outland transportation across the sea) = Diesel emission factor * Diesel consumption of trucks per trip * 10,000 number of round trips/yr * 10 years + Diesel emission factor * Diesel consumption of ships per trip * 20 number of round trips/yr * 10 years = 647,805 tCO₂eq + 306,898 tCO₂eq = 954,703 tCO₂eq

2. SRWC plantation

Plantation. The plantation will be developed only in lands degraded, abandoned, or idled by adverse impacts of climate change, non-sustainable pine plantation practices, or dwindling sugar cane industry. At present, 39,000 ha out of 80,000 ha of sugarcane plantation areas and 30,000 ha of 79,000 ha pine plantation areas are being idled and degraded without any productive activity throughout Fiji²⁴. The Programme makes sure that all the 50,000 ha of SRWC plantation areas in Fiji will be chosen only within those lands. The 5,000ha of SRWC plantation areas in PNG will be chosen in idled and degraded areas of PNG as will be done in Fiji.

Referring to the non-active condition of planned plantation areas, the non-tree living condition is established as the emission baseline for calculation. Also, plantation areas will be divided into four blocks with each carrying out different phase activities of planting, growing, and harvesting. Approximately 25% of plantation areas will always remain filled with SRWC growing. These are also considered for the mitigation impact calculation as follows.

CO₂ reduction = Above ground net volume growth (m³/ha/yr) * wood density (ton/m³)
 * emissions conversion factor (tC/ton) * 44/12 (CO₂eq/C)
 * (20,000ha for Component 1 or 35,000 ha for Component 2) areas (ha) * 25% * 25 years
 = 2,245,833 tCO₂eq for Component 1 or 3,930,208 tCO₂eq for Component 2

Gliricidia is primarily considered for proper plantation species. With the support of external advisory agency, Indufor, the Programme takes relevant figures presented in the table below for potential CO₂ reduction estimates. For the sake of a conservative estimate, 35 m³/ha/yr and 0.5 are applied respectively for representing the figure of the above-

²⁴ Source: Fiji Sugar Corp. and Fiji Pine Ltd.

ground net volume growth and wood density. Calculation sophistication will be added on after conducting more research and developing more detailed plantation management plans, which will be assisted by the GCF grant once the Programme is launched.

Table 5 Figures applied for calculating CO₂ reduction from SRWC plantation

Species	<i>G. Sepium (Gliricidia)</i>
Above-ground net volume growth(m ³ /ha/yr)	36-77
Wood density(ton/m ³)	0.53-0.63
Emission conversion factor(tC/ton)	0.5

Harvesting of SRWC plantation. Each Component regards CO₂ emissions by harvesting SRWC plantation with the usage of diesel-fueled machinery. The amount of diesel consumption and operating hours throughout the project life cycle are considered for calculation as below. 16.53 MJ/hr and 28.94 MJ/hr are applied respectively for Component 1 and 2 after estimating diesel consumption of machinery as for the amount of harvested plantation species.

$$\begin{aligned}
 \text{CO}_2 \text{ emission} &= \text{Diesel consumption of machinery (MJ/hr)} * 2 \text{ units} * 2,400 \text{ hr} \\
 &\quad * \text{Diesel emission factor (0.08764)} * 25 \text{ years} \\
 &= \underline{287,552 \text{ tCO}_2\text{eq}} \text{ for Component 1 or } \underline{383,484 \text{ tCO}_2\text{eq}} \text{ for Component 2}
 \end{aligned}$$

Transportation. Transportation emissions from SRWC plantation are identically considered for calculation as follows;

$$\begin{aligned}
 \text{CO}_2 \text{ emission} &= \text{CO}_2 \text{ emission by trucks (15 km inland transportation)} \\
 &= \text{Diesel emission factor} * \text{Diesel consumption of truck per trip} * \text{number of round trips} \\
 &= \underline{1,312,004 \text{ tCO}_2\text{eq}} \text{ for Component 1 or } \underline{2,032,192 \text{ tCO}_2\text{eq}} \text{ for Component 2}
 \end{aligned}$$

The average distance of transportation is 15 km as SRWC plantation will be done in nearby areas of individual plants in pursuit of transportation efficiency. The average number of round trips per year is determined as 5,700 times for Component 1 and 10,000 times for Component 2.

3. Plant operation

Component 1. The tCO₂eq mitigation impact from biomass power plants is calculated by applying the CDM method as follows:

$$\text{CO}_2 \text{ reductions (tCO}_2\text{eq)} = \text{Baseline Emissions (tCO}_2\text{eq)} - \text{Project Emissions (tCO}_2\text{eq)}$$

Where:

- Baseline Emissions(BE) : Estimated emissions where electricity is generated on the basis of the current national energy mix
- Project Emissions(PE) : Estimated emissions where electricity is generated on the basis of the future energy mix structure changed as a result of the proposed renewable projects

First of all, the Baseline Emissions(BE) are calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where:

- BE_y = Baseline Emissions on a yearly basis (tCO₂)
- $EG_{PJ,y}$ = Quantity of net electricity that is produced and fed into the grid as a result of the project implementation on a yearly basis (MWh)
- $EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation on a yearly basis (tCO₂/MWh)

Secondly, for the Project Emission (PE) is calculated by CDM method which is subject to the methodological tool “Project and leakage emissions from biomass, Version 03.0. (IPCC)” The estimate project emissions including cultivation of biomass is denoted as below:

$$PE = PE_{SOC,y} + PE_{SM,y} + PE_{EC,y} + PE_{BB,y} + PE_{TR,y}$$

- Where:
- PE_y = Emissions resulting from project activities in year y (tCO₂e)
- $PE_{SOC,y}$ = Emissions resulting from loss of soil organic carbon, in year y (t CO₂e)
- $PE_{SM,y}$ = Emissions resulting from soil management, in year y (t CO₂e)
- $PE_{EC,y}$ = Emissions resulting from energy consumption, in year y (t CO₂e)
- $PE_{BB,y}$ = Emissions resulting from clearance or burning of biomass, in year y (t CO₂e)
- $PE_{TR,y}$ = Emissions resulting from transport of biomass, in year y (t CO₂e)

The Project Emissions calculated as above is near zero because site burning, soil-tilling and fertilizer will not be applied, and the emissions from transportation and harvesting of SRWC plantation have already been taken into account in the calculations above. Therefore, CO₂ reductions from biomass power plants are calculated to be approximately equal with the Baseline Emissions (BE) in this sector.

As a result of calculation, expected mitigation per annum and throughout the lifespan (25 years of PPA) are as below:

Table 6 CO₂ mitigation from biomass power plants

Contents	Lifespan (tCO ₂ eq)	Annually (tCO ₂ eq/yr)
Three biomass power plants in Fiji	3,252,993	130,120
One biomass power plant in PNG	1,232,471	49,299
Total	4,485,464	179,419

Component 2. In Fiji, tCO₂eq mitigation impact from the wood pellet plant is caused by surplus electricity from the captive 24MW biomass power plant to national grid in Fiji. After deducting auxiliary power consumption, emissions by fuel consumption of chipping machinery, and transmission loss, available electricity for sale is assumed as 39,736 MWh. Therefore, CO₂ reduction from the pellet plant is calculated as below;

$$39,737 \text{ MWh (electricity for sale)} \times 0.5965 \text{ (emission factor of Fiji)} = 23,703 \text{ tCO}_2\text{eq}$$

Since emissions from biomass transportation have already been taken into account in the calculations above, the CO₂ reduction from the pellet plant in Fiji is 592,575 tCO₂eq for 25 years (annually 23,703 tCO₂eq).

4. Pellet export transportation

The Programme also considers potential emissions by pellet export in CO₂ calculation. Although the Programme targets both SIDS and Asian countries for export, we assume the case of exporting all pellets to Asian countries for a conservative measure.

Table 7 CO₂ Mitigation from Transportation of the Wood Pellet Export

Items	CO ₂ Emissions	Notes
Project emission from shipping to importing countries (A) (Unit : kgCO ₂ eq/MWh)	50.08	- Farthest Distance: 7,800 km - Number of round trip : 10 times (50,000ton/voyage)
Power supply to grid by pellets (B) (Unit : MWh/yr)	772,292	- (125 MW – 15MW(Auxiliary Consumption)) * 81%(efficiency factor) * (1-1%(transmission loss)) * 24 hr * 365 days = 772,292 MWh/yr
Emission from pellet export transportation (Unit : tCO ₂ eq)	<u>966,910</u>	- (A)*(B)*25years/(1000kg/1t)

< Expected increase of energy >

The expected number of households to newly have access to low-emission energy is as below:

$$\text{Number of households with access to low-emission energy} = \frac{\text{Generated electricity (kWh/year)}}{\text{Electricity consumption per capita (kWh/year/person)} \times \text{Average number of persons in one household}}$$

Where:

	Fiji	PNG
Electricity generated from one 12 MW power plant	70,630,000 kWh/year	
Electricity consumption per capita ²⁵	957.892 kWh	462.65 kWh
Number of persons with access to electricity generated from one 12 MW power plant	73,735 persons	152,664 persons
Average number of persons in one household	4.58 persons ²⁶	5.3 persons ²⁷
Number of households with access to electricity generated	48.297 (three power plants)	28,805 (one power plant)

As a result of this calculation, 77,102 households including 48,297 households in Fiji and 28,805 households in PNG will have access to clean and low-carbon electricity from this Programme.

Describe how the project/programme's indicator values compare to the appropriate benchmarks (i.e. the indicator values for a similar project/programme in a comparable context).

The proposed Programme is expected to achieve a higher level of CO₂ mitigation effects than other similar renewable projects. Given similarities in region (Pacific) and investment area (renewable energy), the Programme compares its indicator value with those of the GCF-approved "Pacific Islands Renewable Energy Investment Program" and "Tina

²⁵ IndexMundi, Electricity consumption per capita (kWh per person), <https://www.indexmundi.com/g/g.aspx?v=81000&c=pp&l=en>, the numbers are averages of the years between 2007 and 2014 of the two countries.

²⁶ Fiji Bureau of Statistics, 2013-14 Household Income and Expenditure Survey (2015)

²⁷ PNG National Statistical Office, Population: Summary of Findings, <https://www.nso.gov.pg/index.php/population-and-social/other-indicators>

River Hydropower Development Project.” This Programme is estimated to reduce 4,736,230 tCO₂eq for a total project cost (except grant) of 427 million at a rate of \$90.16 per tCO₂eq. The rate is far more favorable in comparison with \$176 per tCO₂eq and \$290 per tCO₂eq of comparable respective projects. In addition, powered by a renewable energy source, biomass power plants undoubtedly have significant CO₂ mitigation effects compared to fossil fuel power plants in Fiji and PNG.

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/programme investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

Describe how the proposed project/programme's expected contributions to global low-carbon and/or climate-resilient development pathways could be scaled-up and replicated including a description of the steps necessary to accomplish it.

The Programme is expected to trigger paradigm shifts by scaling-up and replication of a single project into a programme. It seeks to enhance sustainability of forestry systems, and transform the current energy infrastructure settings towards more self-sufficient and climate-resilient pathways.

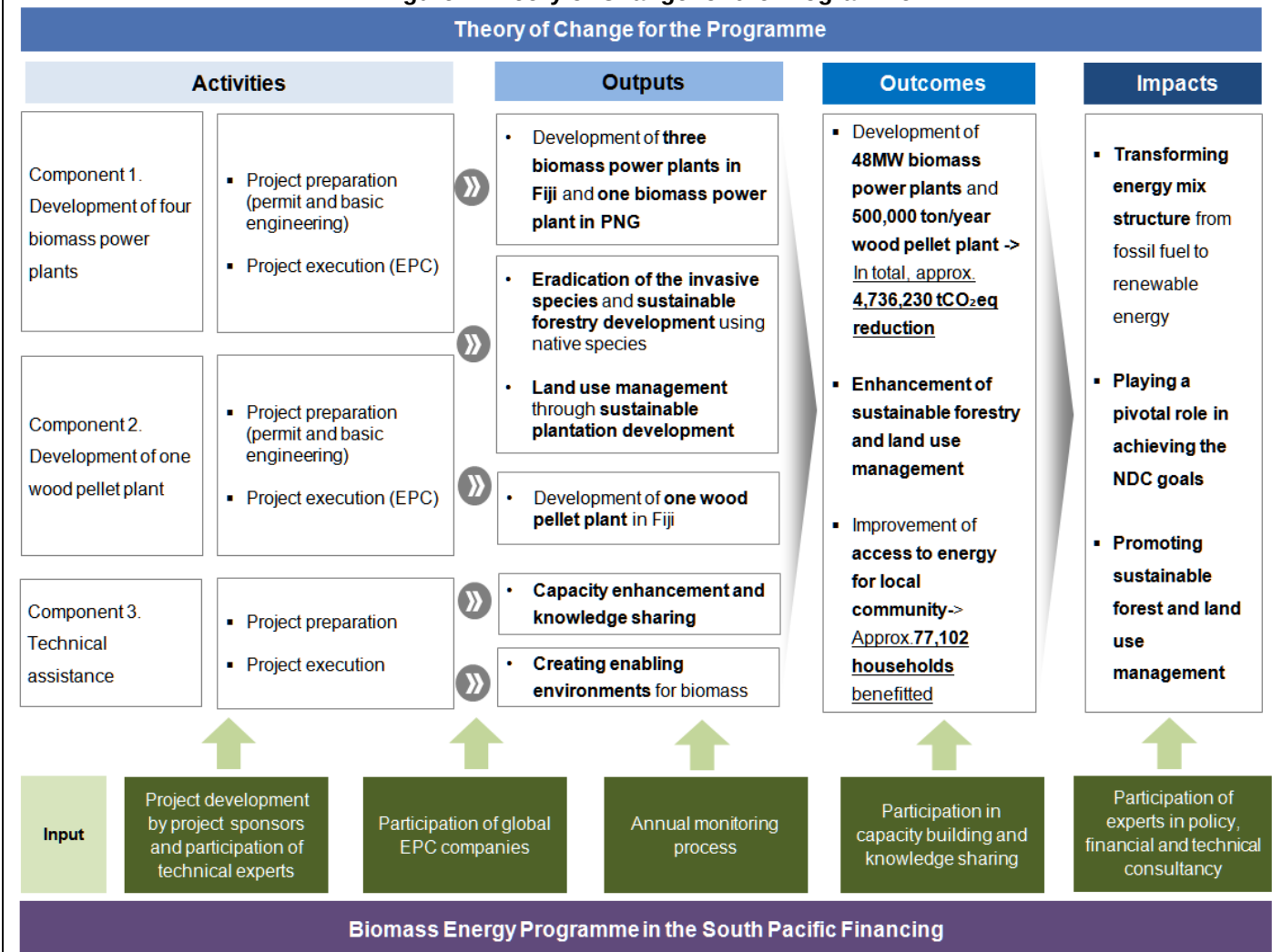
The GCF's financing the first biomass project in Fiji is to materialize and accelerate such intended scale-up and replication. The Programme success bolstered by the GCF co-financing will clearly demonstrate that renewable energy projects may be feasible in the South Pacific realm, and could be replicable to other areas across South Pacific SIDS. The built confidence will contribute to enlarging climate finance from the private sector, which will eventually help the South Pacific community achieve the paradigm shift goal.

Above all, the wood pellet plant to be established in Fiji is expected to play an important role for potential scaling up and replication of the biomass energy system in isolated areas of South Pacific SIDS which cannot afford the main grid for power generation. The distribution of produced pellets with small-scale biomass facilities of 100kW as a package into geographically isolated islands within the South Pacific SIDS is a meaningful milestone towards Pan-Pacific energy initiative, not only in that low-carbon biomass-based electricity is deployed, but also in that marginalized people gain access to electricity for the first time in their life.

The Programme will enable the desired transformation to be extended to the regional level. Firmly anchored in Fiji and PNG and garnering capacities and rich resources of both countries, the Programme will attain a greater potential for scale-up and replication of activities in forestry and land use management and of biomass energy initiatives, which will accelerate paradigm shifts towards low-emission and climate-resilient pathways throughout South Pacific SIDS at last.

Underpinned by a theory of change, each activity of the Programme has been carefully designed as illustrated in the figure below to create synergies between individual activities and achieve paradigm shift.

Figure 7 Theory of Change for the Programme



E.2.2. Potential for knowledge and learning

Describe how the project/programme contributes to the creation or strengthening of knowledge, collective learning processes, or institutions.

TA including diverse capacity building and knowledge sharing activities will act as a lubricant that keeps all the parts of the Programme aligned and moving smoothly without any friction. The requested GCF grant will finance the TA activities to be designed properly integrated into main forestry and land use and biomass projects so that they could contribute to enhancing the potential for scale-up and replication of this Programme.

The TA of the Programme has two sub-components that target Fiji and PNG each, and each sub-component is expected to present substantial potential for knowledge and learning by targeting broader region-wide stakeholders as well as local beneficiaries in countries of operation.

Capacity building and technology transfer. Capacity building sessions will be organized engaging with the Programme's managerial personnel, local engineers, or other public officials in Fiji, PNG and also from abroad. Various trainings will be carried out on technical, environmental, social, and gender capacities as well as project management skills including project monitoring, evaluation, and auditing. All the capacity building trainings will make sure of the participation of a certain proportion of female workforce and each training session will be monitored and evaluated for its effectiveness and the participants' satisfaction. Any feedbacks received from the participants will be reflected in the following training sessions.

For example, a ten-day technical training course can be organized targeting key managerial personnel to increase their capacity by learning more about the global biomass and wood pellet market and research trend, biomass projects development, and power generation market in the South Pacific region. The trainings can include site visits to biomass power plants, biomass processing facilities or research institutions. During the ten-day course, another session can specifically focus on technical skills required for operation, management, administration, and auditing of biomass plants and associated facilities and equipment.

Another tentative capacity building activity is a biennial four-day-session that helps local engineers and administrative staffs enhance their capacity. It aims to avoid a heavy dependence on technology sources from outside, and to gradually increase the share of the local workforce when it comes to running biomass power plants and a wood pellet plant. In short, the Programme pursues to transfer state-of-the-art biomass technology to both target countries.

Additionally, specifically-customized trainings will be provided to the local communities that are involved in the reforestation and plantation management work with a focus on the eradication of the invasive species and on the plantation of selected native species and the fast growing species, *Gliricidia*.

The Spread of knowledge. Knowledge sharing events will also be organized at the local, national, and regional levels. These events aim to share the progress being made and benefits the Programme delivers to South Pacific SIDS as well as the covered countries. Demonstration of the Programme as a best practice will encourage increased awareness of the innovative biomass technology as an attractive alternative, and of financial synergy through the GCF investment.

One example of the knowledge sharing events is a region-wide workshop. TA consultants will organize knowledge sharing workshops targeting local and regional stakeholders to present on the progress of the Programme. This may include the plant's construction and O&M, implementation of the GCF E&S Safeguards, and result of stakeholder engagement such as resettlement or compensation if relevant. The workshop reports that will be produced and distributed by the TA consultants can also be useful for dissemination of the knowledge and the progress made in the region will contribute to a more sustainable future.

Study supporting Mitigation Impacts. The support for study on local ecosystem and CO₂ saving potential from forestry and land use management in Fiji is expected to result in significant potential for knowledge and learning. Estimating carbon emissions reduction potential from The SRWC plantation in Fiji has not been carried out before, and *Gliricidia* requires a sophisticated technical approach to find out its carbon saving potential. The study supported by the GCF grant will produce analysis and findings that are very specific to the Programme activities

E.2.3. Contribution to the creation of an enabling environment

Describe how proposed measures will create conditions that are conducive to effective and sustained participation of private and public sector actors in low-carbon and/or resilient development that go beyond the program.

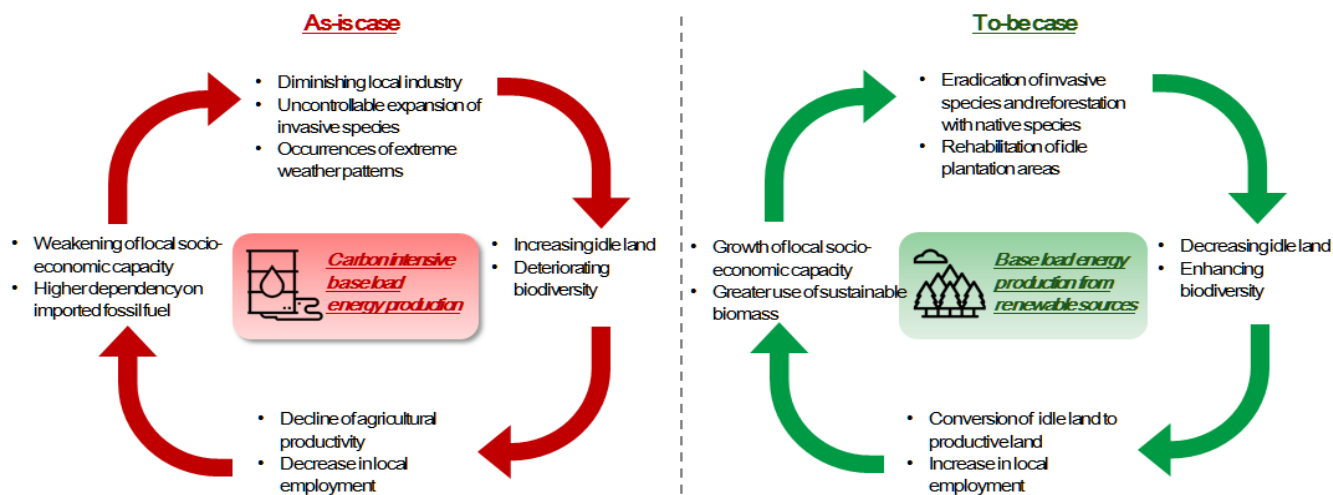
The contribution of the Programme for creating an enabling environment can be demonstrated for how the Programme can turn the vicious cycle of current land use practices into the virtuous cycle as shown in the figure below.

The 'As-Is' scenario is based on the reality where local economy is hugely relying on imported fossil fuel, and the natural environment which has been a traditional source of livelihood has been undermined by invasive trees (African tulip). Fiji has actually found itself confronted by an increase in abandoned land, a decrease in biodiversity, and a further decline in local income creation and socioeconomic capacity. Consequently, local community members have lost control over the land they own, which has led to failure of the government's efforts for effective land use management.

The Programme will achieve transforming the 'As-is' reality into the 'To-be' scenario. Through supplying fuels by developing local plantations and working with community land owners, the Programme can initiate reforestation after eradicating invasive trees, encourage plantation on the idled and degraded land, and restore sources of income for local communities. All the relevant operations will be done in accordance with the Plantation and Harvesting Guidelines and regular inspections of compliance with the guideline will be carried out by the environmental officer in charge (see ESIA for more detail for the first biomass plant in Sabeto). Local community members engaged in reforestation and the plantation management will be direct beneficiaries of the capacity building and knowledge sharing support. As a result, forestry and land use will be effectively managed and under control of the government and land owners. All the Programme activities will support the government's agendas of eradicating invasive trees, preserving local ecosystem, creating jobs for local communities and so on. Eventually, the Programme will create an enabling environment where sustainability and country ownership can be enhanced.

Figure 8 Transformation into Virtuous Cycle

Biomass Energy Programme in the South Pacific



A substantial portion (75%) of grant is to be used for creating an enabling environment in PNG. Concerned about PNG's relative fragile environment for biomass deployment, the Programme plans to assign US\$ 3million (out of US\$ 4million in total) into PNG for supporting the government's making the model framework for renewable IPPs, helping the development of an effective renewable energy governance structure, and assisting the mapping of the nationwide land use patterns and biodiversity dynamics. Such TA activities are expected to establish an enabling model framework and relevant institutional circumstances under which the Programme will successfully implement the biomass power plant project. By extension, the Programme anticipates that such an enabling environment will attract more investments in renewable energy including biomass in the near future. The expected enabling environment will pull the trigger on the whole South Pacific community when it comes to energy paradigm shift towards low-carbon and clean renewable energy structure.

Describe how the proposal contributes to innovation, market development and transformation. Examples include:

- *Introducing and demonstrating a new market or a new technology in a country or a region*
- *Using innovative funding scheme such as initial public offerings and/or bond markets for projects/programme*

The Programme will introduce innovative biomass technology, create new market for biomass energy, and trigger further transformation for renewable energy sector in the South Pacific region. In particular, the development of a wood pellet plant has been a key consideration even from the design stage of this Programme because produced pellets are to be supplied as fuel to other biomass power plants located within the region or outside, which will not only formulate and invigorate the region-wide biomass market development, but also contribute to transforming energy mix to renewable basis within and outside the South Pacific region.

The biomass market development process will be accompanied by TA activities which will play an important role for applying proven innovative technology on the ground and expanding new renewable initiatives across Fiji, PNG, and, beyond that, throughout the South Pacific region.

E.2.4. Contribution to regulatory framework and policies

Describe how the project/programme strengthens the national / local regulatory or legal frameworks to systematically drive investment in low-emission technologies or activities, promote development of additional low-emission policies, and/or improve climate-responsive planning and development.

South Pacific SIDS does not usually have adequate regulatory framework or institutional setup to pursue their renewable energy target through actual project financing. In case of Fiji, although Fiji has an ambitious target for renewable energy deployment, there is no legal framework for Public-Private Partnership (PPP) or IPPs which is probably the reason for only a single pilot project has been tried in Fiji. Especially, the roles and responsibilities of specific entities are rarely defined in a crystal clear manner, which leads to duplication of work and responsibilities, a slow decision-making process, and severe inefficiency. As for PNG, although it has a National Energy Policy 2016-2020, which indicates the need for strengthening renewable energy sector, it lacks a regulatory framework and specific plan for renewable energy.

When it comes to the forest and land use, the Forest Decree (1992) is the main law regulating forest use in Fiji. However, there is no provision for sustainable land use management or participation of local landowners in the management of forest resources. Part V of the Decree recognizes the rights of customary landowners and provides that subsistence forest use recognized by customary law is permitted and should not be restricted by the Decree. This law has been under review for many years and is yet to be amended. The proposed Programme will establish a best practice, which will catalyze further improvement in the forestry sector.

This Programme, based on the experiences with the pilot project and with support from the GCF, will present a successful IPP project where foreign private investors have locked in, and this helps the country to trigger actions for establishing adequate legal framework to promote more investment in renewable energy projects.

The recent NDC Implementation Roadmap clearly demonstrates how the Programme has been contributing to the target country's national strategy development to address climate change challenges. In fact, the Programme has been seriously considered by the Fijian government when it comes to goal setting process of the Roadmap. As a result, the Programme's components are comprehensively integrated through short-, mid-, and long-term mitigation actions of the Roadmap. According to the Roadmap, the South Pacific biomass initiative ensures to achieve 50% of the total mitigation target; exactly speaking, the Programme helps to reduce 212,000 tCO₂/yr out of the total 427,000 tCO₂/yr mitigation target in the energy sector of Fiji.

Another example can be that South Pacific SIDS in the region does not have specific environmental and social assessment guidelines provided by the government, and it was the case for Fiji as well. The project sponsor for the Fiji's Sabeto project had advised the Fijian government during the pilot project period that requirements such as for air pollutants are necessary in order to manage the biomass power plant in a sustainable way, and had voluntarily set

up the guidelines to conduct the relevant assessments. The project sponsor will apply the same approach to this Programme, and work in close partnership with the governments to support them to begin establishing adequate regulatory framework for environmental and social management guidelines.

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

The Programme is anticipated to bring in wider co-benefits to the beneficiaries, local communities and relevant stakeholders, beyond target countries and region.

<Environmental Co-benefits>

- i. Fuels for biomass power plants and the wood pellet plant will be co-supplied mainly from eradicated African tulip and SRWC. The invasive African tulip is native to dry forestry in Africa, and is famous for its invasiveness. According to FAO²⁸, the wide spread of this invasive alien species has become a serious problem in Fiji recently. In that sense, removal of African tulip will help preservation of Fiji's native local ecosystem and biodiversity.
- ii. The local SRWC plantations will be done in lands highly degraded by African tulip or abandoned by ineffective land use practices in association with declining sugarcane, pulp, and banana industries. The poor condition of the lands would be recovered by maintaining optimum frequency of harvesting rounds. Additionally, Gliricidia carefully chosen for plantation is well known for its nitrogen-fixing properties that will help improve soil nourishment and avoid damage from any additional fertilizer.
- iii. Of invaded lands by African tulip, some belonged to natural forest and the other to agricultural areas. All the vacant lands with invasive species harvesting being done will appreciate the previous land use as much as possible not to create an unreasonable land use change. The Programme plans to implement reforestation with selected native species in previous natural forest lands while giving back previous agricultural areas to local landowners. If the landowners wish to make use of their land for plantation, and to have a fuel supplier contract in favor of income generation, the Programme will not disregard local landowners' desires.
- iv. The Programme will contribute to reducing the use of diesel in the region. The Programme will result in improvement of the air quality by reducing air pollutants generated from traditional diesel power generators such as the emissions of nitrogen oxides, carbon monoxides, hydrocarbons and particulate matters. Phasing out diesel will also benefit the local community in terms of lowering exposure to the risks such as fuel spills and water and land contamination while diesel is being transported or stored underground.

<Economic Co-benefits>

- i. Key economic co-benefits are job creation and income generation.
 - Job opportunities will be generated especially for plantation management. In general, in Fiji, more than 90% of the land is owned by local community members so fuel suppliers will sign fuel supply agreements in order to develop plantation. Instead, local landowners will plant and harvest SRWC in their own land every 2-3 year, and provide them to the power plant, which will create benefits of employing landowners and local community members.
 - Eradicating African tulip will provide full-time and part-time employment opportunities in regions affected and financial supports to linked service industries such as transportation and equipment procurement.
 - Biomass power plants are also expected to generate job opportunities. A single biomass power plant is expected to create 420 jobs, and the operation of a wood pellet plant is expected to bring in 560 jobs for the local communities. In total, four biomass power plants and one wood pellet plant can generate jobs for 2,240 local people during the project implementation period.

²⁸ <http://www.fao.org/agriculture/crops/thematic-sitemap/theme/biodiversity/weeds/issues/att/en/>

Table 8 Job creation of this Programme

Project	Phase	People	Unit	Total hired people
Power plant	Construction	200	4	800
	Operating	50	4	200
	Plantation	150	4	600
	Harvesting	20	4	80
Pellet plant	Construction	200	1	200
	Operating	70	1	70
	Plantation	250	1	250
	Harvesting	40	1	40
Total				2,240

- ii. From a macro perspective, reduced use of diesel will help decrease the countries' heavy reliance on the imported fuel and increase energy security. This also implies that the Programme will reduce the countries' vulnerability to the impact of oil price fluctuations. The Pacific countries' GDP growth fell sharply as the 2008 financial crisis hit the world, and the fall has been hugely attributed to surging oil prices²⁹.

<Social Co-benefits>

- i. Income generation from the employment will enhance family welfare. The earned income can help reduce poverty and child labor, and promote health and education attainment of the local population.
- ii. KDB will steadily encourage the project sponsors to support community facilities such as schools and kindergartens so as to further trickle down attained financial profits of the project. In fact, the project sponsor has already been making financial contribution and relevant assistance, for instance, by supporting the kindergarten graduation ceremony and scholarships to local children with the pilot project (Nabou project).
- iii. Capacity building such as trainings and other educational programmes will be provided to employees working at biomass plants. These employees can continue working for the plants but can freely move to other local working places, which spur knowledge and technology transfer.
- iv. Overall, the programme pursues human development goals by capacity enhancement of human resources in Fiji and PNG.

<Gender Co-benefits>

Gender-sensitive social inclusion is at the heart of the overall Programme elements, as gender parity plays a critical role in moving towards a climate-resilient pathway in an inclusive way. In particular, the Programme does not only target women but also take the needs of both women and men into account to achieve genuine gender equality.

Gender inclusive approach is particularly important for the countries such as Fiji and where the male-female ratio of the labor force and the population is nearly equal. The Programme fully recognizes that without gender mainstreaming strategies, the Programme will not be able to reach out to groups discriminated by gender, and thus will fail to achieve its full sustainability potential.

- i. Certain proportion of women will be engaged in all types of consultations conducted during the Programme period, in order for their voices to be heard.

²⁹ ADB, AusAID (2012) Revitalising the Fiji Economy

- ii. The Programme will adopt well-devised gender mainstreaming practices in terms of employing adequate number of female workforce, applying the equal terms to both men and women employees (i.e. remuneration), and providing desirable working environment for women (i.e. sexual harassment policy).
- iii. Various trainings will be provided particularly targeted at female (or male in case of PNG if the project site possesses female-dominated gender dynamics) engineers to help the female labor force to increase the technical capacity and move into senior managerial levels with higher incomes, thus moving up the poverty ladder and achieving social mobility. Women's economic empowerment is expected to accelerate by building a low-emission future in Fiji, PNG and the Pacific as a whole. Plus, their empowerment will help end violence against women and girls – which is one of serious gender-based human rights violations in the region.
- iv. Such practices will showcase the gender mainstreaming approach (best practices in mainstreaming gender throughout the whole business cycle) to other businesses in Fiji and PNG. Such practices can also serve as a momentum for advancement of the National Gender Policy of Fiji and the National Policy on Women Equality and Women Empowerment of PNG, and ignite other gender related initiatives in Fiji and PNG.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

Describe the scale and intensity of vulnerability of the country and beneficiary groups, and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc).

Not Applicable

E.4.2. Financial, economic, social and institutional needs

Describe how the project/programme addresses the following needs:

- *Economic and social development level of the country and the affected population*
- *Absence of alternative sources of financing (e.g. fiscal or balance of payment gap that prevents from addressing the needs of the country; and lack of depth and history in the local capital market)*
- *Need for strengthening institutions and implementation capacity.*

Phasing out diesel dependency in energy production. The economy in Fiji and PNG is characterized by its vulnerability to both internal and external shocks. Truly, the country's economic growth rate has been severely influenced by global financial crises and natural disasters such as cyclones and flooding. Above all, its heavy dependence on imported fuel is a huge burden. The development of biomass power plants is expected to contribute to the countries' financial stability, as the power plants will lessen Fiji and PNG's dependence on imported fossil fuels, in particular, diesel for energy production.

Land use sustainability and livelihood improvement. Fiji's pulp and sugar cane industries have been experiencing economic downturn, and a large number of related businesses have been shutting down their farms. Even worse, recent climate disasters (i.e. Cyclone Winston) have badly damaged the sugarcane plantation farms, and the prices of sugar cane have sharply decreased. The businesses have been forced to close down their farms, and it has resulted in not only large areas of land being abandoned but also thousands of people losing jobs. It has directly affected the local economies leaving the community members economically vulnerable without access to their livelihoods. By planting Gliricidia in abandoned lands and providing jobs, the Programme will achieve enhancing local livelihoods as well as improving forestry and land use systems.

Absence of alternative sources of financing. Though the governments of Fiji and PNG are ambitious to expand their renewable energy capacities, they have limited budget to carry out such projects and their financial markets are

not mature enough to attract private investments. During the pilot project implementation in Fiji, the project sponsor had difficulties in attracting private investment due to the country's financial and political risks. In that sense, therefore, international institutions such as the GCF can play a significant role in countering such hardships, paving a way for further renewable investments, and enabling the Fijian and PNG governments to move forward in achieving their renewable energy targets.

In particular, PNG mentioned in its INDC that the amount of GHG mitigation of the country highly depended on the availability of external funding. That is, in achieving its ambitious goal of achieving 100% electricity generation through renewable energy, it will need to attract sources of finance from outside the country.

Need for strengthening institutional capacity. According to a report prepared by the Fiji Department of Energy and UNDP, efforts in developing IPP market did exist in the past, but such attempts were not successful. Main issues behind such poor performance include³⁰:

- Power purchase tariffs are significantly below avoided cost of FEA's thermal generation, and thus cannot render IPP projects financially viable;
- Unclear situation with regard to Fiji Commerce Commission ruling on minimum tariffs;
- Complex and costly approval procedures;
- Self-regulating of FEA with regards to grid access;
- Difficulties in accessing financing;
- Difficulties with or lack of access to relevant information.

Moreover, in case of PNG, an issue regarding its renewable energy policy is that it is yet to formulate a specific action plan that will realize the goals. By setting a model framework for renewables and enhancing the governmental system by Grant, the local community and government officials will be able to gain insight and knowledge related to renewable energy.

The Programme will be operated and managed by entities that have extensive experience in terms of renewable technology and finance. Once these entities enter Fiji and PNG and transfer their know-hows to local stakeholders through technical assistance, the local governments could build capacity needed for developing IPP markets and improving renewable energy capacity. As outlined in Section E.2.2, various capacity building and training sessions will be provided to government officials regarding development of regulatory framework.

Moreover, KDB, as the accredited entity, will effectively manage the Programme in compliance with the GCF standards by assisting relevant government agencies in strengthening their capacity of implementing PPP projects and biomass IPP projects.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

Please describe how the project/programme contributes to country's identified priorities for low-emission and climate-resilient development, and the degree to which the activity is supported by a country's enabling policy and institutional framework, or includes policy or institutional changes.

The Programme upholds the principle of country ownership by aligning the Programme components with identified policy priorities of target countries for low-emission and climate-resilient development. Both Fiji and PNG governments have emphasized the significance of raising the share of renewable energy on their policies and

³⁰ In case of Fiji, import of fuel accounted for 22.9% of total merchandise imports in 2015 (Department of Energy & UNDP)

national strategies in order to address disastrous climate change impacts. The Programme fits well with the two governments' interests and priorities by introducing low-carbon biomass energy to the South Pacific realm with rich endowments of biomass fuel sources. In addition, the Programme responds to the country needs of promoting sustainable forestry and land use and reducing CO₂ emission by suggesting ways to effectively handle forestry and land use systems.

<Fiji>

Playing a pivotal role in achieving the NDC goals. Expansion of biomass energy has been strongly supported by the Fiji NDC Implementation Roadmap 2017-2030 published by the GGGI. The Roadmap 2017 - 2030 describes that biomass-based power generation including sustainable biomass plantations can make the largest contribution to achieve the Fiji's CO₂ mitigation target through 2030. Increasing renewable power generation capacity through a 12 MW biomass power plant, and developing sustainable biomass to use as fuel have been proposed as a short-term mitigation action (2017 – 2020) for the Fijian Government. Biomass power plants accompanied by sustainable biomass plantation and increasing their capacity are repeatedly highlighted as medium-term and long-term mitigation actions in this Roadmap as a primary consideration for Fiji to achieve the NDC target.

Accordingly, the Roadmap clearly demonstrates that the Programme can make a significant contribution and play a pivotal role in supporting these countries to achieve the CO₂ emissions reduction targets they have ambitiously announced in the NDCs.

Land use management. The Programme contributes to the Fijian government's priority activities for effectively managing forests and land use, which is aligned with climate-resilient development. Fiji National Biodiversity Strategy and Action Plan stipulates that there are a variety of ecosystems present in Fiji with various habitats containing variety of species of fauna and flora that has helped maintain sustainability in terms of biodiversity and ecosystem stability. The focal areas outlined in the Action Plan include: 1) community support – awareness, involvement and ownership; 2) improving knowledge; 3) developing protected areas; 4) species conservation; 5) control of invasive species; and 6) capacity building and strengthening. The 5th focal area details on how to control invasive species that crowds out native species present in Fiji.

Most notably, the Ministry of Fisheries and Forests of Fiji has been making endeavors for the effective use of abandoned land, focusing on the eradication of African tulip which has caused deterioration of forests and consequently abandonment by landowners. As an initiative to eradicate African tulip, the Ministry issued an Expression of Interest (EoI) in 2015 to find interested parties that can assist the Fijian government in eradicating African tulip. The EoI was supporting the UNDP programme titled "Implementing a Ridge to Reef approach to Preserve Ecosystem Services, Sequester Carbon, Improve Climate Resilience and Sustain Livelihoods in Fiji".

Further details of the Fijian government-led initiatives and efforts can be found in Section C.1 and E.5.3. The Programme is designed in line with the Fijian government's significant efforts to create a sustainable and efficient cycle of land use and forest management.

<PNG>

Increasing renewable energy generation. PNG submitted its INDC in September 2015, where it states that "the main mitigation contribution for PNG would be in terms of an indicative replacement of fossil fuelled electricity generation with renewable energy sources."³¹ While the INDC does not specify the amount of contribution of electricity to emissions, it commits to the goal of raising the share of renewable energy for electricity generation to 100% by 2050.

³¹ Fiji NDC Implementation Roadmap 2017-2030(2017)

PNG's National Energy Policy 2016-2020 sets out detailed strategies for ensuring “affordable, reliable, and accessible provision of energy in a manner that is competitive, sustainable and environmentally friendly,”³² and achieving the target of reaching 100% renewable energy sources for electricity generation. In doing so, the Policy spells out the need for maximizing the use of renewable energy and expresses its willingness to draft and formulate a National Renewable Policy, in which biomass is included as a renewable energy source.

Land use management. Despite covering less than 0.5% of the Earth's surface, it is estimated that PNG holds 5-10 % of global biodiversity. The PNG government announced the National Forest Development Guidelines in 2009, which reflects the government's needs for achieving economic recovery and nation-building through the development of the country's natural resources including forest resources, while taking into consideration environmental preservation and social and cultural values of the forests in PNG.

E.5.2. Capacity of accredited entities and executing entities to deliver

Please describe experience and track record of the accredited entity and executing entities with respect to the activities that they are expected to undertake in the proposed project/programme.

<Accredited Entity– KDB>

Since 2007, KDB has financed approximately USD 3.3 billion for about 45 renewable energy projects including solar, wind power, and biomass projects across the globe. In terms of biomass projects, KDB has played a role as the lead arranger for three projects including a biomass project in China (2011).

Table 9 Track Records of KDB

(Unit: USD million)

Year	Project Title	Total Project Cost	Financial Arrangement by KDB
2011	Biomass Power Plant in China	54	36
2014	Jeonju Power Corporation Biomass Power Plant	88	70
2016	Tees Renewable Energy Plant, Teesside, United Kingdom	1,091	87

Based on its employees' expertise on electricity industry and finance, KDB has a separate Project Finance Unit in power sector, which is divided into three teams by region, and is increasing financing for renewable energy projects in developing countries such as Indonesia and Georgia through coordination with international organizations

<Executing Entity– Project SPC(s)>

KDB will select project sponsors for each biomass power plant based on criteria outlined in its Investment Criteria, as done in the first Sabeto project (Sub-component 1.1), in order to establish project SPC(s) which will act as an executing entity. The criteria is as follows:

- Credit Requirement: Companies having credit ratings that are at least “BBB-” by S&P or “Baa3” by Moody's. In case of Korean sponsors, companies having credit ratings of at least “BBB-” by one of NICE Information Service Co, Ltd., Korea Ratings Co, Ltd. or Korea Investors Services, Inc.; or;
- Track Records:
 - (1) Experiences that utilize commercially proven technology and employ experienced management teams which is related to bio energy industries and,
 - (2) Tack records that meet the ongoing operating, monitoring and maintenance requirements for bio energy industries; and,
 - (3) Experience of performing and/or to have performed at least one project in South Pacific SIDS related to bio

³² PNG National Energy Policy 2016-2020 (2015)

energy industries

- Financial investors (financial institutions or private equity funds as the case may be) can also be invited as sponsors. The above credit requirements or track records are not required for financial investors.

Project sponsors tentatively selected by KDB were the project sponsors of the pilot biomass power plant implemented in Nabou, Fiji, and thus are well-equipped with knowledge and experience pertaining to developing IPP projects in Fiji. Please refer to Section C.4 for further information on KDB's capacity in terms of selection of project sponsors, etc.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

Please provide a full description of the steps taken to ensure country ownership, including the engagement with NDAs on the funding proposal and the no-objection letter.

<Engagement with NDAs and Governmental Agencies>

The project sponsors of the first biomass power plant in Sabeto, Fiji (Sub-component 1.1) have been cooperating with the Fijian government for sustainable forestry and land use and economic use of biomass resources. In fact, the Ministry of Fisheries and Forests had realized the need for eradication of African tulip for its invasiveness, and agreed with the proposal that the project sponsor has made to the Ministry for removing African tulip to use for biomass fuels, and planting native species to replace African tulip.

The EoI that the Ministry released in early September 2015 demonstrates the commitment the Fijian government has made to eradicate African tulip. The EoI states that African tulip is an invasive plant in Fiji and has been posing a great threat to Fiji's national biodiversity, and the Ministry intends to identify interested parties who are willing to assist the Ministry to address the issue of African tulip by entering into a long term agreement.

Figure 9 The EoI and the Letter of acceptance for the African tulip eradication project

Expression of Interest Ministry of Fisheries and Forests Eradication of African Tulip (<i>Spathodea Campanulata</i>) African Tulip is an invasive plant in Fiji and has been posing great threat to Fiji's National Biodiversity. The Forestry Department is seeking for interested parties who are willing to assist by engaging in long term collaboration on the Control and Eradication of African Tulips in Fiji. Applicants who wish to apply must do so with a brief proposal. Selection will be based on the following criteria: 1. Applicants must be willing to enter into long term agreements with the Forestry Department 2. African Tulips must be used for an economical purpose 3. Must have sufficiently demonstrated know-how, appropriate technology and experience in the methods and technology they are proposing 4. Activities must lead to the management and towards the complete eradication of African Tulips 5. Proposed areas of responsibility for the applicant 6. Financial bond to be paid Interested Parties are requested to send in their applications to: Conservator Forests – Mr. Samuela Lagitaki Ministry of Fisheries and Forests Box 2218 Government Buildings Takayawa Building, Toorak, Suva Email: sislagitaki@gmail.com Closing Date: Friday 11th September 2015	Fisheries and Forest Headquarters Takayawa Building 360 Toorak Road Suva Telephone: (876) 3316011/3316091/3316093 Fax: (876) 3316092 Ref : _____ Date: 25/09/15 The General Manager ELTECH Limited P.O. Box 3359 Lautoka Post Office Fiji Islands Expression of Interest: Eradication of African Tulip (<i>Spathodea campanulata</i>) Thank you for your Expression of Interest on the above that was advertised in the Fiji Sun on Saturday 29th August 2015. I wish to inform you that the Honorable Minister for Fisheries and Forests had approved your selection. You are requested to make a presentation to the Ministry of Fisheries and Forests which will include other government stakeholders on Thursday 22/10/15 on your plan of operations, and to discuss other details such as the payment of your bond. I congratulate you on your selection <i>Inoke Wainiqolo</i> Permanent Secretary for Fisheries and Forests
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(Source: Ministry of Fisheries and Forests of Fiji)

Since Eltech, a subsidiary of the project sponsor, GIMCO, was selected as a service provider for this EoI, it has been carrying on cooperation with the Ministry to eradicate African tulip. The Ministry has been supportive of Eltech's reforestation and plantation activities.

Following that, the first task force committee has been organized by the Ministry of Fisheries and Forests on the matter of utilization of African tulip. The task force committee includes all the related government departments such as the Department of Agriculture and Department of Rural Development under the Ministry of Agriculture, Rural Development, and National Disaster Management, and Department of iTaukei Affairs under the Ministry of iTaukei Affairs & Sugar Industry. The first task force committee meeting was held in February 2016. In April 2016, the Cabinet of the Fijian government expressed support for developing biomass businesses. In addition, the project sponsor has signed a MoU with the FEA which agrees upon overall cooperation in developing biomass energy generation projects in Feb 2017. The MoU had been drafted based upon a series of consultations with the NDAs and other governmental agencies that have taken place over the last few years. It is evident that the Programme is in line with what the beneficiary governments wish to achieve. In short, country ownership has been well realized as the key principle throughout steps taken.

As a result of in-depth communications between KDB and the NDAs, Fiji and PNG already issued the No-Objection Letter on August 2017 and January 2018 respectively.

Please also specify the multi-stakeholder engagement plan and the consultations that were conducted when this proposal was developed.

<Multi-Stakeholder Engagement Plan and Consultation>

Several hearings and meetings have been and will be carried out to ensure that the proposed Programme is implemented without conflicts with stakeholders. Specifically, the project sponsor are willing to organize several government stakeholder meetings and to invite relevant government officials for them to be assured that the Programme is aligned with vision and strategies of the governments with regards to land use, renewable energy, and benefits the local community will obtain.

In fact, the project sponsor had conducted several consultation meetings with various stakeholders for environmental and social issues of the biomass power plant in Sabeto, Fiji (Sub-component 1.1), particularly focusing on utilizing land and road access, providing employment opportunities, and complying with regulated procedures. Social studies were conducted targeting different focus groups: 1) related departments under the Ministry of Environment, local authorities such as Town Council, Rural Local Authority, and FEA; and 2) community members including women and youth. Interviewing the focus groups has provided opportunities to listen to concerns and issues of the local groups, and to come up with appropriate measures to mitigate the negative impacts. According to landowners who were key stakeholders of the project, the raised concerns and related impacts are expected to be appropriately managed, and to benefit the livelihood of the local people by transforming the abandoned land to the productive land by plantation.

In case of PNG, KDB will make endeavors to establish a constructive partnership with government bodies in relation to the GCF-funded activities such as PPL at an appropriate timing of a certain level of PNG project details being confirmed. Particular project activities including selecting plantation areas will be carefully implemented after conducting thorough feasibility studies and considering feedback from multi-dimensional stakeholders in PNG. In the decision-making process, ideas, opinions, and advices from PNG counterparts will be treasured in order to make a consensus that satisfies all relevant parties.

<Women's Active Engagement>

The Programme recognizes the importance of gender mainstreaming. Prior to the proposal submission, the stakeholder engagement strategy for women is currently being discussed with the Ministry of Women, Social Welfare, and Poverty Alleviation. The discussion entails capacity building activities targeting local women so called 'Biomass Project' – i.e. training local women to be a biomass engineer and garnering their support for distributing small scale biomass power facilities as packages into isolated areas without grid connection.

The Programme is seeking specific ways for effectively implementing the 'Biomass Project' as part of the Programme component 3 by encouraging local entities' active engagement. The Ministry can play a significant role in facilitating the consultation process with local women (both native and indo) and other relevant stakeholders and enabling the Programme gender sensitive.

<Miscellaneous Remarks>

A communication strategy for complaints raised by local communities, stakeholders, and the government will be established. That is, throughout the construction and operation phases of the biomass power plant, a form of "Complaint Register" will be kept on a project site to register complaints regarding the noise, dust and surrounding environmental degradation.

KDB also has an information disclosure system for its projects on its website where stakeholders can have access to information related to the projects. The system was established to enhance communications with borrowers and promptly respond to the complaints.

KDB, as the accredited entity, has attended the GCF Structured Dialogue with the Pacific held in Tonga in July 2017 where it had the opportunity to communicate the progress being made with this Programme, and has collected more information on the demand for biomass across South Pacific SIDS. This consultation played a significant role in designing the Programme to be more demand-driven so that it can reach those that are vulnerable and in need.

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers; providing the least concessionality; and without crowding out private and other public investment.

The financing structure of this Programme is designed to leverage the GCF finance to fill the financing gap, and to induce more private capital to enable the project. At the same time, considering the characteristics of private sector projects, financial terms and conditions for the GCF are suggested with the minimum-level of concessionality.

The market rate presented in Section B.3 indicates that the debt to equity and the minimum simple DSCR are not attractive enough to invite private sector investments in the Programme. The pricing and funding size of the GCF loan plays a decisive role in achieve the minimum level of financial requirements under which commercial banks may consider loan provision.

Please describe the efficiency and effectiveness, taking into account the total project financing and the mitigation/adaptation impact that the project/programme aims to achieve, and explain how this compares to an appropriate benchmark. For mitigation, please make a reference to E.6.5 (core indicator for the cost per tCO₂eq).

< Biomass projects >

Compared to other similar projects, the proposed biomass power plant project is assumed more cost-effective. Information on the comparable projects is as in the table below. The cost per MW of the first Sabeto project is more favorable in comparison with other two biomass energy projects in Sri Lanka and Malaysia.

Table 10 Information on Comparable Transactions

Items ³³	Proposed Project	Comparable Project 1	Comparable Project 2
Project Title	Fiji Biomass Plant	Renewgen Kaduwela Biomass Plant	Felda Global Jenka Biomass Plant
Country	Fiji	Sri Lanka	Malaysia
Source	Wood Chip	Municipal Solid Waste	Palm Oil Residues
Status	Under Contemplation	Completed (2012)	Completed (2012)
Installed Capacity (MW)	12.0	10.0	12.5
EPC Cost (million USD)	36.4	32.6	40.5
Cost per MW (USD/kW)	3,033	3,265	3,242

< Wood pellet plant >

The total investment cost of the pellet production infrastructure (Sub-component 2.1) and capacity increase for existing transportation infrastructure (Sub-component 2.2) is USD 114.8mil (USD 230/pellet-ton) in this Programme. By the way, the investment cost of Highland Pellet s-Pine Bruff in U.S. which started construction in 2017 is USD 300 pellet-ton, while that of SY Energy in Korea which started construction in 2015 is USD 333/pellet-ton. These references show that the cost of the pellet plant under this Programme is more cost-effective in comparison with other two projects.

³³ The feedstock price and operating cost of each comparable project varies in a wide range. For example, Comparable project 1 uses solid wastes with little cost but some tipping fees if applicable. LCOE (Levelized cost of Energy) is hard to compare between individual cases but LCOE is generally lower in biomass than in Waste to energy. (Source : Bloomberg New Energy Finance, 2014).

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

Please provide the co-financing ratio (total amount of co-financing divided by the Fund's investment in the project/programme) and/or the potential to catalyze indirect/long-term low emission investment.

The total Programme size is USD 431.0 million of which the financing from the GCF is USD 106.4 million. The GCF financing accounts for nearly 25% of the entire Programme financing and thus the co-financing ratio is 3.05 (co-financing ratio is 3.17 except the grant for TA).

The investments from KDB (the accredited entity of the Programme), project sponsors (such as GIMCO, GS Power, and Mirae Asset Daewoo, etc.), and financial investors (such as Mirae Asset Daewoo, etc.) present a reasonable balance of private capital for project financing. USD 42.8 million from other investors is not confirmed at this stage, and local financial institutions are currently reviewing their participation.

Meanwhile, all the planned projects under the proposed Programme will contribute to building a successful track record, and these will pave ways for private sector's participation in more renewable energy projects in South Pacific SIDS in the future. Although risks still exist in terms of investing in South Pacific SIDS, private sector will be able to consider investing more comfortably if there is an evident precedent of success.

E.6.3. Financial viability

Please specify the expected economic and financial rate of return with and without the Fund's support, based on the analysis conducted in F.1.

The equity ratio to achieve the minimum requirement of simple DSCR (1.2 x) can increase up to approx. 40% of the total Programme size without the GCF involvement. As the volume of equity increases, project sponsors having limited financial capacity are reluctant to participate in the projects, which is likely to result in the failure of financing. On the other hand, without adjusting the equity ratio, the minimum simple DSCF decreases to 1.1, which makes it difficult to attract private lenders.

Table 11 Minimum Simple DSCR before and after the GCF involvement

Project	With the GCF involvement minimum simple DSCR (A)	Without the GCF involvement		
		minimum simple DSCR (B)	Equity ratio to adjust (B) to achieve (A)	Equity ratio for (B) to achieve 1.2
Biomass Power Plant in Sabeto (Sub-component 1.1)	1.19	1.06	38%	38%
Wood Pellet Plant in Wairiki (Component 2)	1.2	1.07	37%	37%

The figures in the table above do not take into account the currency fluctuation. Risks to the investors may increase once the currency volatility is fully considered in the analysis. Fiji Dollar had depreciated by 20% during the Asian financial crisis in 1998 and the global financial crisis in 2009, respectively. It had appreciated since 2009 to peak at 2011, depreciated again till 2015, and recently, FJ Dollar against US Dollar is 0.48. In PNG, the trade weighted exchange rate fell 10.8% from October 2015 to Oct 2016, with the kina slipping 11.4% against the US dollar over that period (USD 0.3485 end of Sept 2015, USD 0.3155 28 Sept 2016).

Thus, the GCF involvement is critical to reduce the risks upon the private sector, and to attain financial viability of projects with terms and conditions at least equal to or more favorable than private sectors. The GCF, as the anchor investor, will play the role of attracting private sector investments.

Please describe financial viability in the long run beyond the Fund intervention.

Please describe the GCF's financial exit strategy in case of private sector operations (e.g. IPOs, trade sales, etc.).

The Programme will generate profit through the PPA signed for 25 year with the FEA and the PPL, and all loans including that from the GCF will be repaid within the operation period. Please refer to Section D.2 for more details.

E.6.4. Application of best practices

Please explain how best available technologies and practices are considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.

<Reforestation after Eradication of African Tulip>

To promote biodiversity after eradication of African tulip. African tulip spread throughout 30,000ha will be eradicated and the area after eradication will be rehabilitated. The tentative species selected for reforestation include native species. The Programme will appreciate two principles when it comes to rehabilitating the harvested land: 1) consideration of the previous land use; and 2) local landowners' desire. (more specified in Environmental Co-benefits of E.3.)

The potential fuel supplier (Eltech³⁴) has acquired Harvesting License for African tulip from the Fijian government, will obtain approval of environmental impact assessment from both the Ministry of Fisheries and Forest and the Department of Environment, and will eradicate the invasive plant in accordance with the Ministry of Fisheries and Forest's Harvesting Code. After eradicating African tulip, reforestation will be managed according to the guidelines set out by IPCC and UNDP-led programme, "Implementing a Ridge to Reef (R2R) approach to Preserve Ecosystem Services, Sequester Carbon, Improve Climate Resilience and Sustain Livelihoods in Fiji", as well as Fijian regulations. Reforestation in line with strict global standards and best practices will enhance biodiversity and local ecosystem.

<SRWC Plantation in Idled and Degraded Lands>

To select an adequate species for plantation in idle and degraded lands. Local Gliricidia has been selected for the SRWC plantation in abandoned ex-sugarcane lands, given its various advantages. Firstly, Gliricidia is a species native to tropical or subtropical climate and at the same time tolerant to a wide range of soil conditions. Secondly, it is also easy to control the propagation since it does not spread seeds and the main means of reproduction is cutting. Moreover, as a nitrogen-fixing tree, Gliricidia has the capacity of recovering soil fertility, and can play a role as a fallow crop. According to risk assessment of the Pacific Island Ecosystems at Risk (PIER), the local Gliricidia is "highly domesticated" in the Pacific context with an extremely low probability of invasiveness, in spite of its fast-growing properties.

To ensure appropriate intervention in plantation management. All the processes of the plantation management will be in line with the IFC Performance Standards, particularly PS6. Under the final supervision and responsibilities of KDB in accordance with AMA and the practical control of sponsors, the Programme's potential supplier is to precisely figure out vulnerabilities and critical characteristics of the primary supply chains, and to stick to global standards for monitoring and managing the plantation work.

To select the right site for plantation. Selected SPCs will have to search for lands that were once utilized for cropping or planting purposes but now left idle due to declining major industries. For instance, in case of the pilot biomass power plant in Nabou, Fiji, a 5,000ha land, which was previously used for pine tree plantation but now abandoned, has been rehabilitated as the plantation site. After signing direct contracts with landowners who own such lands, the landowners will get a job for land management under the lease contract. Such a win-win model has received positive feedbacks from landowners in relation to the pilot biomass power plant in Nabou, Fiji. The same criteria of converting idle land into productive land to raise the quality of life for residents will be applied for the plantation in Fiji as well as PNG.

³⁴ Eltech is the only entity with access rights to African tulip harvesting, approved by the Fiji government. Based on the country ownership principle, the Programme gives priority to the government choice. In that African tulip is primary fuel, Eltech has been an integral part of the Programme since the early development phase.

<Enhancing Renewable Energy Access>

To maintain high combustion efficiency to reduce greenhouse gases emission from the power plant. ³⁵ A single 12MW biomass power plant can be seen relatively small in size equipped with the combustion system of solid fuels, and the combustion efficiency varies depending on the fuel condition. The combustion efficiency is one of the most sensitive parameters that affect the efficiency of the overall plant operation. To reduce GHG emissions from solid fuel-fired power plants, therefore, a high level of the combustion efficiency is particularly important.

To keep the combustion efficiency high, fuels should be uniform in size, moisture level and type. This project, however, does not coincide with the ideal condition. Biomass fuel sources supplied to the plant are less likely to be uniform due to varying conditions during gathering, delivering, and drying processes.

The moving grate system will be installed to resolve the issue of such non-identical types of fuel condition. If collected fuel is too wet, then the fuel may be moved slowly at the drying area so that the fuel can reach down to the appropriate moisture level for combustion. This is possible because the moving grate system is usually divided into several zones – i.e. drying, burning, mixing, and ash zones – and can control the speed of each zone separately. This mechanism allows constant efficiency to be maintained regardless of the type of fuels.

< Mitigating Vulnerability to Cyclone Events>

Tropical cyclone has destructive impacts on plant facilities and forestation. To reduce such vulnerability to cyclone events, mitigation measure will be developed and employed for each project during preparation, construction, and operation in accordance with national regulations as well as with international best practices as follows.

Preparation period. The mitigation plan will be developed to outline strategic approaches for how to compile and monitor cyclone risk, and how responses are coordinated and implemented before, during and after cyclone events. It may have three components: 1) early warning system; 2) incident response; and 3) communication strategy. Every project company for the Programme will nominate the coordinator, who has a role of monitoring, forecasting, and reporting the extreme weather pattern in relation to cyclones.

Constructional period. EPC contractor should have capability to consider a wind force factor for building designing in line with international best practices. Fiji has experienced cyclone occasions with the maximum speed of 70-75m/s which exceeds 57m/s. Therefore, a wind force factor will be applied in a stricter manner.

Operational period. The mitigation plan for cyclone events during operation phases will be prepared by O&M supplier. The specific responsibilities of staff in charge with cyclone contingency plans include operation and supervision of the warning and response systems, and coordination with governmental agencies and relevant stakeholders.

³⁵ The sponsors of the first power plant are now operating the pilot project (Nabou biomass power plant) at the initial stage. During the ramp-up period, the sponsors are trying to find the optimum operating conditions. The know-how and experience of the sponsors are greatly helpful for subsequent projects in this Programme to maintain high combustion efficiency.

E.6.5. Key efficiency and effectiveness indicators

GCF core indicators	Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)	
	(a) Total project financing(except grant)	US\$427,000,000
	(b) Requested GCF amount(except grant)	US\$102,400,000
	(c) Expected lifetime emission reductions overtime	4,736,230tCO ₂ eq
	(d) Estimated cost per tCO ₂ eq (d = a / c)	US\$90.16 / tCO ₂ eq
GCF core indicators	(e) Estimated GCF cost per tCO ₂ eq removed (e = b / c)	US\$21.62 / tCO ₂ eq
	<i>Describe the detailed methodology used for calculating the indicators (d) and (e) above.</i>	
	Please refer to Section E.1.2.	
	<i>Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.</i>	
	Please refer to Section E.1.2.	
GCF core indicators	Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)	
	Participation of the GCF will mobilize private investment at a total of USD 324.6 million. The indicative breakdown of the USD 324.6 million is as follows:	
	- KDB: USD 42.8 million (loan) - Other commercial investors: USD 153.7 million (loan) - Sponsors : USD 128.1 million (Equity)	
	<i>Describe the detailed methodology used for calculating the indicators above. Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.</i>	
	- Equity to debt ratio: based on 3:7 which is a typical ratio applied for private sector projects in developing countries - Other commercial banks: subject to investment appetites of prospective lenders which are based on market and project conditions investigated by KDB	
Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)		The key economic co-benefit is job creation for 2,240 local persons in biomass power plants and a pellet plant. Considering reforestation and plantation, a greater income generation benefit is expected through additional part-time and full-time job opportunities.

F.1. Economic and Financial Analysis

Please provide the narrative and rationale for the detailed economic and financial analysis (including the financial model, taking into consideration the information provided in [section E.6.3](#)).

GIMCO, one of the project sponsors of the first biomass power plant in Sabeto, Fiji (Sub-component 1.1), has conducted feasibility studies for the Programme. Following that, Korea Ratings, an external firm specializing in feasibility studies, has prepared a draft of the feasibility studies as of June 2017. The feasibility studies conducted financial analysis such as profitability analysis, cash flow analysis of loan repayment, and DSCR Analysis of each activity, and pro forma financial statements have been developed for the 12MW Sabeto Project.

The feasibility study claims that the project sponsors have enough capacity of construction, fuel supply, technology application, and operation and maintenance, in terms of risk management and project implementation. A further due diligence is to be carried out at an in-depth level once legal arrangements such as inter-stakeholder agreements and EPC contract are materialized. As an accredited entity, KDB will monitor a series of procedures that properly employed external agencies with reasonable specialty conduct due diligence in accordance with best practices of project finance.

More details can be found from the feasibility study reports (Annex 2) and financial model (Annex 3) attached in Section I.

Based on the above analysis, please provide economic and financial justification (both qualitative and quantitative) for the concessionality that GCF provides, with a reference to the financial structure proposed in [section B.2](#).

The financing structure of this Programme is designed to leverage finance from the GCF to fill the financing gap and induce more private capital to enable the project. As the anchor investor, the GCF will play the role of stabilizing the investment structure to solicit as much private sector participation as possible. Also, considering the nature of private sector projects, financial terms and conditions for the GCF are suggested with the minimum-level concessionality.

Please refer to Section E.6.3 for more information. Additionally, the financial analysis with the GCF participation can be found from the attached financial model (Annex 3).

F.2. Technical Evaluation

Please provide an assessment from the technical perspective. If a particular technological solution has been chosen, describe why it is the most appropriate for this project/programme.

The best technical solutions for the Programme were selected in both areas of forestry and land use management and renewable energy generation.

Forestry and land use Management. To select adequate species for plantation in idled and degraded lands, invasiveness of selected species, *Gliricidia* as reviewed by the reference of the 'Pacific Island Ecosystem at Risk' (PIER) program'. According to the weed risk assessment undertaken by the PIER, *Gliricidia* in the Pacific is identified as being a low risk of invasiveness³⁶.

For more information related to technical evaluation in forestry and land use management, please refer to Section E.6.4.

Renewable Energy Generation. The technologies and measures employed for the biomass power plants have been chosen for the sake of operational safety of the plant. With the boiler efficiency running at 87.9~88.7% Net Calorific Value (NCV) and the turbine efficiency at 28.95~29.78%, the plant will have the gross efficiency of 25.45~26.35% NCV overall. As the parasitic load is expected to be about 10-12%, the net efficiency will be 22.13%-23.37% NCV.

³⁶ http://www.hear.org/pier/wra/pacific/gliricidia_sepium_htmlwra.htm

Hence, the net maximum continuous rating output will be 10.44-10.85MW in case of a 12MW biomass power plant.

- **Steam Boiler & Grate:** the steam boiler uses LAWI EtaComb® Combustion Systems, supplied by LAWI Engineering and designed to withstand a 30-year lifetime. The boiler uses a stoker type with moving grate, which will enable firing multiple biomass fuels, with the boiler efficiency of 87.9-88.6 % NCV. The grate is fabricated from a high chromium alloy (>26% Cr) which should provide suitable resistance to high temperature.
- **Turbine Generator:** the turbine type will be an SST300 of Siemens has about 20 years assumed lifetime at 50MW. The projected turbine efficiency is 28.95-29.78%, with the rated output at the generator terminals 12,000kW, running at the speed of 7500/1500 rpm (turbine/generator).
- **Condensing & Feed-heating Plant:** the heat balance diagram proposes a water-cooled two pass surface (shell and tube) condenser with stainless steel and carbon steel tube plates, and a rolled steel shell. Two condensate extraction pumps are to be provided and there is a steam jet ejector condenser and a gland steam condenser in the main condensate flow path en-route to the de-aerator.

Once in full operation, the expected overall capacity factor is analyzed to be 81% given 90% availability factor and 90% load factor. The technical due diligence has agreed to this figure.

Please refer to Section E.6.4 for efficiency of the technologies that will be applied in renewable energy generation.

F.3. Environmental, Social Assessment, including Gender Considerations

Describe the main outcome of the environment and social impact assessment. Specify the Environmental and Social Management Plan, and how the project/programme will avoid or mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with the Fund's Environmental and Social Safeguard (ESS) standard. Also describe how the gender aspect is considered in accordance with the Fund's Gender Policy and Action Plan.

<Environmental and Social Management Framework>

As an accredited entity, KDB has the guidelines for environmental and social management that are consistent with the GCF Environmental and Social Safeguards (ESS). The ESMF has been developed for the Programme in accordance with KDB guidelines as well as GCF ESS. Basically, the ESMF presents guidance to the project sponsor on the process of how to screen scope and assess each project within the Programme, and manage environmental and social issues that may arise during construction and operation. It also identifies the roles and responsibilities of the environmental and social management organization of the projects and associated agencies including any capacity building requirements.

Key actions and guidance shown in the ESMF for sound environmental and social management of the Programme are summarized below.

Programme overview. The main components of the Programme are feedstock production, biomass power plants, and a pellet plant. Especially, feedstock areas are considered to be associated facilities in accordance with PS1 of IFC PS on ESS.

Screening and scoping. The ESMF involves screening of anticipated environmental and social impacts of the Programme and defines the risk category accordingly as Category A, B, or C. Currently KDB is accredited to submit a Funding Proposal for a programme that is no higher than Category B. Thus, KDB will identify the potential environmental and social impacts of the projects through screening and will continue to carry out the activities as long as the Risk Category is identified to be lower than B. Before finalizing the Risk Category, KDB will seek for advice from a consultancy with global reputation to review whether the categorization is determined appropriately.

ESIA and ESMP. Depending on the risk category and results of the screening, scope of the Environmental and Social Impact Assessment (ESIA) is determined. It is highlighted that the assessment should present proposed measures to minimize, mitigate, and offset adverse impacts in an appropriate manner in relation to the nature and

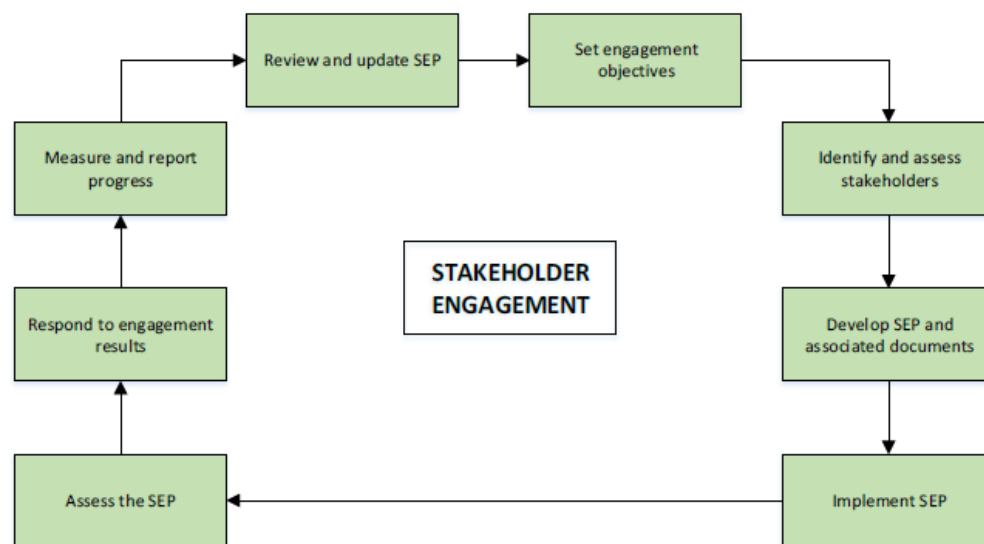
scale of the proposed project. Mitigation and monitoring measures proposed within the ESIA will be detailed within the Environmental and Social Management Plan (ESMP) for each project.

Land Tenor System. In case of Fiji, lands required will be acquired through the iTaukei Land Trust Board (TLTB), which acts as a trustee for the landowners to protect their interests in relation to land lease arrangements. Fiji laws prevent the sale of traditional land but provides for lease agreements of up to 99 years (particularly in relation to idle land) without termination unless the purpose of lease is violated. As for PNG, Customary land which consists of 97% of total areas is held by tribes, clans and land groups with its ownership dictated by local customs and traditionally held values and beliefs. The government is able to acquire or lease customarily owned land, and leases are typically granted for a period of 99 years.

Land requirements will vary on a project by project basis and the impacts of land acquisition will be determined in detail through the ESIA processes for each. Since the land required will be idled and degraded areas without using for agricultural purpose, any tension with the allocation of 'land for energy' and 'land for food' is not expected in this Programme. If the consultation process for the ESIA raises any conflicts or concerns, then appropriate mitigation will be recommended to alleviate them.

Stakeholder engagement. Community and stakeholder engagement is a key component of the environmental and social assessment process throughout the Programme. An outline of a stakeholder engagement process for the specific project is provided as below.

Figure 10 Overview of stakeholder engagement process (SEP = Stakeholder Engagement Plan)



(Source: ESMF)

The programme level stakeholder engagement was conducted at the design stage of the Programme. Representatives from the government – the Ministry Fisheries and Forests, Ministry of Agriculture, Ministry of iTaukei Affairs & Sugar Industry, etc. - attended the consultation meetings, expressing support for the Programme activities including eradication of African tulip, the SRWC plantation, and the biomass and wood pellet plant in Fiji. The details of consultation are summarized in Appendix H of the ESMF attached as Annex 3 in Section I.

Grievance mechanism. The ESMF includes the approach for grievance mechanism. Grievance mechanism is designed to receive and facilitate resolution of concerns and grievances about the project's environmental and social

performance in compliance with the relevant IFC PS 1, 2, 4 and 5. The sponsors will inform the affected communities about the mechanism in the course of the stakeholder engagement process.

More details of the ESMF are elaborated in Section 1~10 of the ESMF attached as Annex 3 in Section I.

<Strategic Environmental and Social Assessment>

The strategic environmental and social assessment (SESA) has been developed as a separate section of the ESMF. The SESA provides relevant institutions and gap analysis, a high level overview of the environmental and social context of the region, and likely impacts in relation to the Programme. Likely impacts at pre-construction, construction and operation phases are described, and key mitigation measures are provided. Since the project screening criteria is designed to avoid diverse, irreversible or unprecedented impacts, the activities of the Programme are likely site-specific and readily addressed with key mitigation measures suggested in the SESA.

More details of the SESA are elaborated in Section 11 of the ESMF attached as Annex 3 in Section I.

<Environmental and Social Impact Assessment of the Sabeto biomass power plant>

The ESIA has been developed for the first 12MW biomass power plant in Sabeto, Fiji. As per the guidance provided in the ESMF, the ESIA presents the areas of influence considering fuel sources, description of the project, the baseline data, environmental and social impacts, management plans, and the result of the consultations.

Land use management. Environmental and social impacts related to fuel sources are likely to be limited and readily addressed through management and mitigation measures. Since plantation areas will be idle and degraded land that is not used for agricultural purposes, there will be no clearance of indigenous habitats, and land use will barely change. In addition, it is assessed that eradication of invasive African tulip will generate beneficial impacts to the biodiversity of areas. The management plan is put in place to avoid likely adverse impacts and ensure compliance with the GCF and IFC safeguards for Forestry and land use management.

Biomass power plant. The physical, ecological, and social and economic impacts were assessed near the site. The anticipated environmental and social impacts related to power plant are generally site-specific, readily addressed with mitigation measures. The mitigation measures are necessarily required during construction and operation in relation to traffic, noise, air quality, etc., and the ESMP is developed into detailed matrix showing responsible institutions, verification measures, cost, etc. All the measures and activities are designed to comply with IFC PS, WB EHS Guidance and Fijian requirements.

Land Acquisition. The site for the power plant is acquired through the TLTB in full agreements with all that have been interested in the land. The plantation areas will be secured by long-term lease agreements or supply contracts. Since these areas have been left idle, landowners are very supportive for making a long-term leasing agreement, anticipating additional economic earnings. In case of supply contracts, the sponsors/suppliers support the land owners to plant the fuel wood trees in their own lands and buy the wooden fuel from them, so that land owners have new income sources without leasing the lands. Therefore, it's unlikely to have conflicts related to land rights.

Consultation. The ESIA contains the results of the consultations that have taken place with various stakeholders at village, local, civil society, local authority, and the central government levels. At the consultation with the community members, questions were raised about potential benefits such as job opportunities during construction and operation, and had discussion over potential impacts related to air quality, health and safety.

A separate consultation meeting was organized for women regarding gender inclusiveness. During the consultation, it was highlighted to the female community members that this project understands the importance of women's participation and the significance of potential roles and benefits for women in the biomass power plant. The project sponsor made a presentation specifying the expected number of jobs for women and briefing how environmental and social impacts will be managed for women, children and other vulnerable groups.

More details are elaborated in the ESIA attached as Annex 3 in Section I.

<Gender mainstreaming>

The Programme values gender mainstreaming, and takes gender sensitive approach. This is particularly important for the countries such as Fiji and PNG which present nearly equal male-female population ratio. However, both countries are still ranked low in Human Development Index, and women's participation in economic and social activities is severely constrained. In case of Fiji, poverty was categorized as one of the critical gender inequality issues by the government³⁷, and for PNG, lacking access to job opportunities and income-generating activities due to limited access to trainings and education restricts women from having equal treatment when compared with men.³⁸

The Programme will encourage gender equality and mainstreaming gender considerations by implementing capacity building activities. Various trainings will be provided particularly targeting female employees to increase technical capacity, and thus as a result, can have diversified income opportunities apart from subsistence farming or any low-paid daily jobs. The trainings will be focused on engineering, O&M of the plants, or administrative works. The Programme aims for a sustainable gender mainstreaming processes that will help female employees to successfully stand as an independent decision maker for their own livelihoods.

Moreover, the Programme will promote hiring female workforce, and make sure female employees are treated in an equal way as male employees in terms of remuneration and working hours. The Programme will also ensure that adequate working environment is provided for women.

F.4. Financial Management and Procurement

Describe the project/programme's financial management and procurement, including financial accounting, disbursement methods and auditing.

<Financial Management>

Financial resources from the GCF will be managed according to the general provisions of the AMA and FAA. In particular, the GCF proceeds will be provided to individual SPCs of the biomass power plants and the wood pellet plant through KDB.

As always named among the top seven PF arrangers in Asia Pacific, KDB Bank has its internal capacity of investment decision with the disciplined risk appetite, and of the systematic credit approval system after a thorough credit due diligence. Equipped with decades of accumulated international capacity, KDB is fully capable to sort out qualified projects with great potential.

In addition, KDB, as an accredited entity, will assure the debt financing repayment by pledging any present or future properties, assets, and revenues of the borrower to the lenders and managing cash flows via an escrow account. The controlling authority may be given to KDB or another qualified independent agent that can represent all lenders in a reasonable manner. Each project SPC would repay debts to KDB which would then repay the debts to the GCF. The expected flow of GCF loans is elaborated in detail in Section C.7.

In compliance with its internal regulation titled 'Project Finance and Syndication Guideline', KDB, will execute financial management of the projects under the Programme. KDB will regularly check whether the Programme is being executed in compliance with requirements such as withdrawals and payments of funds, etc. as outlined in the loan agreements.

KDB will visit the project sites more than once a year to oversee the status of construction or operation, but if the projects are implemented in areas where a regular visit is difficult, KDB will hire a consultant to execute management. Moreover, KDB will make its full efforts to secure project stability by overseeing compliance with affirmative covenants

³⁷ <http://egov4women.unescapsdd.org/country-overviews/fiji/gender-equality-and-women%E2%80%99s-empowerment-priorities-in-fiji>

³⁸ <https://www.adb.org/sites/default/files/linked-documents/cps-png-2011-2015-ga.pdf>

such as maintenance of corporate entity status, submission of financial statements, signing of insurance contracts, implementation of terms related to E&S, etc., and negative covenants such as prohibition of restricted payment, termination and change of contracts, etc. on a regular basis, as outlined in the agreements.

Since the Programme is large-scale and expects a long implementation period, signing of contracts and supervision for several projects are needed over time. Therefore, the SPC may appoint and sign the agreement with an inter-creditor agent bank and/or a security agent bank that will carry out tasks related to the funds and project management. For each sub-component, the agent banks will be in charge of project cash flow management, security management, and withdrawals of loans.

KDB, as the accredited entity of the Programme, will thoroughly supervise all activities related to financial management.

<Procurement>

KDB has a procurement policy that abides by the GCF procurement guidance. The SPCs of each biomass power plant and a wood pellet plant are required to procure and purchase goods and services in accordance with KDB's procurement policy. The method of international open bidding will be strictly applied for selecting the EPC contractor in pursuit of transparency. Such condition will be reflected in legal documents between KDB and the SPCs in order to ensure that each SPC complies with the procurement rules. All the procurement arrangements will be governed by KDB.

More specifically, KDB will ensure the procurement procedures of all projects comply with "the Guidance for GCF-Funded Projects (hereinafter the "Guidance")" established on September 5, 2016. Chapter 2 of the Guidance states the regulations pertaining to borrowers' purchase and procurement procedures. Items relevant to the Programme are as below:

- Article 5. Prevention of Fraud and Corruption
- Article 6. Procurement Method
- Article 7. Pre-qualification of Bidders
- Article 8. Time Interval
- Article 9. Bid Evaluation
- Article 10. Publication of Contract Award

In accordance with Article 5 of the Guidance, in case that the SPC or any entity relevant to the Programme (a company and/or individual) is discovered to have participated in fraudulent or corrupt activities related to procurement, KDB can ban one from participating in the Programme, or can annul the funding. KDB also ensures that any entity relevant to the Programme complies with international regulation on AML/CFT. For prevention of fraud and corruption, the SPC of each (sub) component is obligated to be investigated by KDB for its account books and other documentations regarding contracts, and to submit materials that the GCF requests based on the AMA and FAA.

For matters not mentioned in the Guidance, KDB's internal regulations titled "Contract Regulations" will be applied for observance. KDB's strict procurement oversight will be arranged before, after, and in between bidding procedures according to the Guidance.

Also, the Programme ensures the fair and appropriate procurement process in selecting proper agencies for implementing Component 3 (TA). For a good use of the GCF grant, agencies specialized in capacity building, knowledge sharing, and other TA activities will be selected according to Regulation on Contracts (hereinafter 'Regulation') which provides guidance for procurement activities of KDB, not a borrower. The Regulation was also completely screened by the Fund as part of accreditation due diligence. A specified procurement plan for using grant is attached as Annex 11 of the proposal.

G.1. Risk Assessment Summary

Please provide a summary of main risk factors. Detailed description of risk factors and mitigation measures can be elaborated in G.2.

Several assessments carried out for this Programme have identified some risk factors. Financial, technical, and environmental and social aspects have been thoroughly appraised and this has resulted in classification of seven main risk factors. The technical risks include construction delay, unreliability of the fuel supply, and risks associated with potential off-takers. The analysis indicates that the main financial risk is the case that the construction cost exceeds the estimated investment (cost overrun).

The key risk factors are elaborated with the level of impact and probability of risk occurring below. Mitigation measures for each risk factor are also described in G.2.

G.2. Risk Factors and Mitigation Measures

Please describe financial, technical and operational, social and environmental and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures.

Selected Risk Factor 1

Description	Risk category	Level of impact	Probability of risk occurring
Unreliability of fuel sources	Technical and operational	Medium (5.1-20% of project value)	Low
Mitigation Measure(s)			

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

A stable supply of fuel sources is prerequisite to a stable operation of four biomass power plants and a pellet plant. In order to ensure sustainable fuel supply, biomass fuels will be obtained from various sources: 1) harvested invasive plants; 2) the SRWC plantation; 3) sawmill residues; and 4) other harvesting and forestry residues.

Selected Risk Factor 2

Description	Risk category	Level of impact	Probability of risk occurring
Delay in construction	Technical and operational	Medium (5.1-20% of project value)	Low
Mitigation Measure(s)			

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

It is unlikely that construction delay caused by complaints during the construction period and/or authorization/licensing process is unlikely occur in that 1) the plant is a renewable power plant with environmentally friendly properties, and in that 2) the positive impacts are shared upon the establishment of the plant when it comes to reducing reliance on imported energy sources and improving balance between supply and demand in the power market.

Selected Risk Factor 3

Description	Risk category	Level of impact	Probability of risk occurring
Development of an E&S Category A project	Social and environmental	Medium (5.1-20% of project value)	Low
Mitigation Measure(s)			

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

Based on the ESMF, possible E&S impacts will be thoroughly assessed in the initial stage of project assessment. Risk categorization will be determined for each project in order to confirm Category B projects and at the same time, key mitigation measures developed for each project will be evaluated. The assessment will be conducted by a third party professional organization to obtain objectivity, so the risk probability pertaining to likely E&S impacts is expected to be low.

Selected Risk Factor 4

Description	Risk category	Level of impact	Probability of risk occurring
Cost overrun during the construction phase	Financial	Medium (5.1-20% of project value)	Medium

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

In order to control the risk of cost overruns and cash deficiency during the construction period, the EPC contract will be signed as a Fixed Price Lump-Sum Turnkey model. Accordingly, the risk related to cost overrun is mostly transferred to the EPC contractor. The EPC cost of the project is similar to those of other biomass power plants, and the tentative project sponsors are well-prepared for the risk factor because they already have experienced the pilot project in Nabou, Fiji.

Selected Risk Factor 5

Description	Risk category	Level of impact	Probability of risk occurring
Price fluctuation	Financial	Low (<5% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The sales of electricity will be determined by the PPA tariff rate which will be quarterly adjusted to inflation rates of Fiji's Consumer Price Index (CPI) in order to avoid financial risks induced by price fluctuation.

Selected Risk Factor 6

Description	Risk category	Level of impact	Probability of risk occurring
Market and policy risks	Other	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

The project is designed to sell electricity at a fixed price to FEA through a long-term PPA. The agreement of fixing the price will accordingly mitigate possible risks pertaining to changes in the supply and demand structure of the Fijian electricity market and relevant policy change.

Selected Risk Factor 7

Description	Risk category	Level of impact	Probability of risk occurring
Off-taker risk	Other	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

Please describe how the identified risk will be mitigated or managed. Do the mitigation measures lower the probability of risk occurring? If so, to what level?

FEA, the sole revenue source of the project, stands in authority of regulating power generation, transmission and distribution, and controlling the electricity market in Fiji. The FEA owns approximately 70% of power plants nationwide, and generates about 740 GWh/year (approximately 86.5% of power generation nationwide). Given that the revenue stream of the FEA has been stable and the EBITDA margin has been over 30%, it is assessed that the FEA's profitability is at an excellent level.

In case of PNG, quasi-governmental authorities or the government itself such as PPL will be selected as an off-taker in accordance with the investment criteria.

H.1. Logic Framework.

Please specify the logic framework in accordance with the GCF's [Performance Measurement Framework](#) under the [Results Management Framework](#).

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level ³⁹						
Paradigm shift objectives						
Shift to low-emission sustainable development pathways	The Programme aims to support a paradigm shift to low-emission sustainable development pathways of South Pacific SIDS by developing biomass power plants and a wood pellet plant, which will make significant contribution for the target countries to achieve the NDCs. Simultaneously, the Programme is accompanied by technical assistance that will provide capacity building and knowledge sharing for local stakeholders.					
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Fund-level impacts						
M1.0 Reduced emissions through increased low-emission energy access and power generation	Volume of finance leveraged by Fund funding	Annual performance reports	0	-	324.6 million USD	
	1.1 *Tons of carbon dioxide equivalent (t CO2eq) reduced or avoided as a result of Fund-funded Projects/programmes –gender-sensitive energy access power generation (sub-indicator)	Annual performance reports including CO2 monitoring and validation by independent specialists	0	2,055,565 tCO2eq	4,111,130 tCO2eq ⁴⁰	Life span: 25 years Average annual energy production of a single 12MW biomass power plant: 72,712 MWh Emission factor of Fiji: 0.5965 tCO2eq/MWh Emission factor of PNG: 0.6780 tCO2eq/MWh Emission from pellet export transportation considered (966,910 tCO2eq)
M4.0 Reduced emissions from land use, reforestation, reduced deforestation, and through sustainable forest management and conservation and enhancement of forest carbon stocks	4.1 * Tonnes of carbon dioxide equivalent (t CO2eq) reduced or avoided (including increased removals) as a result of Fund-funded projects/programmes – forest and land-use sub-indicator	Annual performance reports including CO2 monitoring and validation by independent specialists	0	312,550 tCO2eq	625,100 tCO2eq	Life span: 25 years

³⁹ Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that some indicators are under refinement):

http://www.greenclimate.fund/documents/20182/239759/5.3_-_Performance_Measurement_Frameworks_PMF_.pdf/60941cef-7c87-475f-809e-4ebf1acbb3f4

⁴⁰ Please note that the mid-term and final amount of emission reduction is an indicative one based on brief analysis, and more detailed analysis will be conducted with TA grant provided by the GCF.

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level						
Expected Result	Indicator	Means of Verification (MoV)	Base -line	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/programme outcomes	Outcomes that contribute to Fund-level impacts					
M6.0 Increased number of small, medium and large low-emission power suppliers	6.2 Number of households and individuals (males and females) with improved access to low-emission energy sources	Annual performance reports, Reports from recipients	0	n/a	Approximately 77,102 households (48,298 in Fiji and 28,804 in PNG)	373,869 persons (221,205 in Fiji and 152,664 in PNG) including 183,609 females ⁴¹
	6.3 MWs of low emission energy capacity installed, generated and/or rehabilitated as a result of GCF support	Annual performance reports, Reports from recipients	0	12 MW	72 MW	24MW of the 72MW will be used for the wood pellet plant's consumption
M9.0 Improved management of land or forest areas contributing to emissions reductions	9.1 Hectares of land or forest under improved and effective management that contributes to CO ₂ emission reductions	Annual performance reports, Reports from recipients	0	Reforestation of 5,000ha of land after eradication of African tulip and development of 5,000ha of plantation	Reforestation of 30,000ha of land after eradication of African tulip and development of 55,000ha of plantation	
Project/programme outputs	Outputs that contribute to outcomes					
Component 1. Development of four biomass power plants						
1. 12MW biomass power plant #1 in Fiji	MW of power plant	Annual performance reports, Reports from recipients	0	12MW	12MW	
	Hectares of Reforestation	Annual performance reports, Reports from recipients	0	5,000ha ⁴²	5,000ha	
	Hectares of plantation	Annual performance	0	5,000ha	5,000ha	

⁴¹ More details are elaborated in Section E.1.2.

⁴² The target of reforestation (areas of African tulip eradication) is subject to change in accordance with surrounding circumstances. The targets of Hectares of Reforestation under Project/programme outputs 1-3 and 5 in mid-term and final are indicative and to be fixed with in-depth assessments.

		reports, Reports from recipients				
2. 12MW biomass power plant #2 in Fiji	MW of power plant	Annual performance reports, Reports from recipients	0	-	12MW	
	Hectares of Reforestation	Annual performance reports, Reports from recipients	0	-	5,000ha	
	Hectares of plantation	Annual performance reports, Reports from recipients	0	-	5,000ha	
3. 12MW biomass power plant #3 in Fiji	MW of power plant	Annual performance reports, Reports from recipients	0	-	12MW	
	Hectares of Reforestation	Annual performance reports , Reports from recipients	0	-	5,000ha	
	Hectares of plantation	Annual performance reports , Reports from recipients	0	-	5,000ha	
4. 12MW biomass power plant #4 in PNG	MW of power plant	Annual performance reports , Reports from recipients	0	-	12MW	
	Hectares of plantation	Annual performance reports , Reports from recipients	0	-	5,000ha	
Component 2. Development of one wood pellet plant						
5. Wood pellet production infrastructure in Fiji	Number of wood pellet production infrastructure supplying 500,000 ton per annum of wood pellets	Annual performance reports , Reports from recipients	0	-	1	
	Hectares of Reforestation	Annual performance reports, Reports from recipients	0	-	15,000ha	
	Hectares of	Annual	0	-	35,000ha	

	plantation	performance reports, Reports from recipients				
6. Capacity increase for existing transportation infrastructure in Fiji	Number of wood pellet transport infrastructure	Annual performance reports, Reports from recipients	0	-	1	
7. 24MW biomass power plant for the wood pellet plant in Fiji	MW of power plant	Annual performance reports, Reports from recipients	0	-	24MW	
Component 3. Technical Assistance						
8. Capacity enhancement and knowledge sharing in Fiji	Number of stakeholders participating in capacity building and knowledge sharing events	Annual performance reports, Reports from consultants	0	75	150	Main stakeholders are local staffs and contractors, government officials,, local community, , CSO including women.
9. Development of CO2 calculation methodology in Fiji	Number of CO2 savings assessment reports	CO2 savings assessment reports	0	0	1	
10. Capacity enhancement and knowledge sharing in PNG	Number of stakeholders participating in the capacity building and knowledge sharing events	Annual performance reports, Reports from consultants	0	100	200	Main stakeholders are local staffs and contractors, government officials,, local community, , CSO including women and indigenous people
11. Creating enabling environments for renewables in PNG	Number of reports for setting the model framework	Reports for setting the model framework	0	0	1	
	Number of training for enhancing institutional governance system	Annual performance reports	0	0	3	Public officials and other stakeholders
	Number of reports for mapping the land use patterns	Reports for mapping the land use patterns	0	0	1	
12. Development of CO2 calculation methodology in PNG	Number of CO2 savings assessment reports	CO2 savings assessment reports	0	0	1	
Activities	Description				Inputs	
Component 1. Development of four biomass power plants						
Sub-Component 1.1. Biomass power plant #1 in Sabeto, Fiji						
1.1.	Project Preparation (Permit and Basic Engineering) for the				Project	USD

	development of 12MW biomass power plant #1 in Fiji	development by project sponsors and participation of technical experts	50.8 mil
1.2.	Project Execution (EPC) for the development of 12MW biomass power plant #1 in Fiji	Participation of global EPC companies	
Sub-Component 1.2. Biomass power plant #2 in Fiji			
2.1.	Project Preparation (Permit and Basic Engineering) for the development of 12MW biomass power plant #2 in Fiji	Project development by project sponsors and participation of technical experts	USD 57.7 mil
2.2.	Project Execution (EPC) for the development of 12MW biomass power plant #2 in Fiji	Participation of global EPC companies	
Sub-Component 1.3. Biomass power plant #3 in Fiji			
3.1.	Project Preparation (Permit and Basic Engineering) for the development of 12MW biomass power plant #3 in Fiji	Project development by project sponsors and participation of technical experts	USD 57.7 mil
3.2.	Project Execution (EPC) for the development of 12MW biomass power plant #3 in Fiji	Participation of global EPC companies	
Sub-Component 1.4. Biomass power plant #4 in Fiji			
4.1.	Project Preparation (Permit and Basic Engineering) for the development of 12MW biomass power plant #4 in PNG	Project development by project sponsors and participation of technical experts	USD 57.8 mil
4.2.	Project Execution (EPC) for the development of 12MW biomass power plant #4 in PNG	Participation of global EPC companies	
Component 2. Development of one wood pellet plant			
Sub-Component 2.1. Pellet production infrastructure			
5.1.	Project Preparation (Permit and Basic Engineering) for the development of wood pellet production infrastructure in Fiji	Project development by project sponsors and participation of technical experts	USD 91.7 mil
5.2.	Project Execution (EPC) for the development of wood pellet production infrastructure in Fiji	Participation of global EPC companies	
Sub-Component 2.2. Capacity increase for existing transportation infrastructure			
6.1.	Project Preparation (Permit and Basic Engineering) for capacity increase for existing transportation infrastructure in Fiji	Project development by project sponsors and participation of technical experts	USD 23.1 mil
6.2.	Project Execution (EPC) for capacity increase for existing transportation infrastructure in Fiji	Participation of global EPC companies	

Sub-Component 2.3. Biomass power plants for the wood pellet plant			
7.1.	Project Preparation (Permit and Basic Engineering) for the development of biomass power plant for the wood pellet plant in Fiji	Project development by project sponsors and participation of technical experts	USD 88.2 mil
7.2.	Project Execution (EPC) for the development of biomass power plant for the wood pellet plant in Fiji	Participation of global EPC companies	
Component 3. Technical Assistance			
Sub-Component 3.1. Technical Assistance in Fiji			
8.1.	Project Preparation (Consultant procurement) for capacity enhancement and knowledge sharing in Fiji	Consultant procurement process managed by the AE	USD 1 mil
8.2.	Project Execution for capacity enhancement and knowledge sharing in Fiji	Participation of capacity building and knowledge sharing experts	
9.1.	Project Preparation (Consultant procurement) for development of CO2 calculation methodology in Fiji	Consultant procurement process managed by the AE	
9.2.	Project Execution for development of CO2 calculation methodology in Fiji	Participation of experts in CO2 calculation	
Sub-Component 3.2. Technical Assistance in PNG			
10.1.	Project Preparation (Consultant procurement) for capacity enhancement and knowledge sharing in PNG	Consultant procurement process managed by the AE	USD 3 mil
10.2.	Project Execution for capacity enhancement and knowledge sharing in PNG	Participation of capacity building and knowledge sharing experts	
11.1.	Project Preparation (Consultant procurement) for setting enabling environments for renewables in PNG	Consultant procurement process managed by the AE	
11.2	Project Execution for setting enabling environments for renewables in PNG	Participation of experts in policy, financial and technical consultancy	
12.1.	Project Preparation (Consultant procurement) for development of CO2 calculation methodology in PNG	Consultant procurement process managed by the AE	
12.2.	Project Execution for development of CO2 calculation methodology in PNG	Participation of experts in CO2 calculation	

H.2. Arrangements for Monitoring, Reporting and Evaluation

Besides the arrangements (e.g. semi-annual performance reports) laid out in AMA, please provide project/programme specific institutional setting and implementation arrangements for monitoring and reporting and evaluation. Please indicate how the interim/mid-term and final evaluations will be organized, including the timing.

Please provide methodologies for monitoring and reporting of the key outcomes of the project/programme.

<Monitoring, Reporting and Evaluation>

Monitoring, reporting and evaluation arrangement will comply with requirements established between the GCF and KDB in the AMA. KDB shall provide to the Fund the following reports prepared in a form and manner compliant with the practices and procedures of the Fund :

- (a) Annual performance reports on the status of each project throughout the implementation period, including a narrative report on implementation progress based on the logical framework submitted in H.1 (above) and considerations on the ongoing performance of the project against the Fund's investment framework criteria, and a report on ESS as well as gender. Upon the completion of the implementation period, KDB shall provide a performance report about the Programme results and mitigation impacts on a semi-annual basis⁴³.
- (b) Interim and final evaluation reports, setting out any necessary corrective measures (in the case of interim reports), an assessment of the performance of this Programme against the Fund's investment framework criteria, including financial/economic performances as part of the Funded Activity efficiency and effectiveness criterion, as well as the sustainability and scalability of results and impacts and lessons learned.

Also, KDB shall provide the financial information on an annual basis, which will be audited and reported on by external auditors in accordance with ISA. The financial information including but not limited to the following content will be included;

- the dates and amounts of disbursements for the period reported and cumulative amounts up to the period, and compliance with financial covenants;
- the actual expenditures for the period reported and cumulative amounts up to the period;
- a statement of investment income earned on GCF proceeds, as well as the amount of such investment income paid to the Fund

In preparing the reports, KDB may hire an external expert in order to secure expertise and objectivity. The team which is in charge of matters regarding the GCF is a separate team from the Programme implementing team within KDB, and will review the report and evaluate its appropriateness prior to submission to the GCF.

<Calculation of CO₂ Emission Reduction>

Upon implementation of each project, KDB will assess the reduced amount of CO₂ emission independently through an external organization specialized in the calculation, and estimate and monitor the real amount of reduction on a yearly basis over the lifetime of the Programme. KDB's environmental specialist will review the propriety of the overall process of assessment, estimation, and monitoring of CO₂ emission.

⁴³KDB will conduct reporting the audited Fund-level impact indicator of M 4.0 (H.1) on an annual basis.

I. Supporting Documents for Funding Proposal

- ☒ NDA No-objection Letter (*Annex 1*)
- ☒ Feasibility Study (*Annex 2*)
- ☒ Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable) (*Annex 3*)
- ☐ Confirmation letter or letter of commitment for co-financing commitment (If applicable) (*Annex 4*)
- ☒ Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.) (*Annex 5*)
- ☒ Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan (If applicable) (*Annex 6*)
- ☐ Appraisal Report or Due Diligence Report with recommendations (If applicable) (*Annex 7*)
- ☐ Evaluation Report of the baseline project (If applicable) (*Annex 8*)
- ☐ Map indicating the location of the project/programme (*Annex 9*)
- ☐ Timetable of project/programme implementation (*Annex 10*)
- ☒ Indicative procurement plan for grant (*Annex 11*)
- ☒ CO2 calculation logics (*Annex 12*)
- ☒ Gender & Social Inclusion Analysis and Action Plan (*Annex 13*)

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*



MINISTRY OF ECONOMY

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 Website: www.economy.gov.fj, Email: EconomyInformation@economy.gov.fj
 Ro Lalabalavu House, 370 Victoria Parade, Suva

10 August 2017

Mr Howard Bamsey
 Executive Director
 The Green Climate Fund
 Songdo Business District
 Republic of Korea

Dear Executive Director

Funding Proposal by the Korea Development Bank on Biomass Energy Programme in the South Pacific

We refer to the programme 'Biomass Energy Programme in the South Pacific' in the Republic of Fiji as included in the funding proposal submitted by the Korea Development Bank to us on 29 June 2017.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- a) The Fijian Government has no-objection to the programme as included in the funding proposal;
- b) The programme as included in the funding proposal is in conformity with Fiji's national priorities, strategies and plans;
- c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations; and
- d) Institutions involved (Korea Development Bank and the Korea Consortium comprising of GIMCO Co. Ltd; Mirae Asset Daewoo Co. Ltd; and GS Power Co. Ltd) will undertake consultations with the Fijian Government in developing a commercially viable power purchase agreement for the proposed biomass plant projects.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Thank you.

Yours sincerely

Makereta Konrote
Permanent Secretary for Economy
National Designated Authority for the Green Climate Fund



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Climate Change and Development Authority
 Office of the Managing Director

Avara Annex Building, Brampton St, Port
 Moresby CBD. Papua New Guinea

P O Box 4017, Boroko 111, NCD.
 Papua New Guinea

Date: 19th January, 2018

MR. HOWARD BAMSEY
 Executive Director
 Secretariat of the Green Climate Funds
SONGDO
 Republic of Korea

Dear Mr. Howard,

Subject: No objection letter for the Project Proposal prepared by the Korean Development Bank (KDB) titled 'Biomass Energy Programme in the South Pacific'

We refer to the revised project proposal on the Biomass Energy Programme in the South Pacific submitted by KDB to us on the 1st of September, 2017.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the Project Proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Papua New Guinea has no-objection to the revised project proposal
- (b) The project proposal is in conformity with Papua New Guinea's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Yours Sincerely,

MR RUEL YAMUNA
 Managing Director
 Climate Change and Development Authority of Papua New Guinea



CC:

1. Hyo Woon Eom, Head of Global PF Strategy Team
2. Suwon Yi, Senior Manager. Korean Development Bank

Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Biomass Energy Programme in the South Pacific
Accredited entity	Korea Development Bank
Environmental and social safeguards (ESS) category	Category B

Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	2018-01-18
Language(s) of disclosure	English
Link to disclosure	https://www.kdb.co.kr/ih/simplejsp.do?actionId=ADIHGIGI30002&menuId=IHIHEN0021&localeCode=en&orgC=001&notiNo=5 Project: A ESIA provides information related to the construction and operation of the Sabeto biomass power plant in Fiji and associated fuel supply chain. Programme: A Environmental and Social Management Framework (ESMF) provides description of the programme-level environmental and social assessments of risks and impacts.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	2018-01-18
Language(s) of disclosure	English
Link to disclosure	https://www.kdb.co.kr/ih/simplejsp.do?actionId=ADIHGIGI30002&menuId=IHIHEN0021&localeCode=en&orgC=001&notiNo=5 Project: A ESMP for the Sabeto biomass power plant in Fiji, consistent with the requirements for a category B project is contained in the Environmental and Social Impact Assessment (ESIA). Programme: The Environmental and Social Management Framework (ESMF) consistent with the requirements for the programme.
Resettlement Policy Framework (RPF) (if applicable)	
Date of disclosure on accredited entity's website	2018-01-18
Language(s) of disclosure	English
Link to disclosure	https://www.kdb.co.kr/ih/simplejsp.do?actionId=ADIHGIGI30002&menuId=IHIHEN0021&localeCode=en&orgC=001&notiNo=5 The RPF is integrated in the Environmental and Social Management Framework (ESMF).
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Not applicable